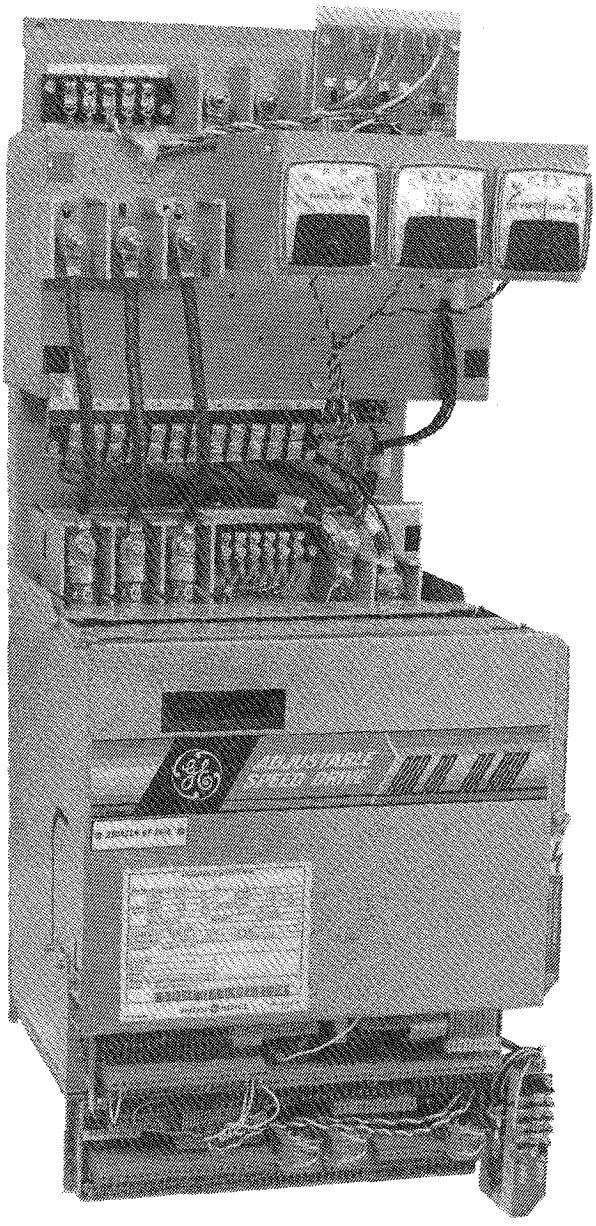
GE Drive Products Operation

INSTRUCTIONS



DFM-300 Digital Field Module

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

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1 – SAFETY PRECAUTIONS

WARNINGS, CAUTIONS, AND NOTES AS USED ON EQUIPMENT

WARNING

Denotes operating procedures and practices that may result in personal injury or loss of life if not correctly followed.

When a warning label is provided it will be red in color with black or white lettering.

CAUTION

Denotes operating procedures and practices that, if not strictly observed, may result in damage to, or destruction of the equipment.

When a caution label is provided it will be amber in color with black lettering.

NOTE: *Notes merely call attention to information that is especially significant in understanding and operating the equipment.*

When a note label is provided it will be white in color with black lettering.

The following paragraphs list some general safety reminders and safety recommendations to be followed when operating or installing this equipment.

WARNINGS

Personal Injury Hazard: Always read the complete instructions prior to applying power or troubleshooting the equipment. Follow the start-up procedures step by step.

Personal Injury Hazard: Read and heed all WARNING, CAUTION and NOTE labels posted on the equipment.

Personal Injury Hazard: Improper lifting practices can cause serious or fatal injury. Lift only with adequate equipment and trained personnel.

Burn and Shock Hazard: Electric shock can cause personal injury or loss of life. Circuit breakers, if supplied as part of the total system, may not disconnect all power to the equipment. See system elementary diagrams. Whether the AC voltage supply is grounded or not, high voltage to ground will be present at many points.

WARNINGS

Burn and Shock Hazard: When instruments such as oscilloscopes are used to work on live equipment great caution must be used. The oscilloscope's chassis should be grounded and a differential amplifier input should be used. Care should be used in the selection of probes and leads and in the adjustment of the oscilloscope so that accurate readings may be made. See oscilloscope manufacturer's instruction book for proper operation and adjustments to the oscilloscope.

Burn and Shock Hazard: Some controllers are furnished with partial enclosures open at the top and bottom. These are intended only for mounting in another enclosure or in a control room having access by qualified personnel only.

Explosion and Fire Hazard: Explosions or fires might result from mounting drive controllers in hazardous areas such as locations where flammable or combustible vapors or dusts are present. Drive controllers should be installed away from hazardous areas, even if used with DC motors suitable for use in these locations.

Shock Hazard: All motor bases and equipment enclosure housings should be connected to the factory or facility earth grounding systems.

CAUTIONS

Do not remove connections or printed circuit cards from the equipment while power is applied. This can damage the equipment.

Do not remove input power from the drive until it has fully executed a stop sequence, as this can damage the drive system.

Installation wiring must be in accordance with the national electrical code and be consistent with all local codes. secondaries of 115 volt control transformers typically have one side fused and the other grounded or available for grounding by the user.

CAUTIONS

The elementary diagrams supplied with the controller identify some signal connections as "low level analog and or digital signals" and indicate special wiring practices. These signals should be isolated from all other wiring (particularly any power wires) and run in a separate conduit or wire run to prevent possible misoperation of the controller.

Meggering can damage electronic components. Do not megger or hi pot without consulting **GE Drive Systems.**

Do not connect any external circuits other than those shown on the elementary diagram. Connection of devices such as ammeters on the shunt or voltmeters on the tachometer may degrade the performance of the drive system.

Do not use power factor correction capacitors with this equipment without consulting **GE Drive Systems.** Damage may result from high voltages generated when capacitors are switched.

2 – RECEIVING AND STORAGE

RECEIVING

The controller should be placed under adequate cover immediately upon receipt as packing cases are not suitable for out-door or unprotected storage. Each shipment should be carefully examined upon arrival and checked with the packing list. Any shortage or damage should be reported promptly to the carrier. If required, assistance may be requested from **GE DRIVE SYSTEMS, ERIE, PA.** When seeking assistance please use serial number, requisition number, model number and drive code to identify the equipment **(Telephone 814-875-3851).**

STORAGE

If this controller is not to be installed immediately, it should be stored in a clean, dry location at ambient temperatures from -20°C (-4°F) to 55°C (131°F). The surrounding air must be free of chemical and electrically conductive or corrosive contaminants.

Precautions should be taken to prevent condensation from forming within the equipment enclosure. If the storage environment exceeds a 15°C (27°F) drop in temperature at 50% humidity over a 4-hour period, a space heater should be installed inside each enclosure to prevent condensation. (A 100 watt lamp can sometimes serve as a substitute source of heat). Higher humidities with smaller temperature changes can also cause condensation.

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3 – INSTALLATION GUIDELINES

LOCATION

SCR controllers are suitable for most factory areas where industrial equipment is installed. They should be installed in well ventilated areas with ambient temperatures ranging from 0°C (32°F) to 40°C (104°F) and relative humidities up to 90%. It should be recognized, however, that since the life expectancy of any electronic component decreases with increased ambient temperature, reduction of the ambient temperature will bring about extended component life. For example, longer component life should be expected if the ambient temperature is held between 20°C (68°F) and 30°C (87°F).

Proper performance and normal operational life can be expected by maintaining a proper environment for the drive system.

WIRING LEVELS

1. Wiring should be run in separate conduits or wireways for signal, control and power wiring levels.
   a. Signal – Low level analog and digital signals
      – Speed and/or position signals
      – Power supplies and logic signals (See system schematics and notes to identify signal level wires)
   b. Control – AC or DC control circuits
   c. Power – Field leads, armature leads
      – Brakes, 115 VAC circuits
2. Signal wiring and power wiring may cross at right angles with a minimum one-inch separation. Avoid parallel runs between signal level wires and power or control wires. If signal wires must be run in parallel to control or power wires, a minimum of a four-inch separation must be maintained between the wires.

GROUNDING

1. Drive common (PCOMX) should be grounded at **only one** point. If the reference is supplied by numerical control or process instrument with a grounded common, the drive common should not be grounded separately.
2. If an isolation transformer is used and must be grounded, it is recommended that a high resistance ground be used.
3. Shields for shielded and twisted, shielded wire must be grounded on one end only. Provision is made to tie shields to chassis ground at the drive I/O. It is preferred to have the shield grounded only at the drive end.

COMMONS

Connection of reference, meter or any other external, commons can only be made at 3TB-PCOMX, unless otherwise indicated by the system elementaries.

SUPPRESSION

Control system relays, solenoids or brake coils can produce erratic drive behavior due to electrical noise transients. To eliminate this possibility, an RC suppressor should be added in parallel with the coils of these devices. A 220 ohm, 2 watt resistor in series with 0.5 mfd, 600 volt capacitor can be used.

TIGHTEN ALL CONNECTIONS

1. Incoming AC line connections.
2. Power connections to motor armature and motor field.
3. 1TB, 2TB, 3TB, 4TB and MATB if furnished.

ADDITIONAL CONSIDERATIONS

Attention should be given to the National Electrical Code and any applicable local codes when installing any drives. Wire size and insulation type, conduit sizing, enclosures, etc., should be determined per these codes.

Environments which include excessive amounts of one or more of the following characteristics should be considered hostile to drive performance and life:

1. Dust, dirt or foreign matter.
2. Vibration or shock.
3. Moisture or vapors.
4. Rapid temperature excursions or high ambient temperatures.
5. Caustic fumes.
6. Power line fluctuations.

7. Electromagnetic interference or **"noise"** introduced by:
   a. Radio frequency signals, typically from portable transmitters used in the vicinity of the equipment or its wiring.
   b. Stray high voltage or high frequency signals as might be provided by arc welders, or by the operation of unsuppressed relay, or contactor, or brake coils as a part of, or in the vicinity of, drive control circuits.

MOUNTING

Totally enclosed controllers should be positioned to permit heat radiation from all surfaces.

A wall-mounted enclosure may be placed side by side with another enclosure. Clearance at least to the width of the enclosure doors should be available in front so that the door may be fully opened for easy access.

Wall-mounted controllers may be mounted on any firm, reasonably flat, vertical surface.

4 – MAIN CONTROLLER ELEMENTS AND THEIR FUNCTIONS

GENERAL DESCRIPTION

The DFM-300 Digital Field Module is a digitally programmable, microprocessor based controller designed specifically to provide regulation in a system utilizing a Motor-Generator set driving a DC motor armature. The DFM-300 controller provides generator field control, regulating the generator output to the DC motor armature. Motor field excitation and regulation is provided by a coordinated motor field control (MFC) package. Overall drive performance is monitored and controlled with protection provided for the prime mover, generator and DC motor. Drive parameters such as minimum and maximum speed, acceleration and deceleration rates, etc. are stored digitally in memory for drift free accuracy. Figures 4-1, 4-2 and 4-3 show the DFM-300 controller.

The controller provides over 100 diagnostic messages which can be decoded via an LED diagnostic readout or an optional Programmer module with alpha-numeric readout. Diagnostic messages can also be read via an optional coded 8-wire output or over the optional RS422/232C communication link. Drive operating parameters stored in the EEROM may also be examined and changed via the RS422/232C link. In addition, the DFM-300 controller provides diagnostics which can check the SCRs, and provide a power-up self check down to the printed circuit board level.

The controller incorporates an outer speed regulator loop with an inner current regulator (or optional voltage loop). In situations where speed accuracy is not a requirement, the speed regulator can be converted to a CEMF regulator. Individually timed linear acceleration and deceleration is adjustable over rates of 0 (current limit acceleration) to 999.9 seconds. Optional extended linear time and "S-Curve" are also available.

Analog voltage or optional current, frequency or digital inputs can be used as the references. Likewise, a wide variety of tach generators can be interfaced to the drive including both AC or DC analog tachs, or pulse tachs.

Output metering signals for armature current, armature voltage, field current, speed, torque and horsepower are also available from the DFM-300 controller. In addition, all signals can be brought out as analog signals via one of three optional digital-to-analog converters or digitally via the optional RS422/232C serial link. (These signals and others can be programmed to activate optional programmable signal level detectors.)

POWER CIRCUITS

The block and "flow" diagrams in Figures 4-4 and 4-5 show the operation of a basic DFM-300 drive. Power is fed through the AC fuses (or an optional circuit breaker), current transformers and MA contactor, and enters the power conversion SCR modules where it is converted to adjustable DC voltage. DC current is fed though shunt and a DC link fuse to the DC generator field. The generator output is in turn controlled and connected through contactor MD to the DC motor armature.

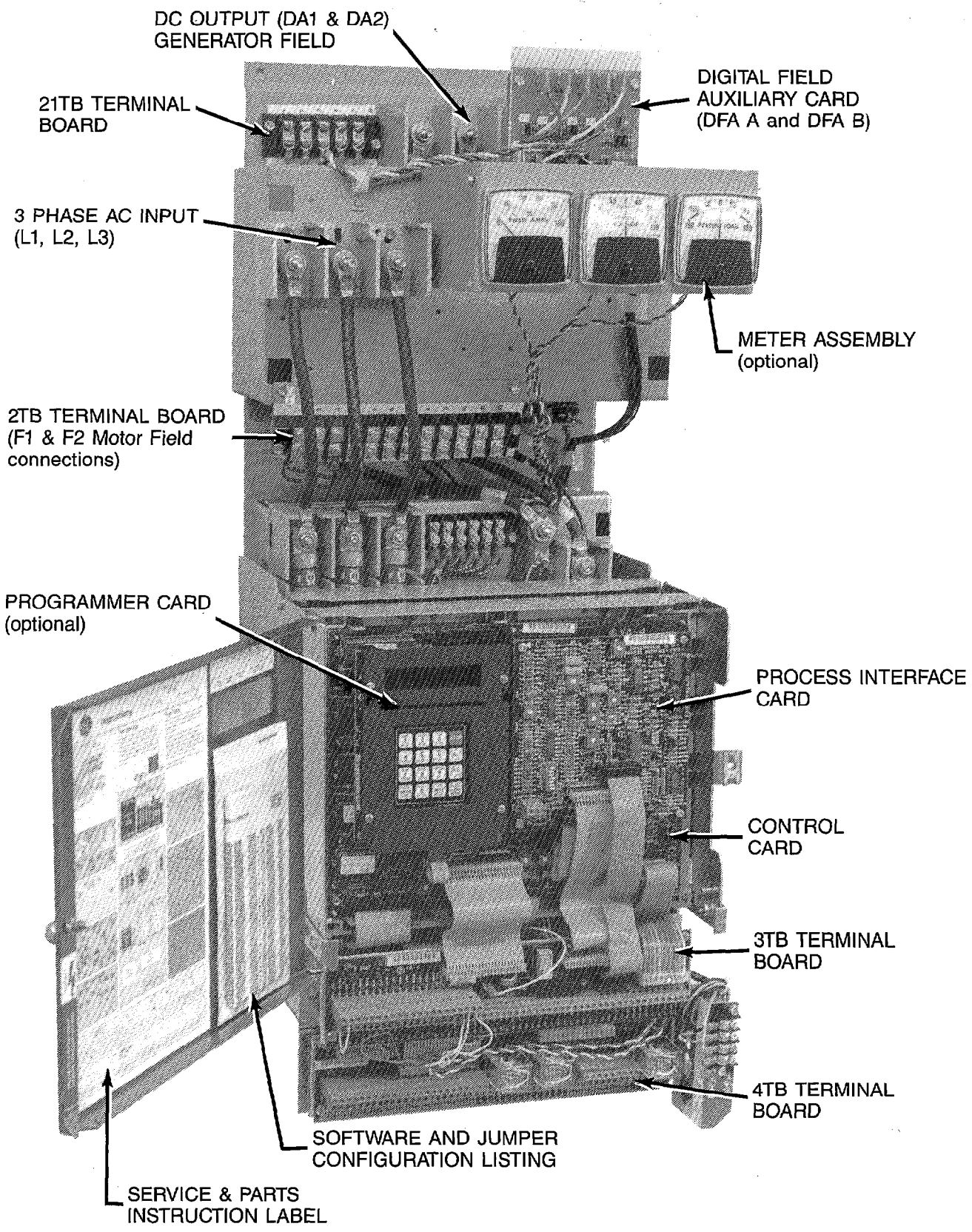


FIGURE 4-1
DFM-300 CONTROLLER WITH CARD DOOR OPEN

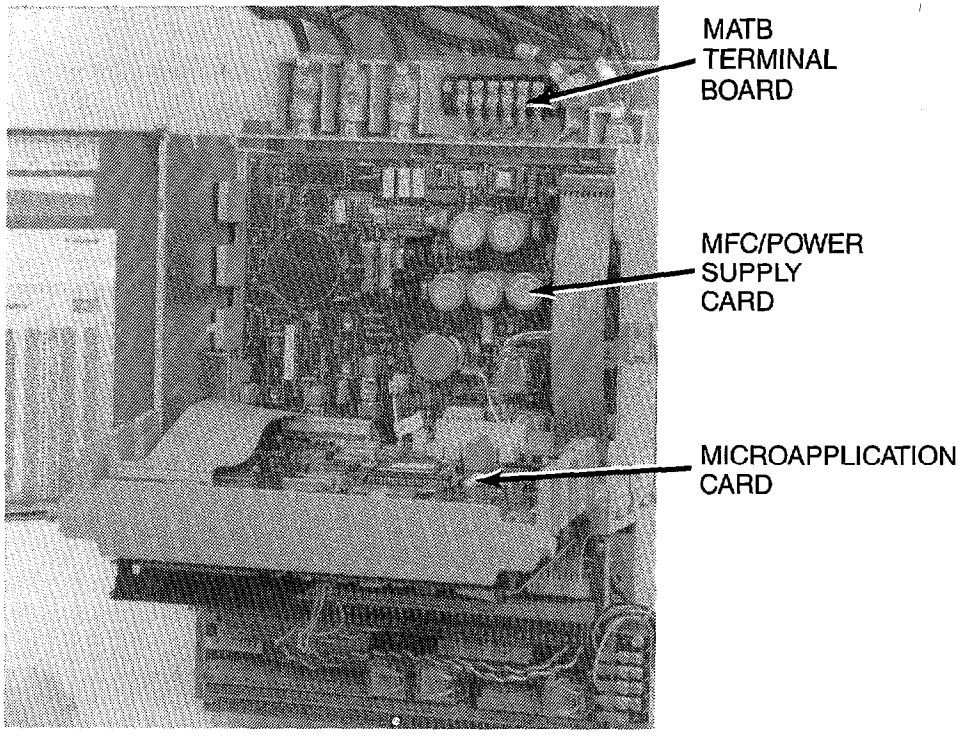


FIGURE 4-2
CONTROLLER WITH FRONT CARD CARRIER LOWERED

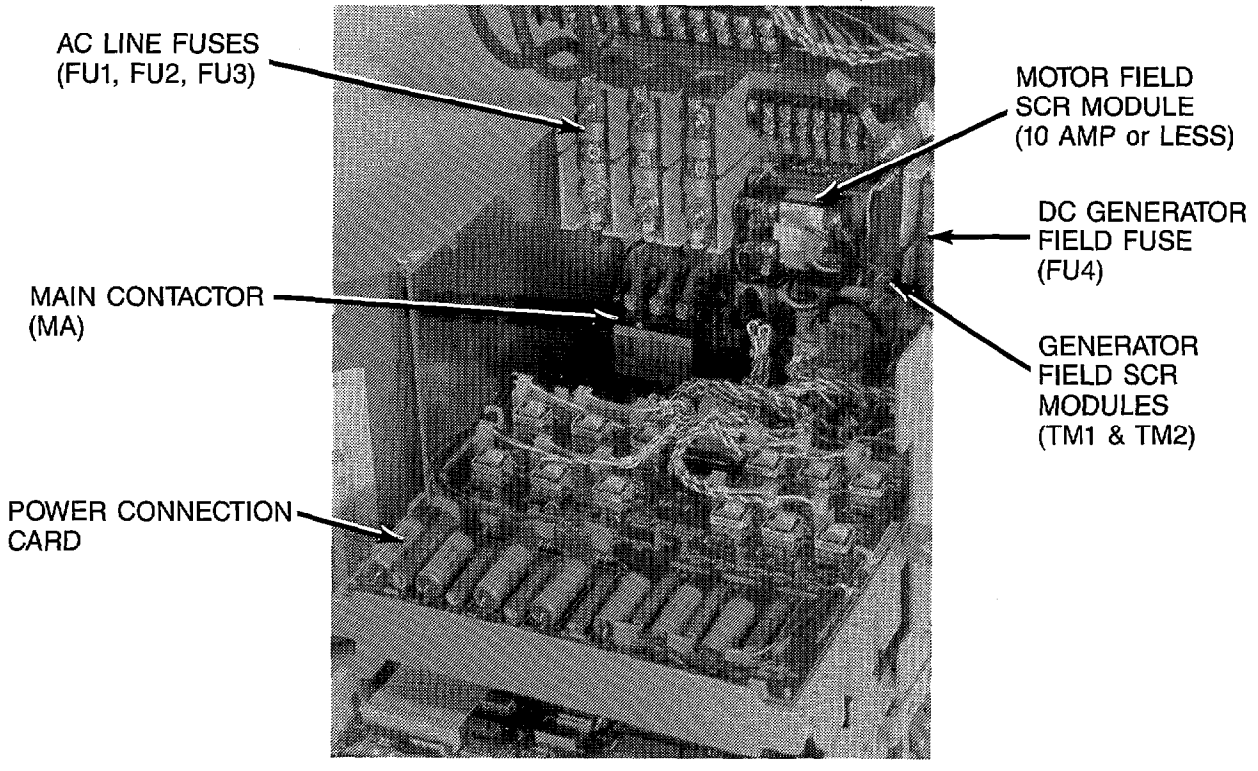
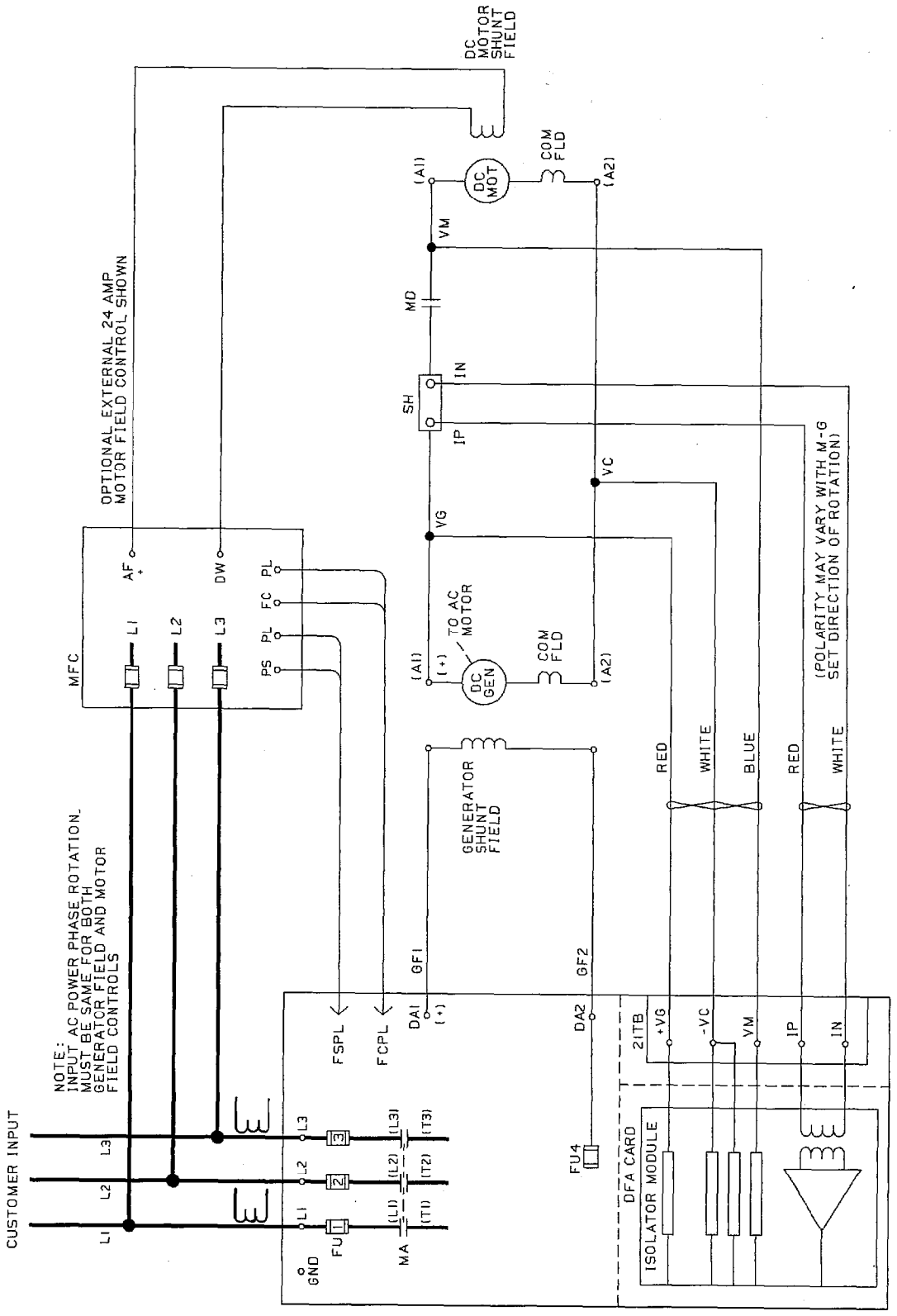


FIGURE 4-3
CONTROLLER WITH REAR CARD CARRIER LOWERED



* See Elementary Diagrams, Figures 4-6 and 4-7 for different version DFA cards.

FIGURE 4-4
TYPICAL POWER CONNECTION

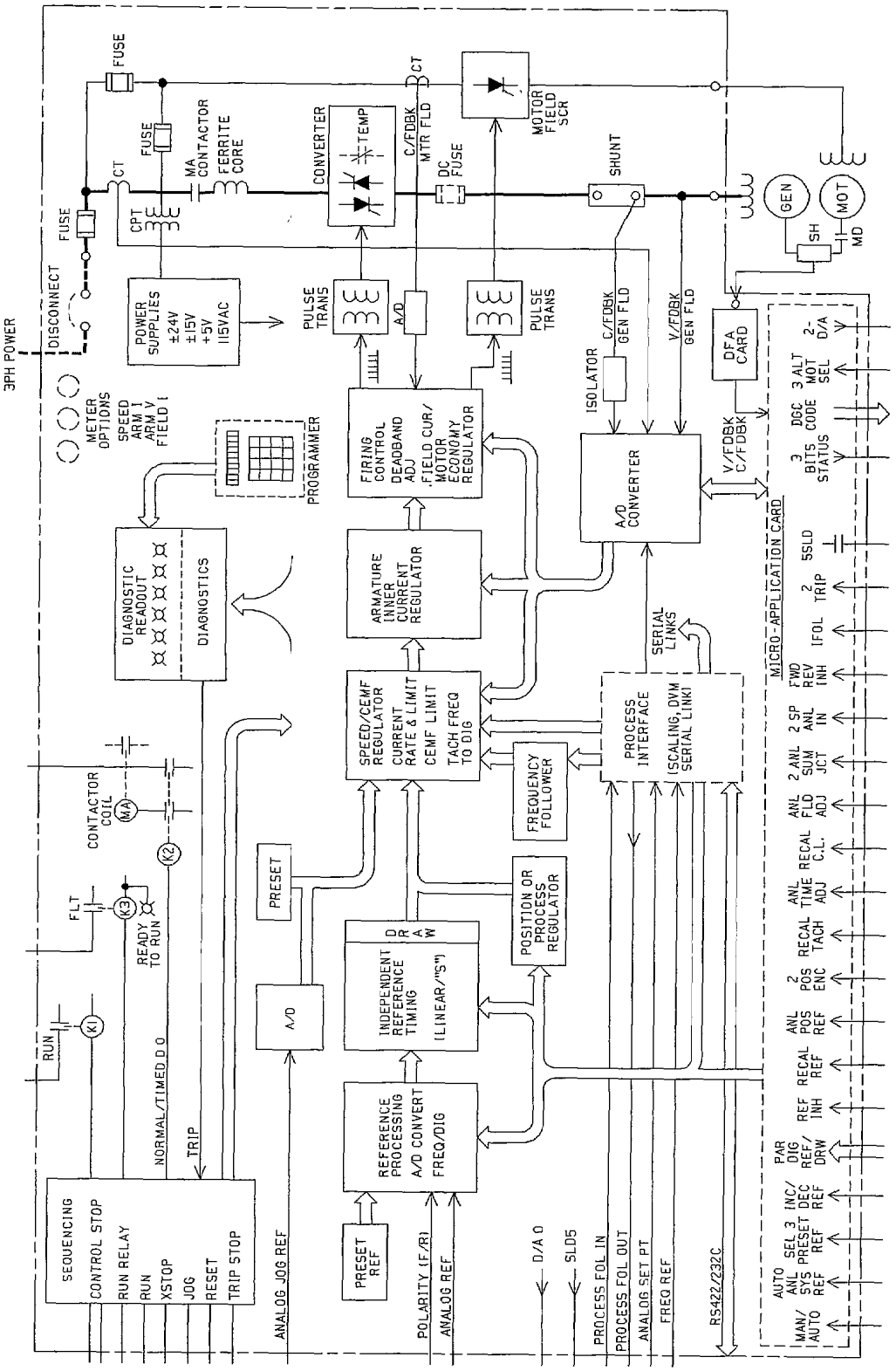


FIGURE 4-5
BLOCK DIAGRAM

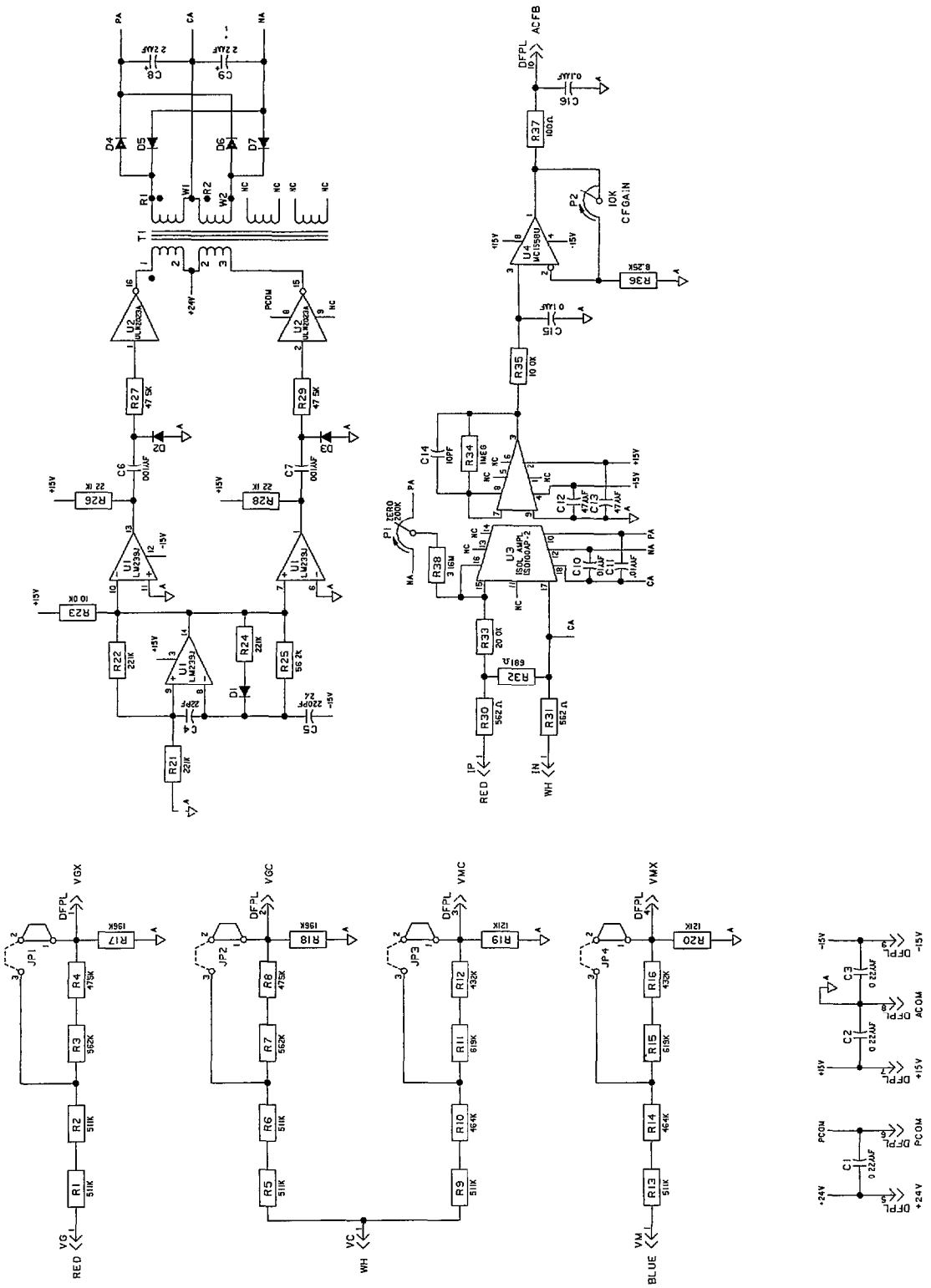


FIGURE 4-6
DIGITAL FIELD AUXILIARY 531X196DFAA__G
ELEMENTARY DIAGRAM

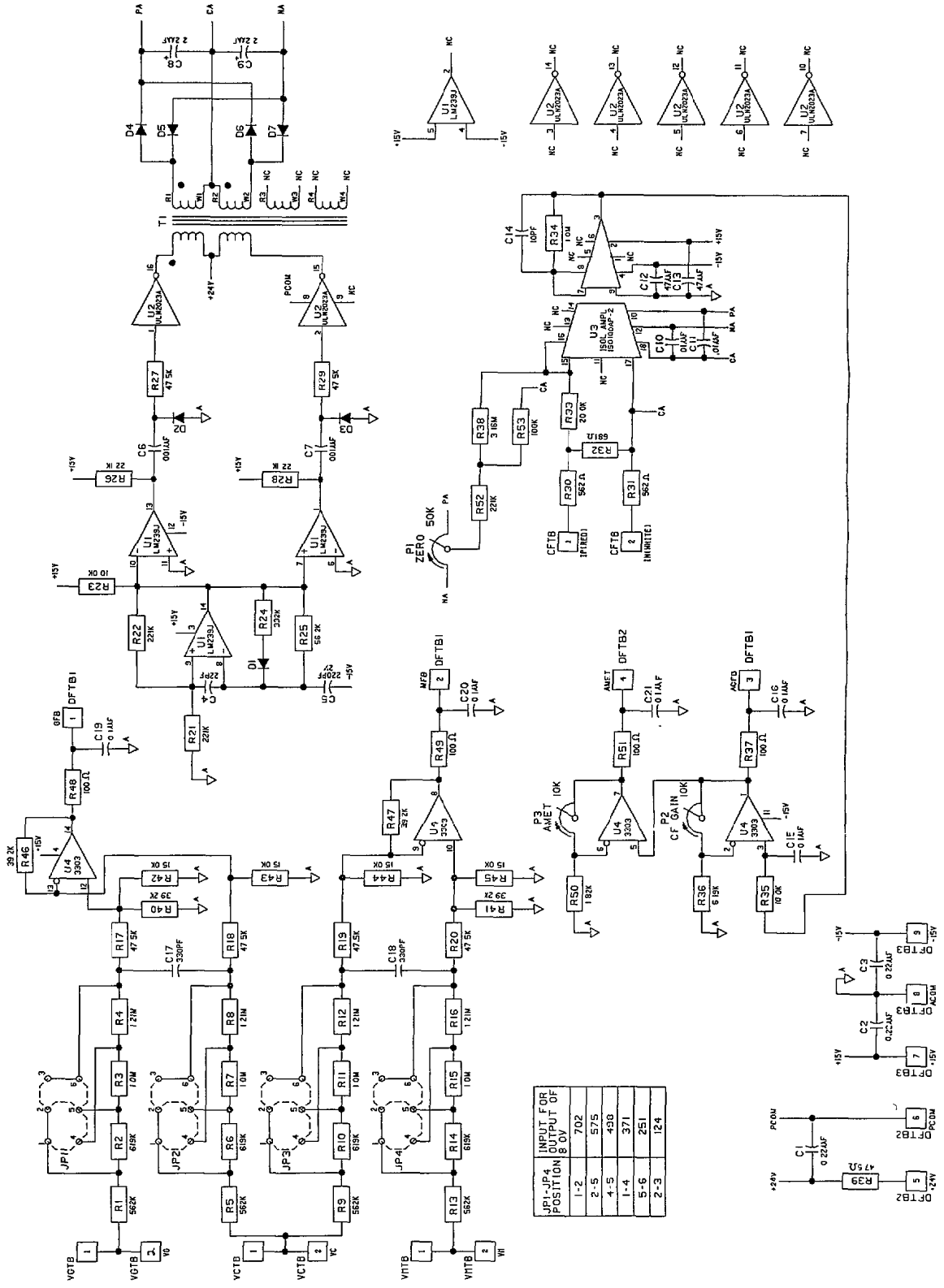


FIGURE 4-7
DIGITAL FIELD AUXILIARY 531X196DFAB__G

MOTOR FIELD CONTROL

The motor field control (MFC) may be provided by an internal 10 Amp maximum field supply (Figure 4-8), an optional 24 Amp field supply or optional external field supplies rated to 72 Amps (Figures 4-9 and 4-10).

For the 10 Amp MFC, the Power Supply card contains the field supply fuses (CFU1, CFU2 and CFU3), metal oxide varistors (MOV), field current feedback transformer (T3), and line reactor (L4). The field power module is mounted on the main controller heatsink.

The 24 Amp MFC provides a separate heatsink with field supply fuses (CFU1-1, CFU2-1 and CFU3-1), line reactor (L4), field power module and MFC Suppression card containing the MOVs and field current feedback transformer (T1).

The 72 Amp external MFC also provides a separate heatsink (with fuses), reactor, power modules and suppression card.

NOTE: *There is a variation in fuse and transformer designations between the three motor field control versions.*

The operation of the MFC may be best understood by reading the following while referring to the appropriate figure. Field power is supplied through fuses CFU1, 2 and 3 (CFU1-1, 2-1 and 3-1 on the external MFC). MOVs protect from line spikes and transients.

The power to the motor field control is "two-thirds" wave rectified via the field power module. The SCR in the field power module controls field current as follows: When AC line L1 is high with respect to L2 and L3, the SCR is fired and current passes through the diode between DX and DW to the motor field. The circuit is completed through the SCR and returns through the line reactor to AC line L3. When L2 becomes high, current flows through the diode between DY and DW to the field and back through the SCR, line reactor and L3. When L3 becomes high, the SCR is reversed biased and the energy stored in the field winding circulates current through the diode between DZ and DW (acting as a free wheeling diode) until L1 becomes high again and the SCR is fired.

Motor field current is sensed by the current feedback transformer (T1 or T3 depending on the MFC rating) and fed into the field gain select circuit on the Power Supply card (through FCPL with external MFC) and to the Control card.

GENERATOR FIELD CONTROL

The SCR Power Conversion Bridge is shown in Figure 4-11. Three phase power is delivered to the conversion bridge through AC line fuses (FU1-3) or an optional circuit breaker. Two line current transformers (CT1 and CT3) provide a current feedback signal for commutation failure indication and the MA contactor and reactors provide protection against AC line transients.

RC snubber networks, located on the Power Connection card, provide protection for the SCRs and are connected across a forward/reverse SCR pair. The SCR firing pulses are generated through pulse transformers located on the Power Connection card in the firing order 1-6-2-4-3-5 for both forward or reverse polarity operation.

DC power is supplied to the generator field through the shunt and DA1 and returns through DA2 which is fused. An RC network across P1 and DA2 aids latching of the SCRs. The voltage feedback signal is in parallel with the snubber and current feedback is derived from the shunt. An optional thermal switch mounted on the heatsink is also available.

AC POWER REQUIREMENTS

The DFM-300 drive is available in either 230 VAC +10%, -5% or 460 VAC +10%, -5%, three phase power configurations. Nominal line frequency is 50/60 Hz with tolerance of ±2 Hz.

AC LINE FUSES (FU1-3)

AC line fuses provide short circuit protection for the SCRs, and the wiring inside the controller. A molded case disconnect or circuit breaker are also available as options. Additional overcurrent protection is provided by both instantaneous and timed overcurrent functions.

FERRITE CORE ASSEMBLY (REACTORS)

The ferrite core assembly is provided on the AC power lines between the AC line fuses and the SCRs to protect the SCRs from possible misoperation due to transient currents and voltages.

MAIN CONTACTOR (MA)

The MA contactor provides a controllable disconnect between the AC power line and the power conversion bridge. The contactor picks up when running mode is selected and no fault exists in the drive. The MA contactor drops out when a fault condition occurs.

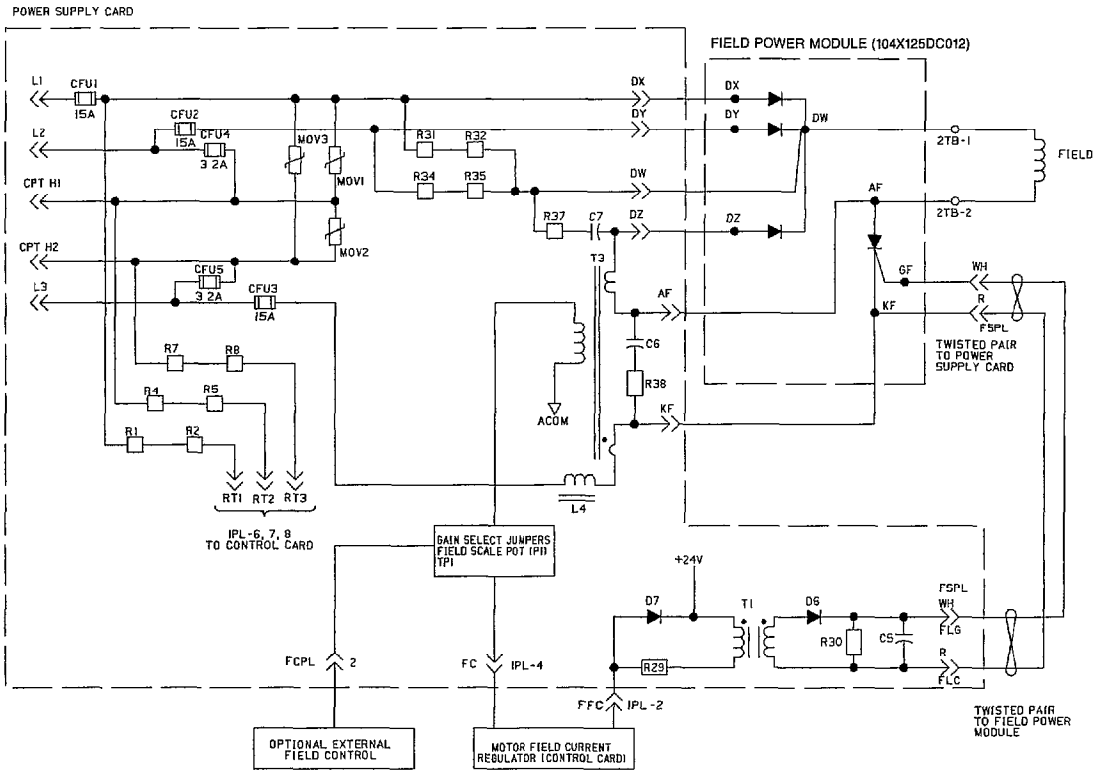


FIGURE 4-8
MOTOR FIELD CONTROL (10 AMP)

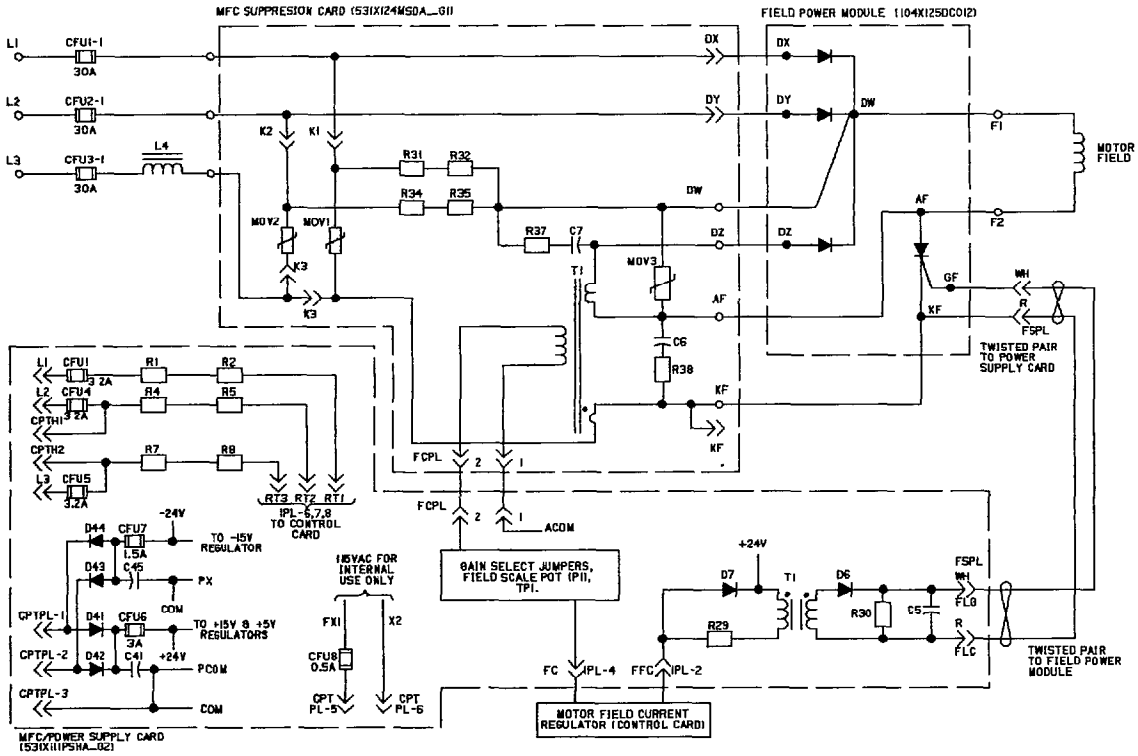


FIGURE 4-9
MOTOR FIELD CONTROL (24 AMP)

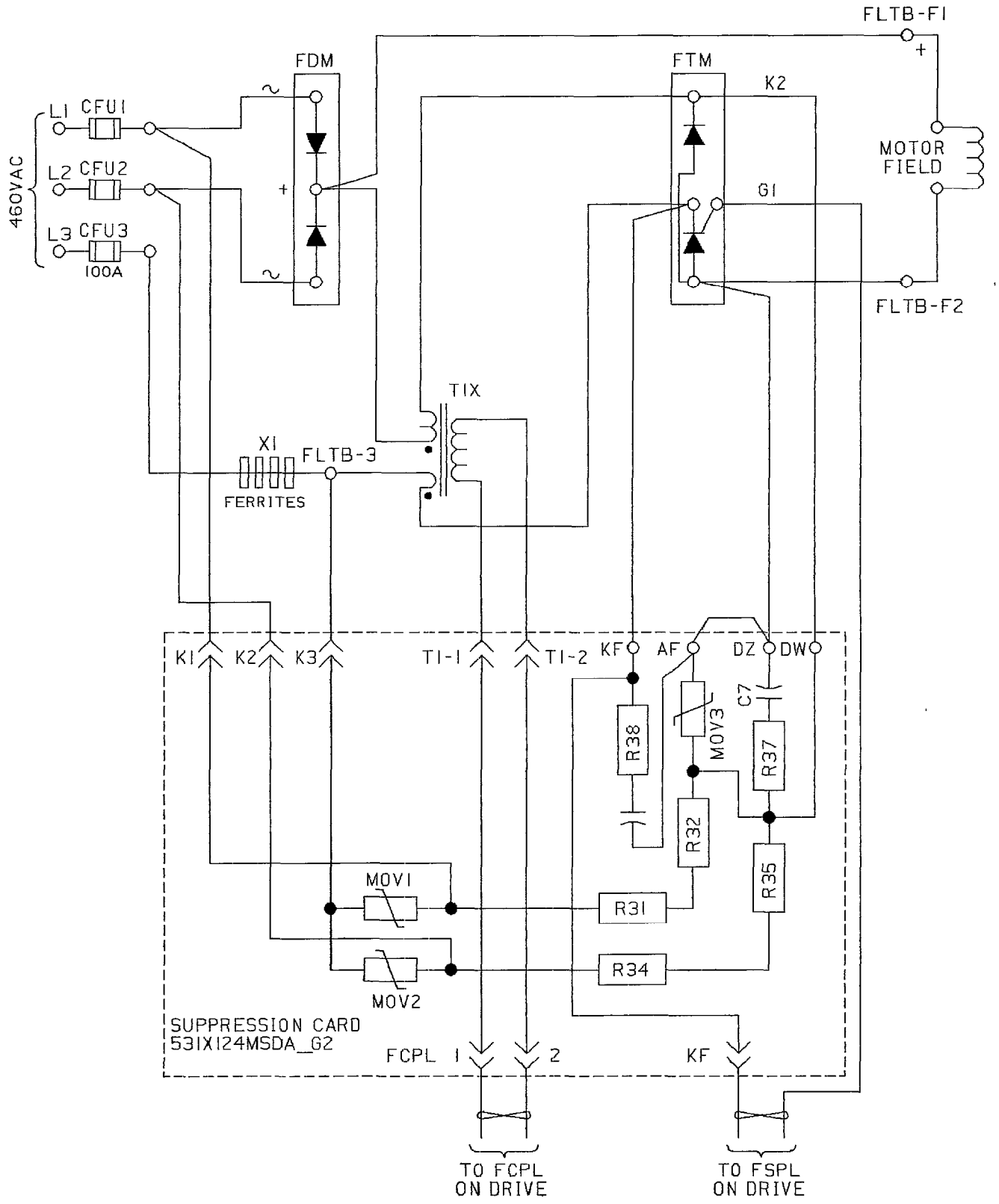


FIGURE 4-10
MOTOR FIELD CONTROL (70 AMP)

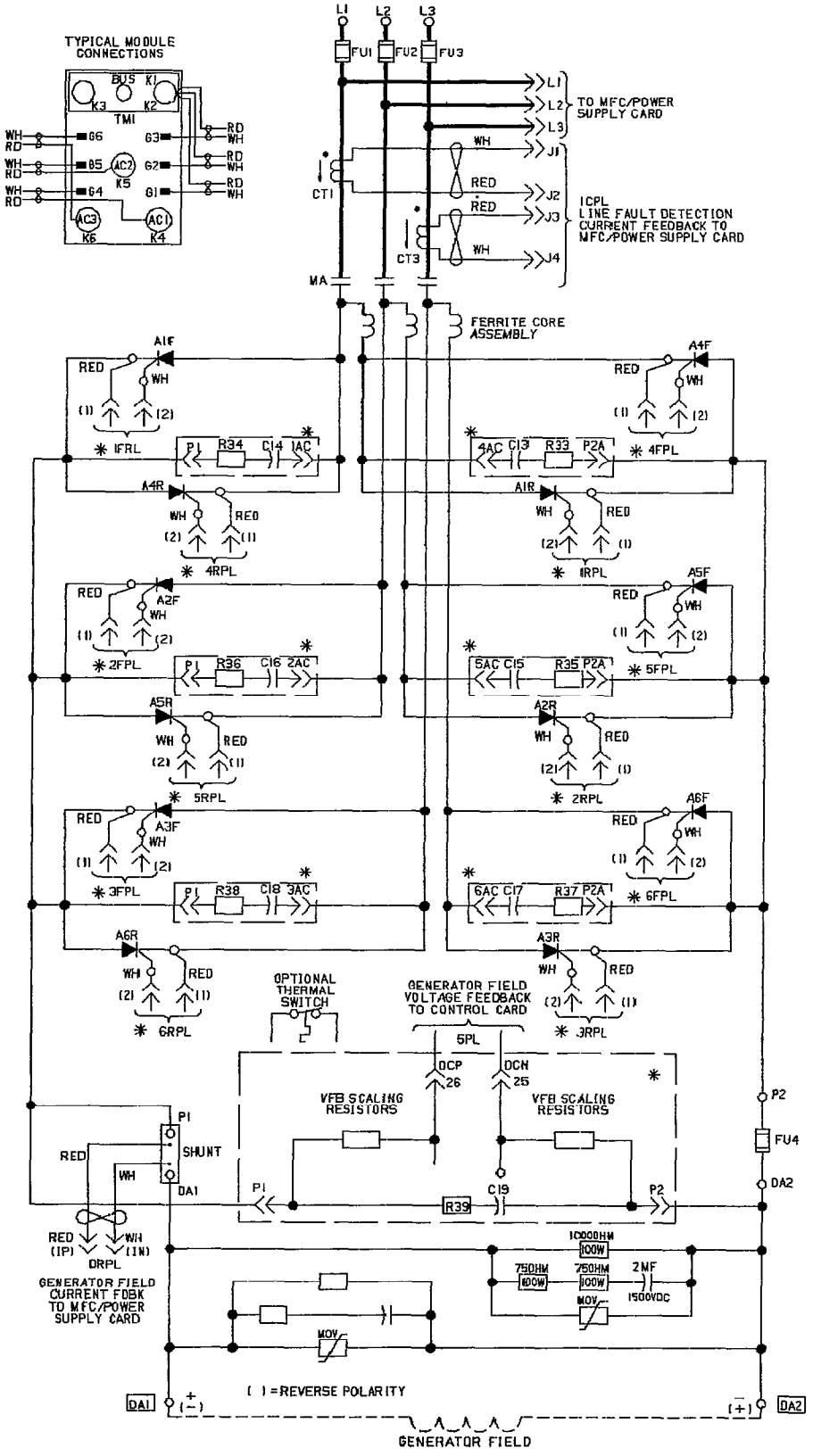


FIGURE 4-11
DFM-300 SCR POWER CONVERSION BRIDGE

DC LINE FUSE (FU4)

A DC line fuse is provided in the DC output line to the generator field and protects the SCRs and field from overcurrent.

CONTROL POWER TRANSFORMER (CPT)

This transformer is energized through its primary fuses (CFU4 and 5) on the Power Supply card. The secondary winding provides two isolated voltages:

1. 115 VAC to operate the coil of the main (MA) contactor, RUN (K1) Relay.
2. 38 VAC center tapped to the Power Supply card to generate the DC control voltages necessary for controller operation.

SHUNT (SH)

The shunt provides the armature current feedback signal to the control circuit. A nominal 100mV output signal is generated at the current rating stamped on the shunt.

CURRENT TRANSFORMER ASSEMBLY (CT)

The current transformer assembly is used to provide AC line current feedback signals that are used for circulating current fault detection.

CIRCUIT CARDS

The DFM-300 controller requires only five printed circuit cards: Control card, Power Supply card, Power Connection card, Microapplication card and DFA* auxiliary card. Optional cards include the Process Interface card, Programmer card, and Terminal Board cards (3TB and 4TB). The Programmer can be purchased in a Local or Hand-held version which plugs into the Control card for inspection and adjustment of parameters when tuning up the drive, as well as providing diagnostic functions.

* *Two different type DFA cards exist and they are* ***not*** *backward compatible.*

- *531X196DFAA__G1 used iso amps that were mounted on Microapplication card.*
- *531X196DFAB__G1 has iso amps mounted on same board.*

See Figures 4-5 and 4-6.

CONNECTION DIAGRAM (See Figure 4-12)

A connection diagram for the DFM-300 controller is shown in Figure 4-12. This graphically describes the internal connections for all plugs and cables. This diagram aids control hook up as well as troubleshooting when connection integrity of one of the cables is in question. The functions of the connectors are described in the "Electrical Connection" section.

CONTROL CARD (See Figure 4-13)

This card provides the intelligence for the DFM-300 drive controller. It contains three microprocessors and performs most of the regulating functions, SCR burst firing, diagnostic and protective functions for the drive along with many customer and motor interfacing functions. The functions performed on the card are implemented in hardware and/or software.

The control card (Figure 4-13) contains six Diagnostic LEDs to display the status of the DFM-300 drive. The LEDs which are lit are interpreted as a binary number indicative of the fault. See the troubleshooting section for listing of Fault codes. Note that during proper controller operation, the LEDs will blink sequentially right to left when the drive is stopped and sequentially in pairs when the drive is running.

MFC/POWER SUPPLY CARD (See Figures 4-14, 4-15, 4-16 and 4-17)

This power supply card provides the necessary +5 VDC, ±15 VDC, and ±24 VDC power used throughout the controller. The generator field current feedback isolation circuit and three relays, "RUN" (K1), "MAX" (K2) and "FAULT" (K3) are also located on this card. Additionally, the motor field control is provided by this card. The version G1 card is used with motors requiring field current up to 10 amps and contains the pulse transformer, choke, MOVs and fuses for the motor field circuitry as previously described. The version G2 card is used in conjunction with the external field control assembly.

A 24 Amp Motor Field Control Assembly (Figure 4-16) is used with the version G2 MFC/Power Supply card for controllers requiring up to 24 amps for the motor field. The assembly contains the fuses (CFU1-3), power module, current feedback transformer, choke and MOVs, which provide power module protection.

The 72 Amp external MFC assembly (Figure 4-17) is also used with the G2 MFC/Power Supply card. This assembly contains fuses, power modules, reactors, MOVs, and a current feedback transformer, similar to the 24 Amp assembly.

The Power Supply card provides input voltage to the control power transformer through CFU4 and 5 via terminals CPTH1 and CPTH2. RT1, 2 and 3 are the AC line sync signals to the Control card. (See Figures 4-8 and 4-9.)

Fuses CFU6, CFU7 and CFU8 are located on the Power Supply card. The internal DC power supplies are protected by CFU6 and CFU7. The 115V AC internal supply is protected by CFU8. These circuits are fed from the control power transformer through plug CPTPL.

POWER CONNECTION CARD (See Figure 4-18)

This card contains SCR and DC output resistor-capacitor (RC) snubbers and the gate pulse transformers for the SCRs. The DC voltage feedback isolation resistors are also mounted on this card.

MICROAPPLICATION CARD (See Figure 4-19)

This card contains a combination of analog and digital interfaces between the Control card and external equipment. These interfaces allow the core drive to be expanded to handle functions required in applications such as material handling cranes and digital field motor control. The card contains the following optional functions:

- Eight Analog Inputs
- Eight Digital Inputs
- Sixteen Bit Parallel Word Input
- Five Programmable Signal-Level-Detector Relays (SLDs)
- Seven-Bit Digital Fault Status Output
- Three Programmable Status Bit Outputs
- Two Encoder Interfaces
- Two Analog Outputs
- Two General Purpose Amplifiers (Version APM card only)

The Microapplication card, in addition to the above features, also contains the DFM-300 Local Area Network (LAN) feature for communicating over a single pair of wires with up to 30 other drives and to a Series Six™ Programmable Controller. The Microapplication card and LAN are discussed in the Local Area Network User's Guide, GEK-85789.

DIGITAL FIELD AUXILIARY CARD (See Figures 4-20 and 4-21)

The DFA* card (Digital Field Auxiliary card) accepts inputs from the Generator-Motor Armature circuit of Generator output voltage, Motor armature terminal voltage and armature current. Circuitry on the DFA card provides isolation of the DFM-300 electronics from the armature power circuit. Low level signals are output from the DFA card, providing feedback of Generator Voltage, Motor Voltage and Armature Current to the DFM-300 regulator.

OPTIONAL PROCESS INTERFACE CARD (See Figure 4-22)

This optional card provides a mixture of analog and digital interfaces between the Control card and the equipment. The optional functions on this card are:

- Analog (AC or DC) tachometer interface.
- Analog speed meter driver (analog tachs only).
- Digital (reluctance or square wave) tachometer interface.
- Frequency (pulse train) reference interface.
- Process Follower (current or voltage reference) interface.
- Digital voltmeter probe.
- RS232C and RS422 serial link interface.

* *Two different type DFA cards exist and they are* ***not*** *backward compatible.*

- *531X196DFAA__G1 used iso amps that were mounted on Microapplication card.*
- *531X196DFAB__G1 has iso amps mounted on same board.*

See Figures 4-5 and 4-6.

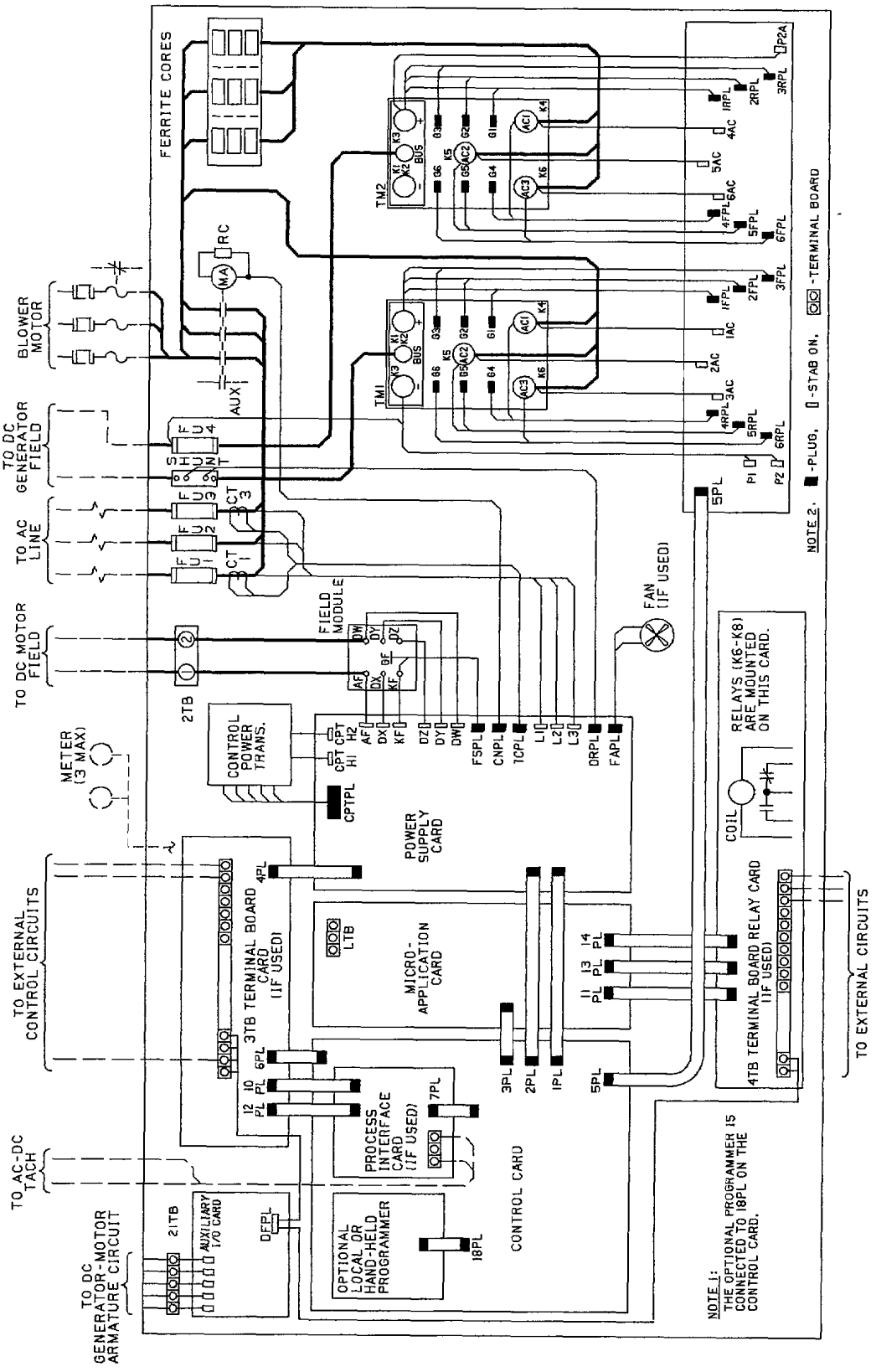


FIGURE 4-12
CONNECTION DIAGRAM

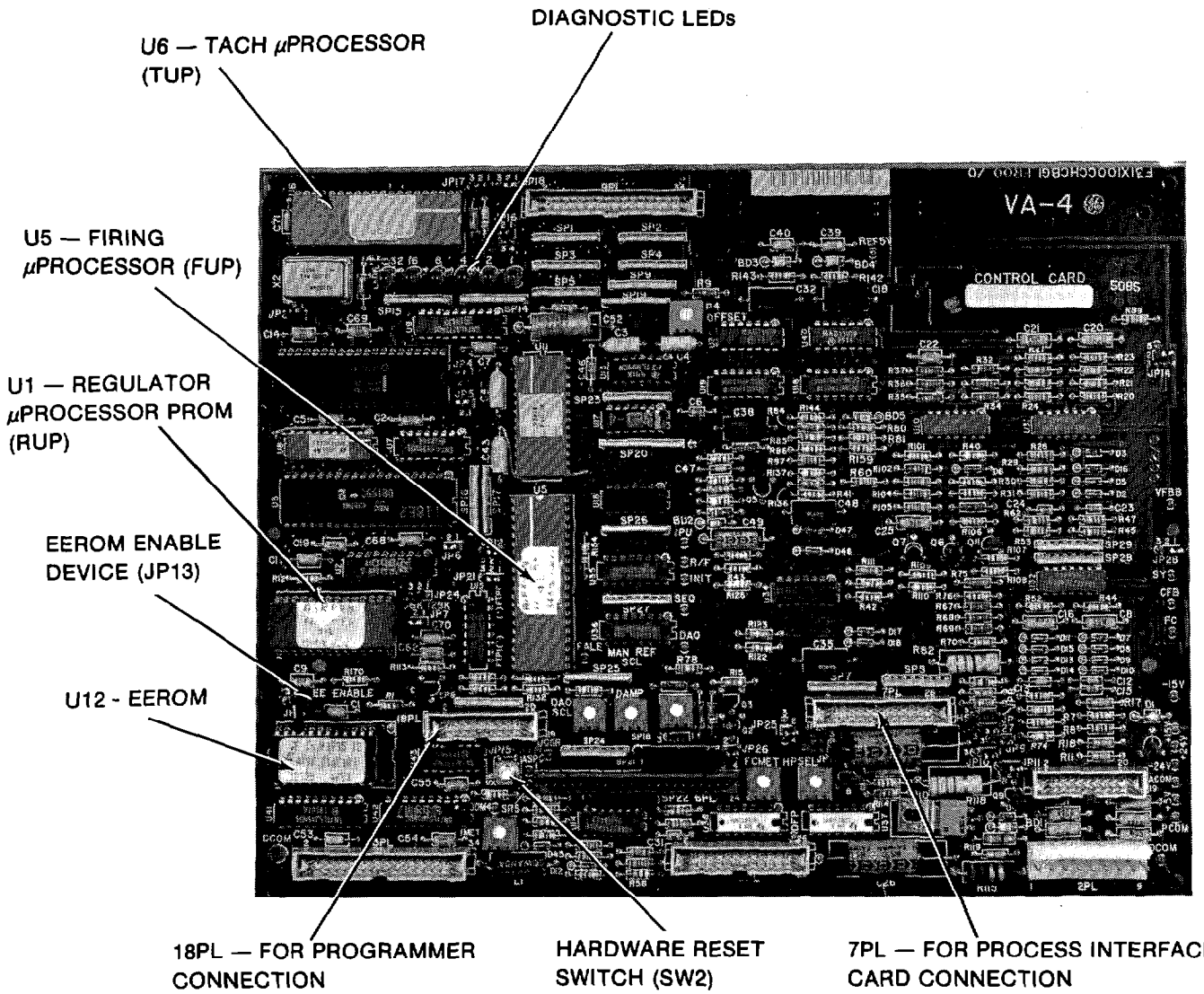


FIGURE 4-13
CONTROL CARD

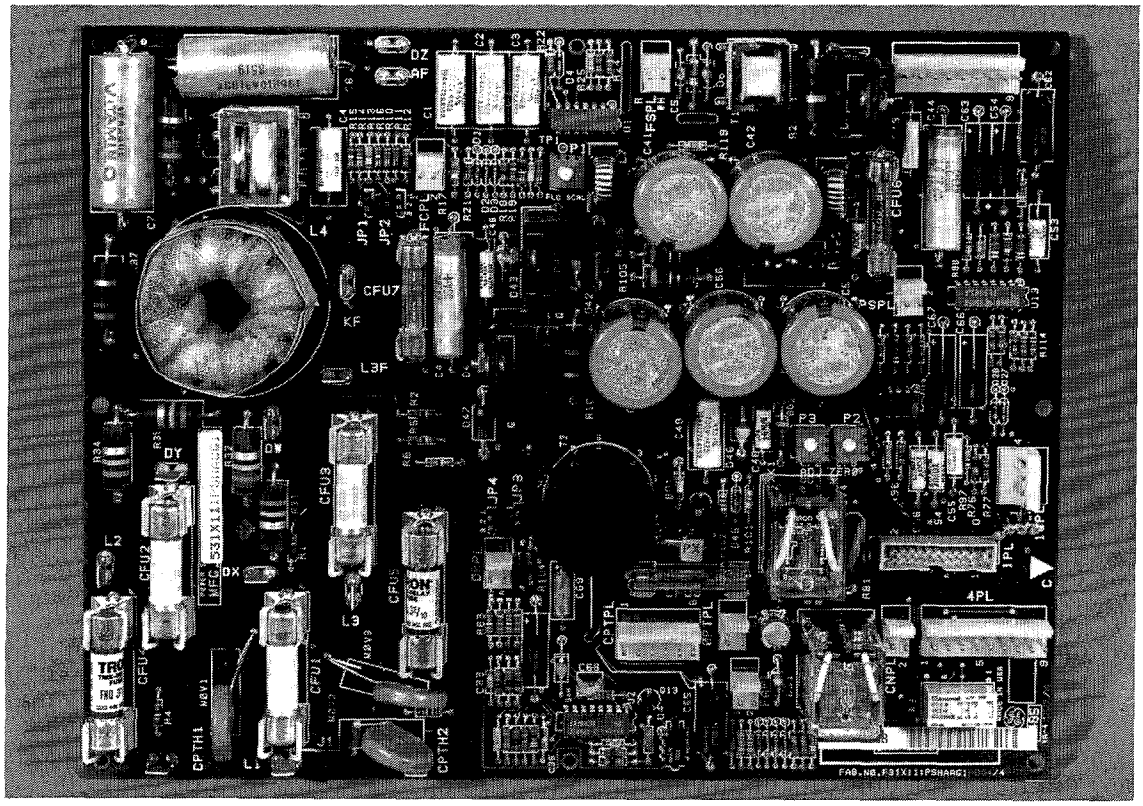


FIGURE 4-14
MFC/POWER SUPPLY CARD (VERSION G1)

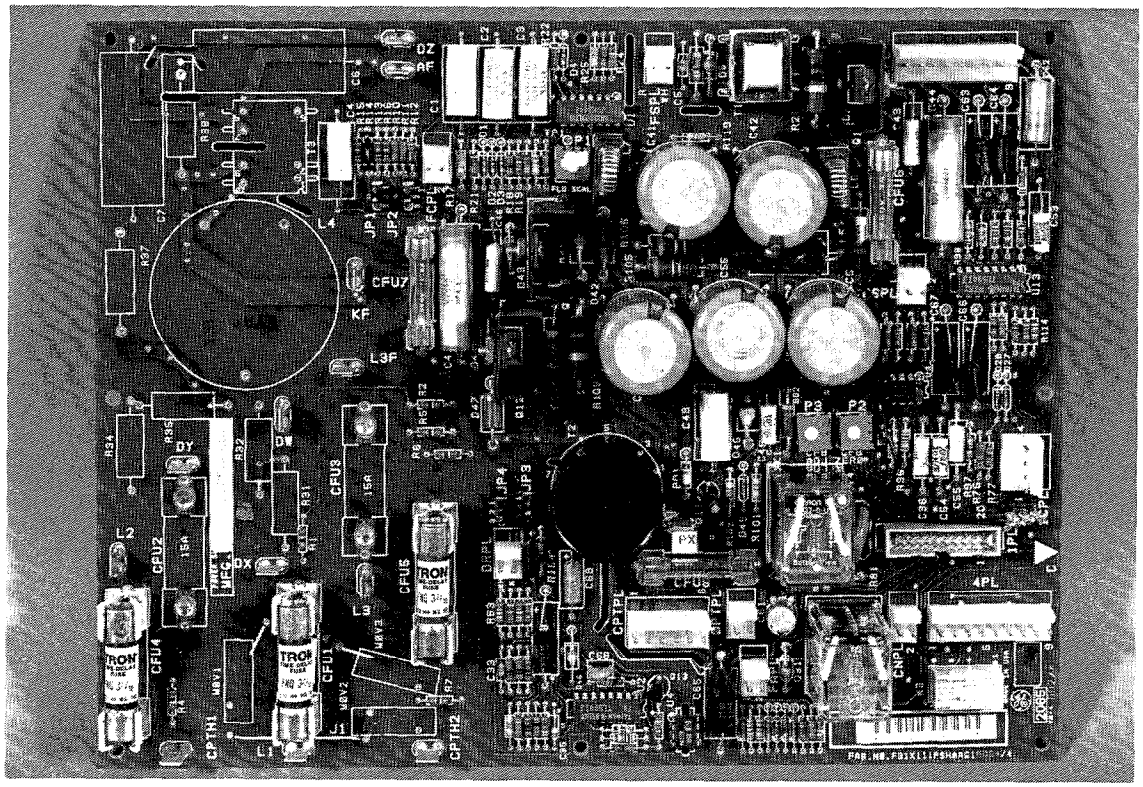


FIGURE 4-15
MFC/POWER SUPPLY CARD
(VERSION G2 REQUIRES EXTERNAL MFC ASSEMBLY)

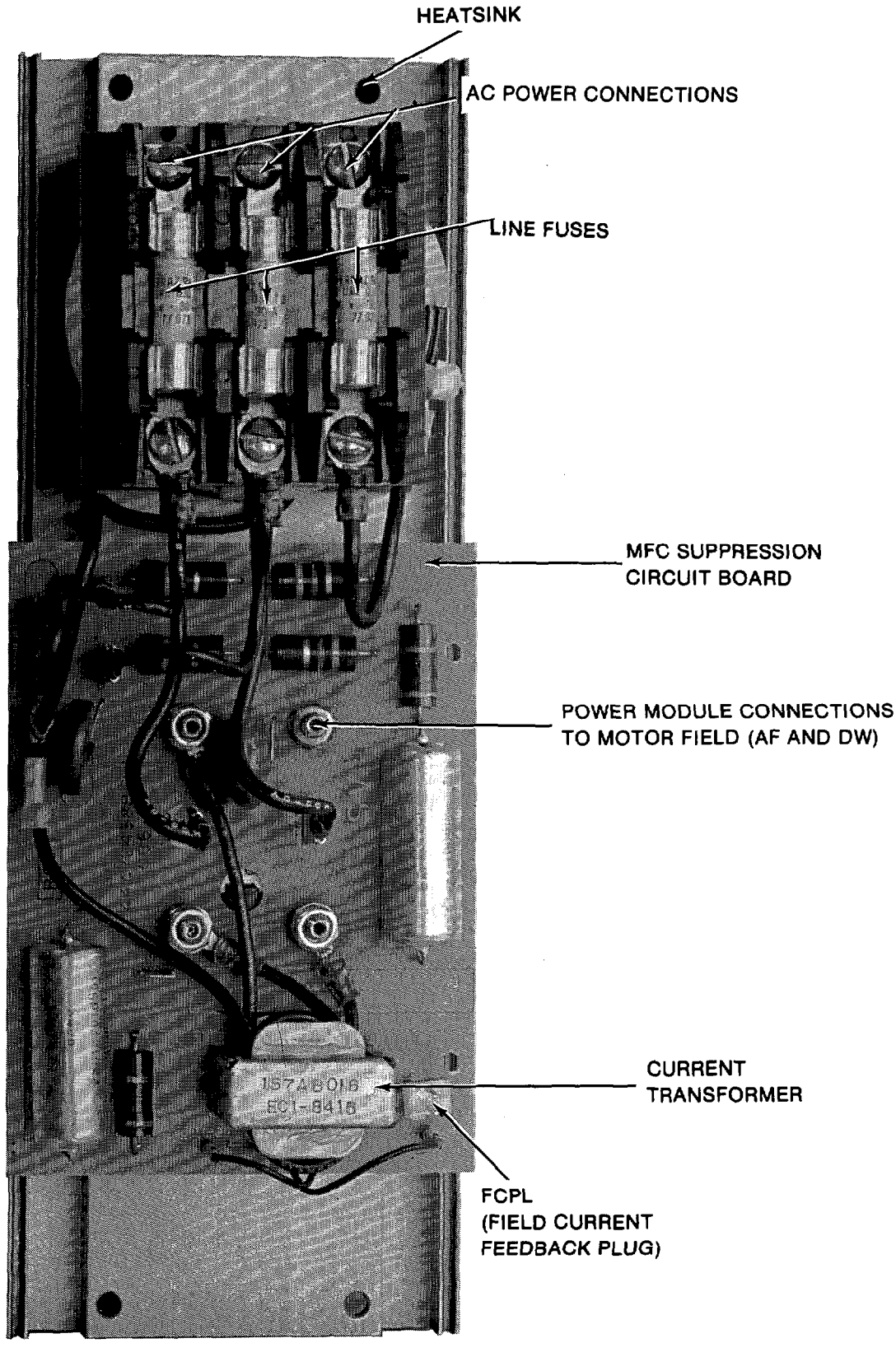


FIGURE 4-16
24 AMP (EXTERNAL) MOTOR FIELD CONTROL ASSEMBLY

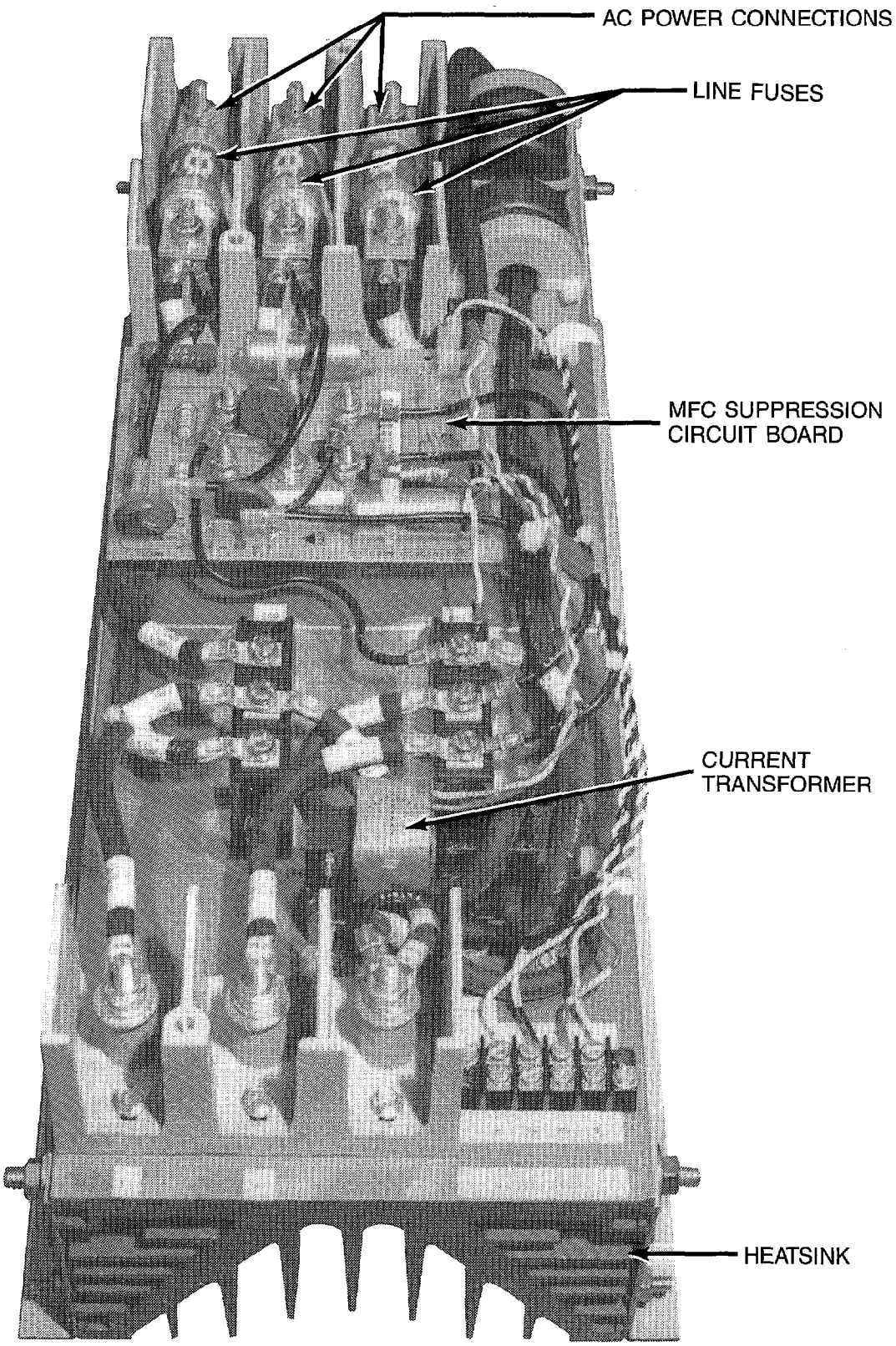


FIGURE 4-17
70 AMP (EXTERNAL) MOTOR FIELD CONTROL ASSEMBLY

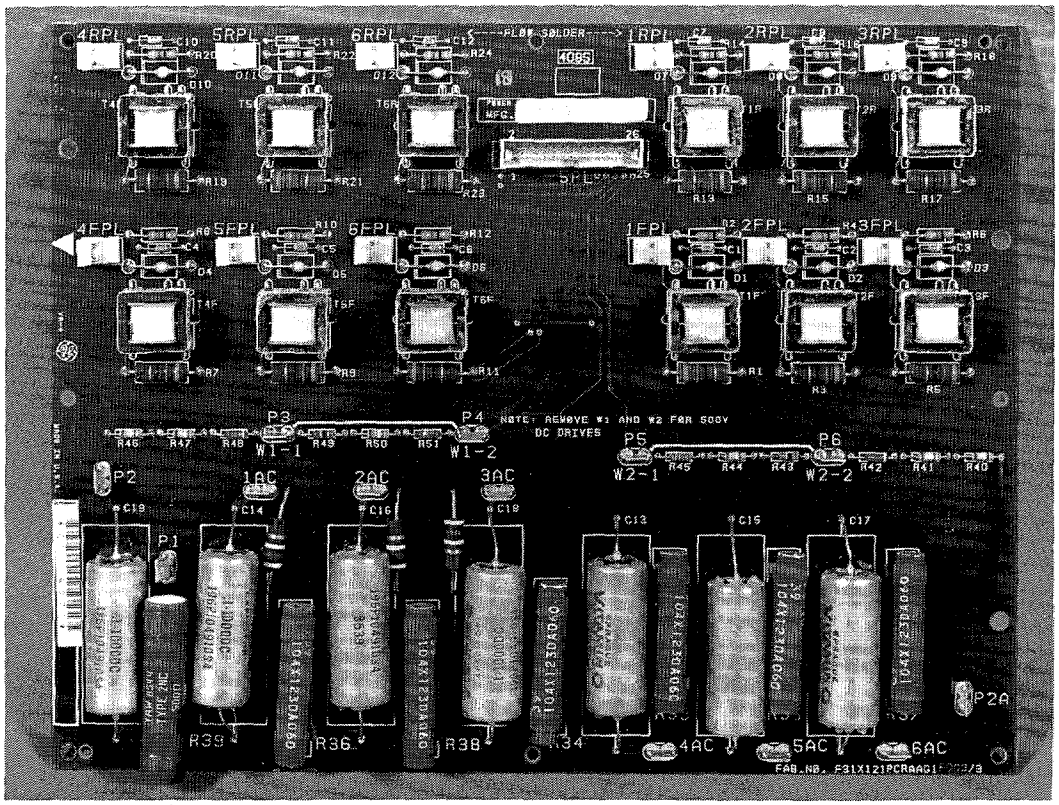


FIGURE 4-18
POWER CONNECTION CARD

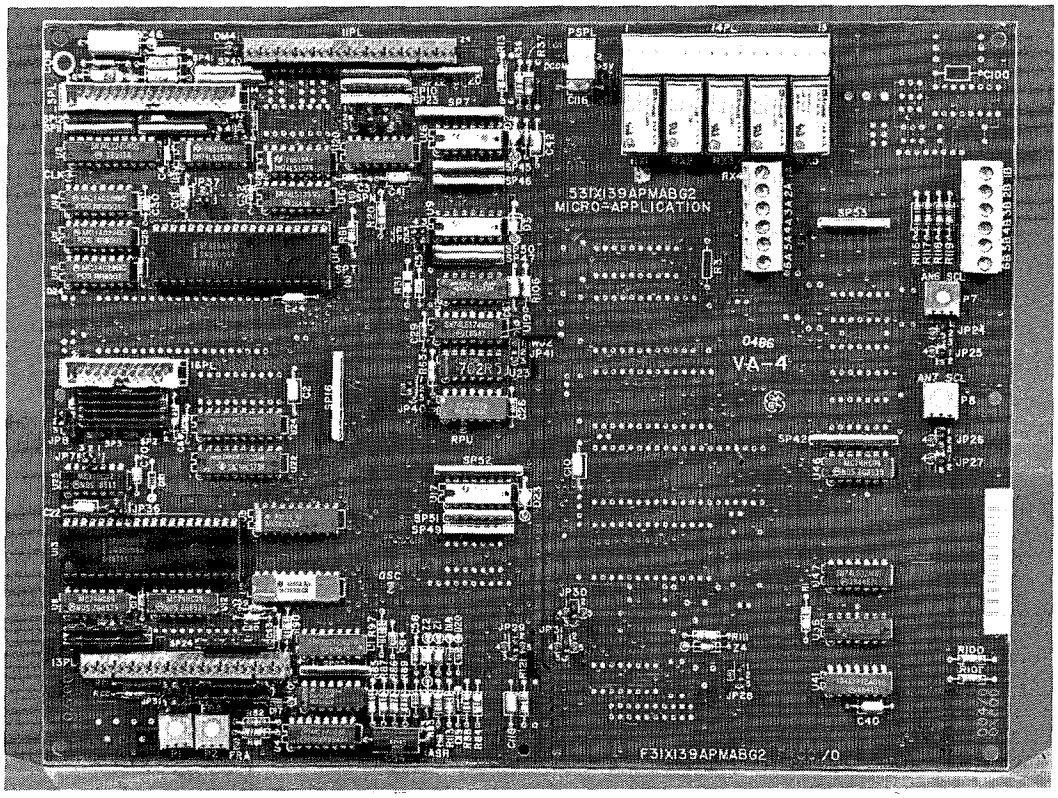


FIGURE 4-19
MICROAPPLICATION CARD

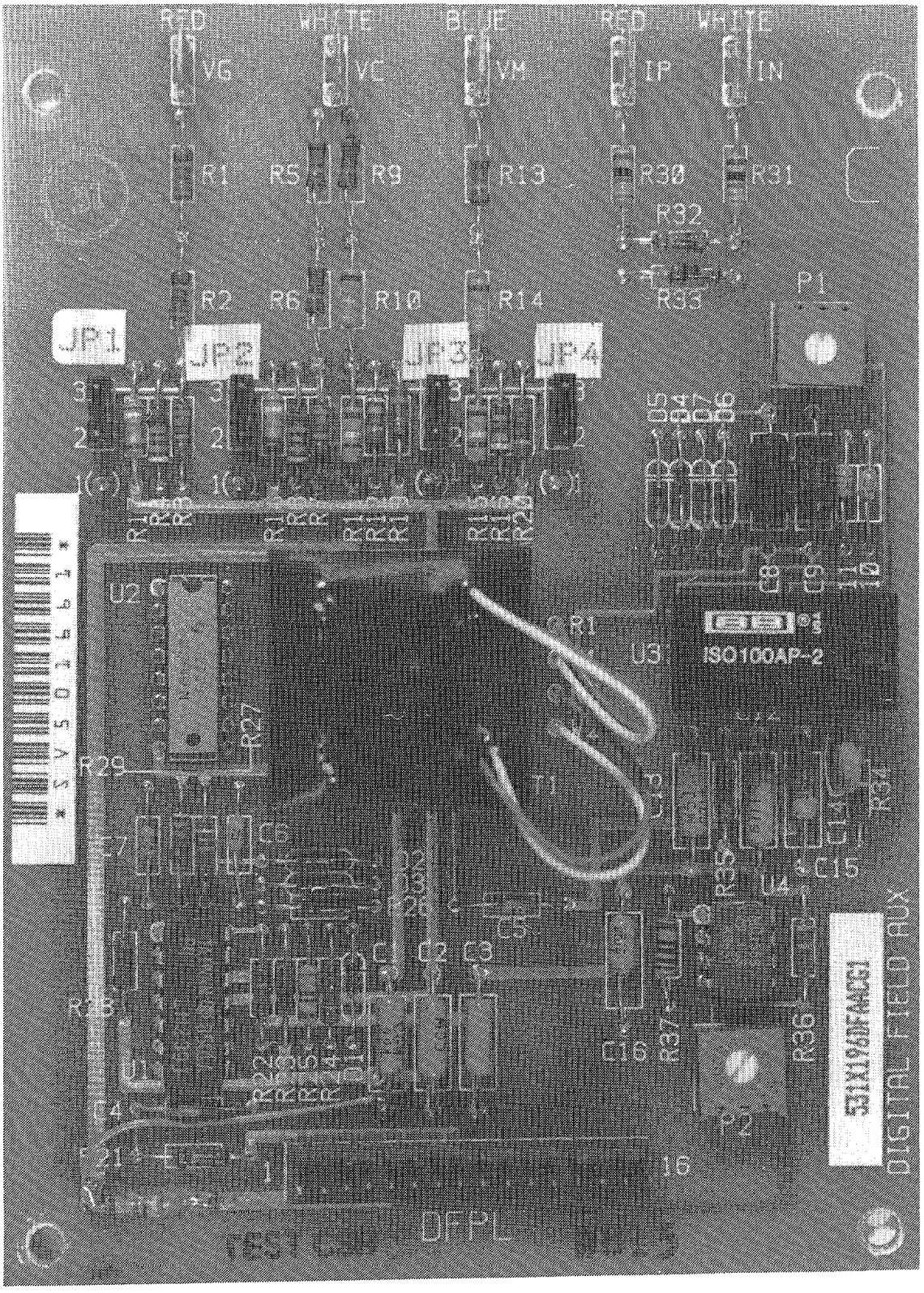


FIGURE 4-20
531X196DFAA__G VERSION DIGITAL FIELD AUXILIARY CARD

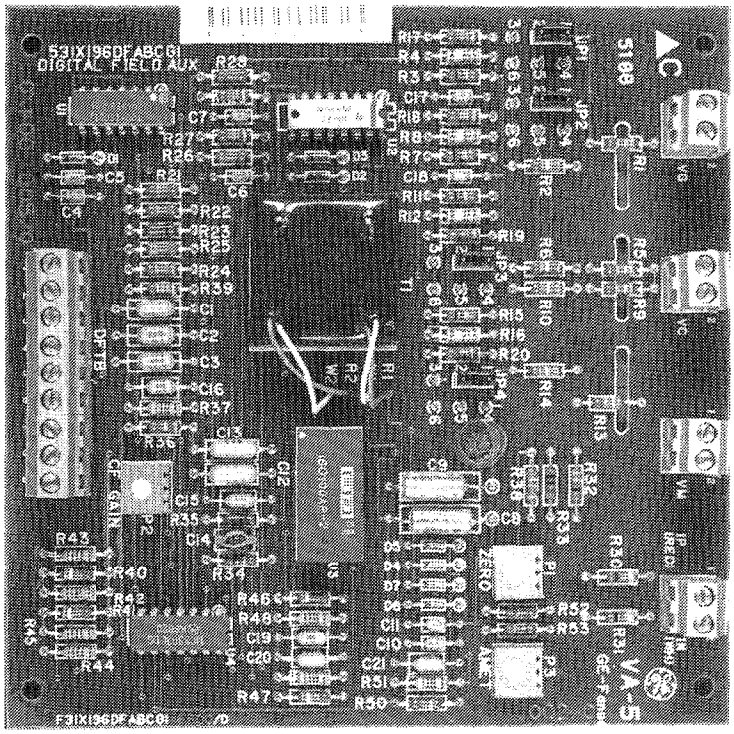


FIGURE 4-21
531X196DFAB__G VERSION DIGITAL FIELD AUXILIARY CARD

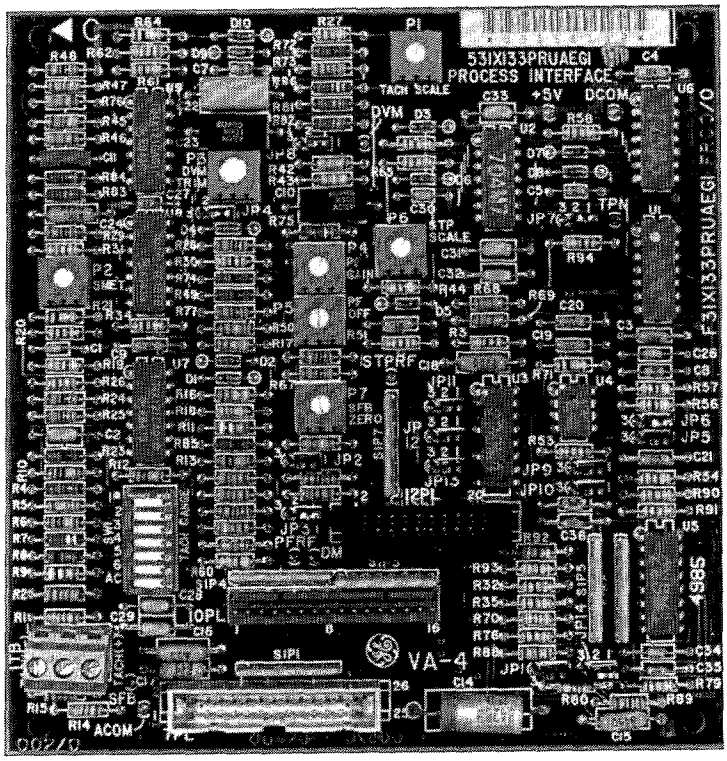


FIGURE 4-22
OPTIONAL PROCESS INTERFACE CARD

OPTIONAL TERMINAL BOARD CARDS (3TB and 4TB) (See Figures 4-23 and 4-24)

These optional cards provide convenient termination points for control interface to the drive. Connections to the Control, Power Supply and optional Process Interface cards are made thru 3TB, which also contains an optional signal-level-detector (SLD5). Connections to the optional Application card are made thru 4TB. In addition, three 115 VAC relays are furnished on the 4TB card.

OPTIONAL PROGRAMMER (See Figures 4-25 and 4-26)

The Programmer is available as either a "Local" Programmer, mounted on the Control card, or as a "Hand-held" Programmer, which is plugged into 18PL on the Control card. The programmer contains a 10-digit alphanumeric display and keypad for communicating with the drive.

The Programmer has three modes:

1. **Operate** mode provides all necessary functions to run the drive and displays the drive status including fault messages.
2. **Parameter** mode allows examination and changes, if necessary, of the system operating parameters.
3. **Diagnostic** mode allows the user to monitor drive values while in operation, self test the controller, and perform a number of drive tests. Refer to "Use of Optional Programmer Module" section for further information.

SERVICE AND PARTS INSTRUCTION LABEL (See Figure 4-27)

Each drive contains, inside the door, a multi-color label similar in appearance to that shown in Figure 4-27. This label provides information on connection point and component locations essential in connecting or troubleshooting the drive. The label contains the following information in color-coded format:

- Card location
- Thyristor module locations
- Fuse information and nomenclature
- Plug locations
- Incoming control connections at 3TB and 4TB
- Power connections
- Factory "hot line" telephone number **(1-814-455-3219)**

INTERFACE CONNECTIONS (See Figures 4-28 and 4-29)

The Interface Diagrams (Figures 4-28 and 4-29) show I/O interconnections for a typical DFM-300 controller. These diagrams are provided for reference only; diagrams specific to a particular application are included in the Custom Instructions.

Interconnection plugs and associated terminal points are described in the pages following the Interface Diagrams. Again it should be noted that these are general descriptions and the Custom Instructions should be consulted for the specific application.

OPERATION SEQUENCE

There are two operating sequences, selectable by bit 15 of ADR298-DFMJ1, generally used. Consult the Custom Software Description provided with the drive for details of the sequence and variations provided for the specific application.

1. BIT 15 = 1
   a. A signal to run (6PL-12) picks up the MA contactor and preconditions the DFM-300 Controller.
   b. A signal to C11 (13PL-10) starts the generator field control.
   c. Either a bit output or an SLD (signal level detector) output from the DFM-300 Controller provides a permissive signal for MD contactor in the motor-generator armature loop.
   d. A control interlock from the MD contactor provides a signal to C10 (13PL-9) and permits the controller to run.
2. BIT 15 = 0
   a. A signal to RUN (6PL-12) picks up MA and MD contactors.
   b. A control interlock from the MD contactor provides a signal to C10 (13PL-9) and permits the controller to run.

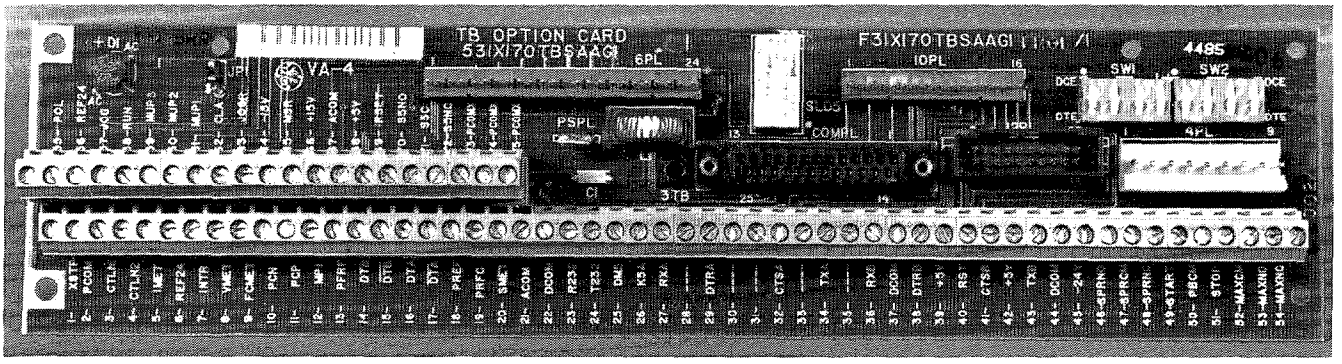


FIGURE 4-23
OPTIONAL TERMINAL BOARD (3TB)

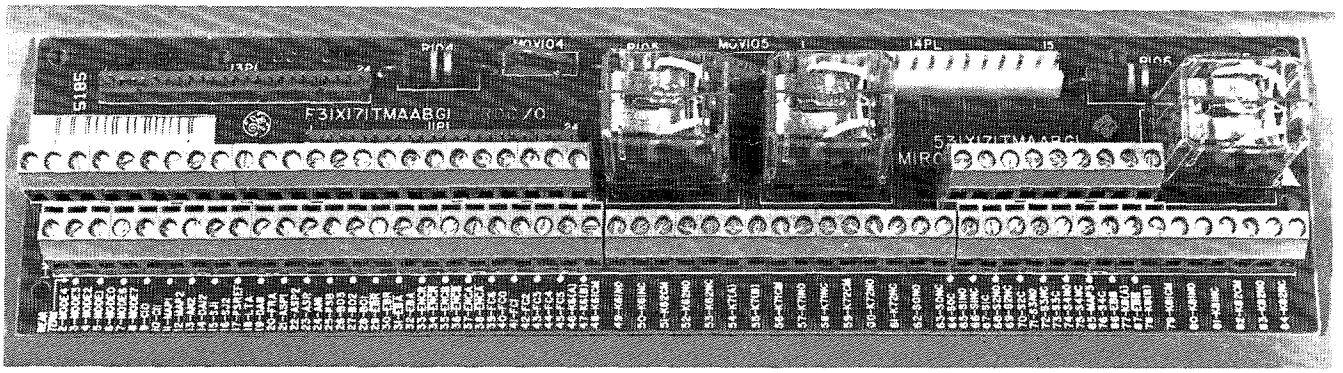


FIGURE 4-24
OPTIONAL TERMINAL/RELAY CARD (4TB)

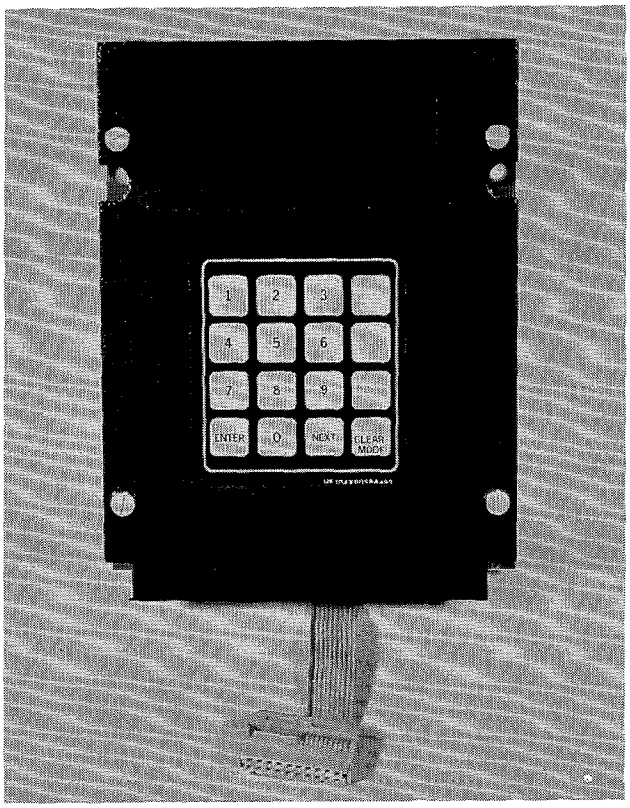


FIGURE 4-25
"LOCAL" PROGRAMMER

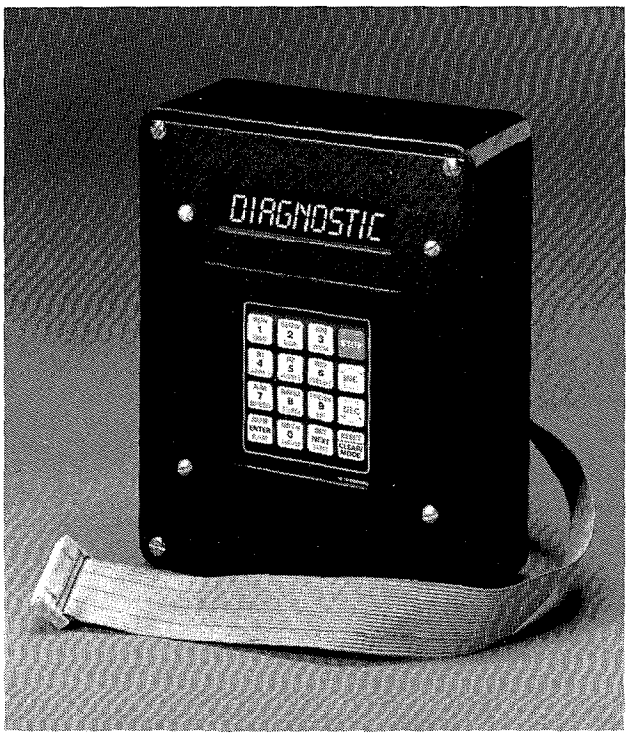


FIGURE 4-26
"HAND-HELD" PROGRAMMER

Instructions

For Service

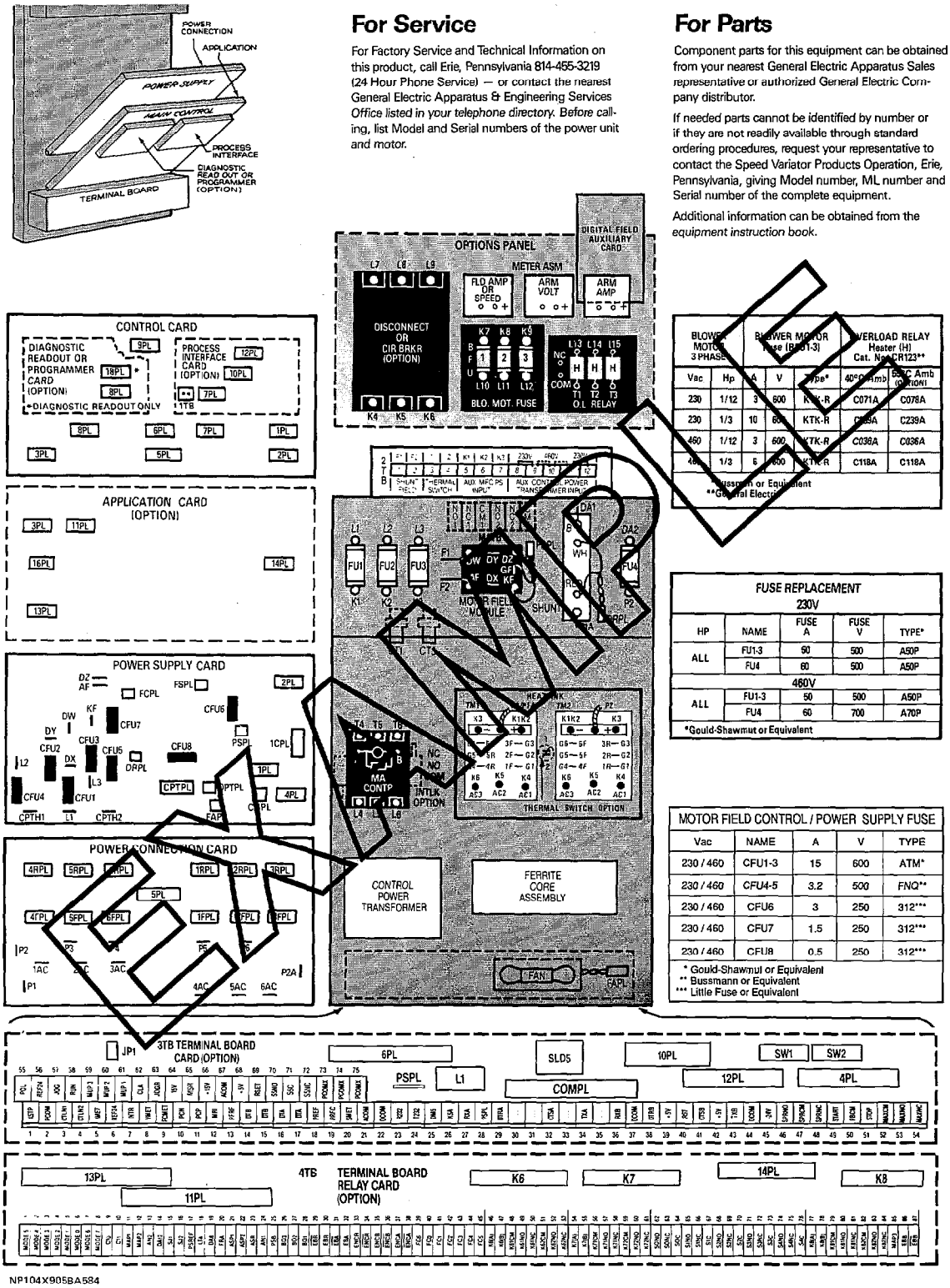
For Factory Service and Technical Information on this product, call Erie, Pennsylvania 814-455-3219 (24 Hour Phone Service) — or contact the nearest General Electric Apparatus & Engineering Services *Office listed in your telephone directory. Before calling*, list Model and Serial numbers of the power unit and motor.

For Parts

Component parts for this equipment can be obtained from your nearest General Electric Apparatus Sales representative or authorized General Electric Company distributor.

If needed parts cannot be identified by number or if they are not readily available through standard ordering procedures, request your representative to contact the Speed Variator Products Operation, Erie, Pennsylvania, giving Model number, ML number and Serial number of the complete equipment.

Additional information can be obtained from the *equipment instruction book.*



BLOWER MOTOR 3 PHASE			BLOWER MOTOR Fuse (FU1-3)		OVERLOAD RELAY Heater (H) Cat. No. CR123**	
Vac	Hp	A	V	Type*	40°C Amb	55°C Amb (Option)
230	1/12	3	600	KTK-R	C071A	C078A
230	1/3	10	600	KTK-R	C118A	C239A
460	1/12	3	600	KTK-R	C036A	C036A
460	1/3	6	600	KTK-R	C118A	C118A

*Bussmann or Equivalent
**General Electric

FUSE REPLACEMENT 230V				
HP	NAME	FUSE A	FUSE V	TYPE*
ALL	FU1-3	90	500	A50P
	FU4	60	500	A50P
460V				
ALL	FU1-3	50	500	A50P
	FU4	60	700	A70P

*Gould-Shawmut or Equivalent

MOTOR FIELD CONTROL / POWER SUPPLY FUSE				
Vac	NAME	A	V	TYPE
230 / 460	CFU1-3	15	600	ATM*
230 / 460	CFU4-5	3.2	500	FNQ**
230 / 460	CFU6	3	250	312***
230 / 460	CFU7	1.5	250	312***
230 / 460	CFU8	0.5	250	312***

* Gould-Shawmut or Equivalent
** Bussmann or Equivalent
*** Little Fuse or Equivalent

NP104X905BA584

FIGURE 4-27
DRIVE CONTROLLER INFORMATION LABEL

NOTE: *Not all functions may be available. Consult elementary diagrams and Custom Instruction Book for the specific drive functions provided and the associated interface connections.*

/ = NOT i.e. /C = NOT C.
* = LOW LEVEL SIG WIRE.

CONNECTION KEY:

- ◇ 6PL CONTROL CD
- △ 13 PL
- ⊠ 14 PL
- □ 11 PL
- ⬡ 16 PL

APPLICATION CARD

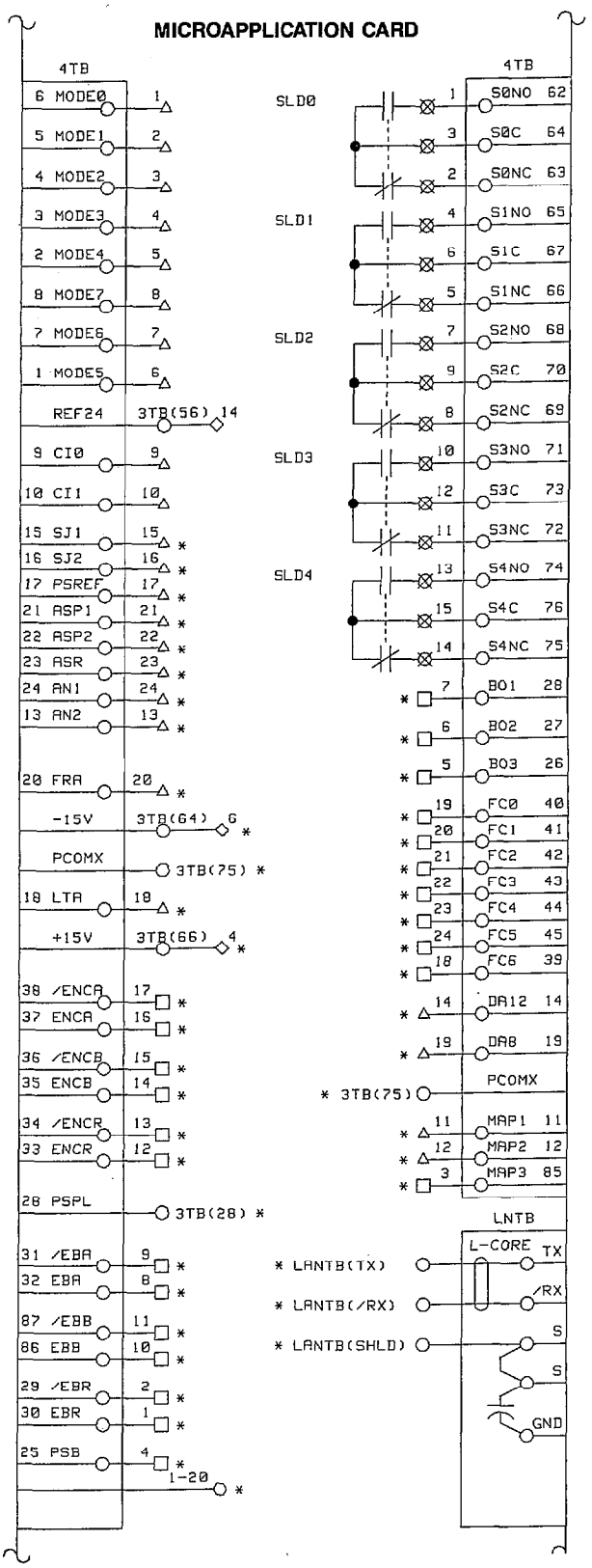


FIGURE 4-28
TYPICAL INTERFACE DIAGRAM

NOTE: *Not all functions may be available Consult elementary diagrams and Custom Instruction Book for the specific drive functions provided and the associated interface connections.*

/ = NOT i.e. /C = NOT C.
* = LOW LEVEL SIG WIRE.

CONNECTION KEY:

- ◇ 6PL CONTROL CD
- ∅ 13 PL
- ■ 14 PL
- ◉ 11 PL
- ▲ 16 PL

APPLI-CATION CARD

CONTROL AND POWER SUPPLY CARDS

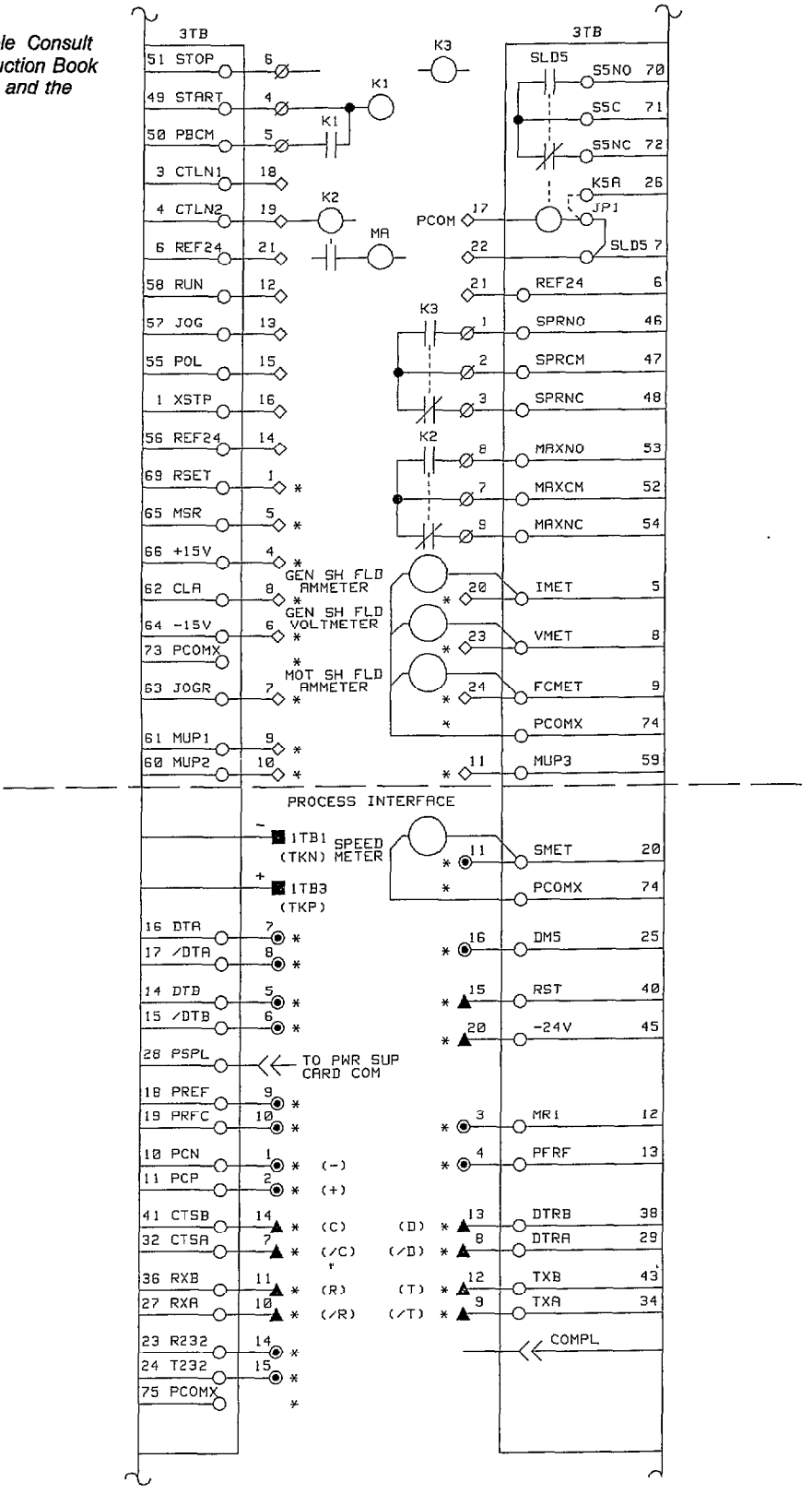


FIGURE 4-29
TYPICAL INTERFACE DIAGRAM (cont.)

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5 – ELECTRICAL CONNECTOR PIN CHARTS

This section contains electrical connector pin charts for all connectors in the controller. The major connectors are listed in numerical order, and any additional smaller connectors are listed at the end of the section under "ADDITIONAL CONNECTORS."

CONNECTOR 1PL — BETWEEN CONTROL CARD AND POWER SUPPLY CARD (PSC)

PIN	NOMENCLATURE	DESCRIPTION
1	DCOM	Digital Common.
2	FFC	Motor field firing control – drives the field gate pulse transformer.
3	DCOM	
4	FC	Motor field current feedback from PSC.
5	DCOM	
6	RT1	AC phase 1 input from PSC.
7	RT2	AC phase 2 input from PSC.
8	RT3	AC phase 3 input from PSC (RT1, RT2, & RT3 are resistance isolated).
9	FRC	Fault relay control – drives the 24-volt fault relay on PSC.
10	-24V	Unregulated -24V return path for FRC and RP.
11	RP	MA contactor pilot relay driver (MAX - K2 on PSC).
12	RUN	Drive run signal from RUN (K1) relay contact on PSC.
13	REF24	Supplies +/-24V to RUN (K1) relay on PSC depending on setting of jumper JP20.
14	W5V	Watchdog reference – provides 5V reference from PSC to undervoltage detector on Control card.
15	J1	Input from line phase 1 CT.
16	J2	Input from line phase 1 CT.
17	J3	Input from line phase 3 CT.
18	J4	Input from line phase 3 CT.
19	DCOM	
20	CFB	Generator field current feedback input from PSC.

CONNECTOR 2PL — BETWEEN CONTROL CARD AND POWER SUPPLY CARD (PSC)

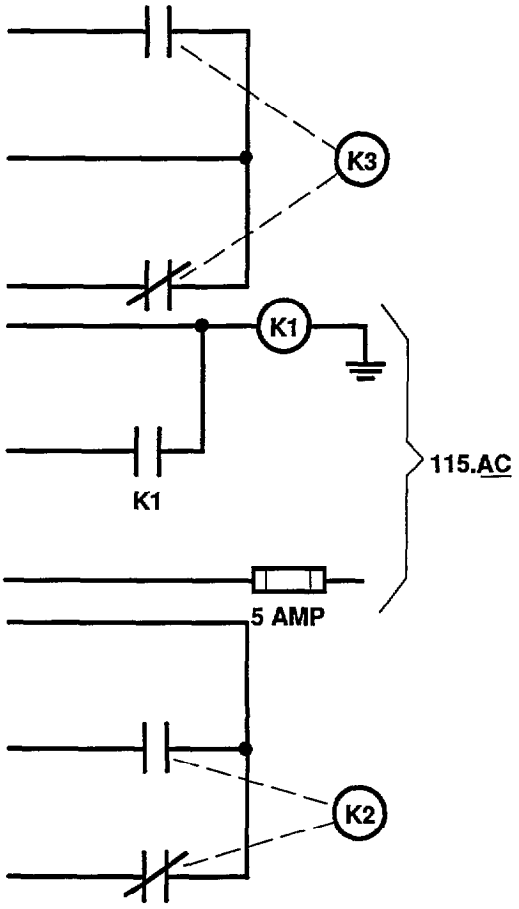
PIN	NOMENCLATURE	DESCRIPTION
1	ACOM	Analog signal return for +/-15V.
2	-15V	-15V regulated analog power supply input.
3	+15V	+15V regulated power supply input.
4	+24V	+24V unregulated raw power supply input.
5	-24V	-24V unregulated raw power supply input.
6	+5V	+5V regulated digital power supply.
7	DCOM	Digital signal return for +5V.
8	+5V	
9	PCOM	Raw power supply return for +/-24V.

CONNECTOR 3PL — BETWEEN CONTROL CARD AND MICROAPPLICATION CARD

PIN	NOMENCLATURE	DESCRIPTION
1	D0	U4 data bus bit 0 (least significant bit).
2	D1	U4 data bus bit 1.
3	DCOM	Digital common.
4	D2	U4 data bus bit 2.
5	D3	U4 data bus bit 3.
6	D4	U4 data bus bit 4.
7	DCOM	
8	D5	U4 data bus bit 5.
9	D6	U4 data bus bit 6.
10	D7	U4 data bus bit 7.
11	DCOM	
12	A8	U4 address bus bit 8.
13	A9	U4 address bus bit 9.
14	A10	U4 address bus bit 10.
15	DCOM	
16	A11	U4 address bus bit 11.
17	SEL	Signal from U3. Used for Microapplication card synchronization.
18	ALE	Address latch from U4.
19	DCOM	
20	/RD	Read control line from U4.
21	/WR	Write control line from U4.
22	A15	U4 address bus bit 15, used for Microapplication card I/O select.
23	RST	Card reset signal to Microapplication card.
24	+5V	+5V regulated digital supply to Microapplication card.
25	+5V	
26	+5V	
27	PCOM	+/-24V power supply return.
28	+24V	+24V non-regulated raw power supply to Microapplication card.
29	-24V	-24V non-regulated power supply to Microapplication card.
30	DM4	Provides "dummy" connection to Microapplication card.
31	+15V	+15V regulated analog supply to Microapplication card.
32	-15V	-15V regulated analog supply to Microapplication card.
33	ACOM	Analog signal return for +/-15V.
34	OPTA	Input to A-D converter from Microapplication card multiplexer.

CONNECTOR 4PL — BETWEEN POWER SUPPLY CARD (PSC) AND OPTIONAL 3TB

PIN	3TB TERM #	NOMENCLATURE	DESCRIPTION
1	46	SPRNO	Fault Relay Normally Open Contact
2	47	SPRCM	Fault Relay Common Connection
3	48	SPRNC	Fault Relay Normally Closed Contact
4	49	START	Connection Point for Start Pushbutton
5	50	PBCM	Common Point for Start and Stop Pushbuttons
6	51	STOP	Connection for Stop Pushbutton
7	52	MAXCM	MA Auxiliary Relay Common Connection
8	53	MAXNO	MA Auxiliary Relay Normally Open Contact
9	54	MAXNC	MA Auxiliary Relay Normally Closed Contact



NOTE: *Contact rating K2 relay is 1.2 amp at 115 volts AC or 28 volts DC.*
Contact rating K3 relay is 1 amp at 115 volts AC or 28 volts DC.

CONNECTOR 5PL — BETWEEN CONTROL CARD AND POWER CONNECTION CARD

PIN	NOMENCLATURE	DESCRIPTION
1	A6F	Drives cell 6F gate pulse transformer.
2	DFP	Delayed firing power for A6F and A5F.
3	A5F	Drives cell 5F gate pulse transformer.
4	PCOM	Shield wire.
5	A4F	Drives cell 4F gate pulse transformer.
6	DFP	Delayed firing power for A4F and A3F.
7	A3F	Drives cell 3F gate pulse transformer.
8	PCOM	Shield wire.
9	A2F	Drives cell 2F gate pulse transformer.
10	DFP	Delayed firing power for A2F and A1F.
11	A1F	Drives cell 1F gate pulse transformer.
12	PCOM	Shield wire.
13	A1R	Drives cell 1R gate pulse transformer.
14	DFP	Delayed firing power for A1R and A2R.
15	A2R	Drives cell 2R gate pulse transformer.
16	PCOM	Shield wire.
17	A3R	Drives cell 3R gate pulse transformer.
18	DFP	Delayed firing power for A3R and A4R.
19	A4R	Drives cell 4R gate pulse transformer.
20	PCOM	Shield wire.
21	A5R	Drives cell 5R gate pulse transformer.
22	DFP	Delayed firing power for A5R and A6R.
23	A6R	Drives cell 6R gate pulse transformer.
24	PCOM	Shield wire.
25	DCN	Resistance isolated generator field voltage.
26	DCP	Resistance isolated generator field voltage.

26	DCP	Resistance isolated generator field voltage.

CONNECTOR 6PL -- BETWEEN CONTROL CARD AND OPTIONAL 3TB

PIN	3TB TERM #	NOMENCLATURE	DESCRIPTION
1	69	RSET	A hard reset of drive is initiated when RSET is connected to +15V. Must be open to run. RSET stops firing of all SCRs, removes delayed firing power, drops out the MA contactor and fault relay, and holds all microprocessors in a reset state until RSET is opened. The drive will not automatically restart if reset while running. Do not apply RSET while running.
2	68	+5V	Not for use external to drive.
3	67	ACOM	Analog signal return for +/-15V power supplies to Auxiliary I/O card. See also connector DFPL.
4	66	+15V	Signal voltage for System Reference pot, RSET circuit, Auxiliary card and others. See also connector DFPL.
5	65	MSR	Manual system reference input to drive, ±15V max. Positive for forward motoring with POL at zero or open.
6	64	-15V	Signal voltage for system reference pot, Auxiliary I/O card and others. See also connector DFPL.
7	63	JOGR	Jog reference input (can be selected to bypass linear time), ±14V = Top speed. Programmable to 7V = top speed for fast update rate. Refer to Custom Software Description.
8	62	CLA	Current limit adjust - adds to software setting of current limit. +10V increases current limit by 100% of rated, -10V decreases it by 100%.
9	61	MUP1	Multiuse point 1 - connected to DFPL-3 (V_{MC}) for motor armature voltage feedback common signal at 3TB. Connects to Microapplication card operational amplifier #2 via JP26 and DM4. See connector DFPL this section, and Table 1. Not used if control equipped with 531X196DFAB__G revision DFA card.
10	60	MUP2	Multiuse point 2 - connects to the connection test point MUP2, which may be jumpered to another connection point.
11	59	MUP3	Multiuse point 3 - connected to +24V power supply by jumper JP28 on the Control card in 2-3 position. Supplies control power to Auxiliary I/O card. See also connector DFPL this section, and Table 1.
12	58	RUN	Alternate run command input in place of "RUN" relay. Do not pull high if run relay is also being used. Drive runs when connected to 3TB56 (6PL-14), REF24. A signal to run (6PL-12) picks up the MA contactor and REMOVES PRECONDITIONING from the controller.
13	57	JOG	Jog command input point. Drive jogs when connected to 3TB56 (6PL-14), REF24. Polarity controlled same as RUN above.
14	56	REF24	Selectable + or -24V supply for RUN, JOG, POL or XTSP input. Polarity set by JP10 and JP20 on Control card.
15	55	POL	Control input which reverses the direction of the drive for any applied reference; e.g. activating POL low is equivalent to reversing

CONNECTOR 6PL -- BETWEEN CONTROL CARD AND OPTIONAL 3TB (continued)

PIN	3TB TERM #	NOMENCLATURE	DESCRIPTION
		TRIP STOP	Not Recommended for DFM DRIVES Unless D.C. Contactor (MD) is opened first.
16	1	XSTP	Must be activated to enable drive. Activated/deactivated same as RUN above. Causes a normal, quick or trip stop when open (see ADR004-STPJP in the Custom Software Description).
17	2	PCOM	Not for use external to drive.
18	3	CTLN1	Control on - must be connected to CTLN2 to allow the drive to run. Opening this connection unconditionally drops MA contactor after a 50 millisec attempt by microprocessor to precondition the drive.
19	4	CTLN2	See CTLN1.
20	5	IMET	Analog load meter output proportional to generator field current, adjustable for 1 to 8V DC output at rated load current. See P2 and JP15 on the Control card (Table 1) for details. Load not to exceed 1000 ohms/volt.
21	6	REF24	Selectable ±24V supply for RUN, JOG, POL or XSTP input. Polarity set by JP19 and JP20 on the Control card.
22	7	INTR	Programmable 24V drive from Control card to the optional signal level detector SLD5 on 3TB terminal board. Controls contact at 3TB70, 71 and 72. See JP1 on the 3TB Terminal Board card (Table 1) for details.
23	8	VMET	Analog generator field voltmeter output. This output is bipolar, normally 4 volts at rated voltage output. (See P8 on the Control card, Table 1, for details.) Maximum load is 10 milliamps.
24	9	FCMET	Analog motor field current meter output, 1 to 5 VDC output. See P5 on the Control card (Table 1) for details. Load not to exceed 1000 ohms/volt.
--	26	K5A	Optional SLD5 control input from REF24 for external control of SLD5. Controls contact at 3TB70, 71, and 72. See JP1 on the 3TB Terminal Board card (Table 1) for details.
--	70	S5NO	Optional SLD5 normally open contact.
--	71	S5C	Optional SLD5 common contact.
--	72	S5NC	Optional SLD5 normally closed contact.

CONNECTOR 7PL — BETWEEN CONTROL CARD AND OPTIONAL PROCESS INTERFACE CARD

PIN	NOMENCLATURE	DESCRIPTION
1	ACOM	Analog signal return.
2	DVM	Output from digital voltmeter.
3	SFB	Output from analog tach circuit.
4	ASFB	Amplified signal from analog tach circuit, for improved resolution at low speed operation.
5	ACOM	
6	-15V	-15V supply to analog circuits on Process Interface (P.I.) card.
7	+15V	+15V supply to analog circuits on P.I. card.
8	+5V	+5V supply to logic circuits on P.I. card.
9	+5V	
10	TRS	RS422 driver tri-state command to P.I. card.
11	DCOM	Digital signal return.
12	DIR	Signal from digital tach interface signalling direction of motor rotation when a quadrature track is used.
13	TPN	Net pulse train signal from digital tach or AC analog tach interface used to determine tach speed.
14	PRF	Pulse train from digital reference interface used to determine desired reference when used.
15	DCOM	
16	DCOM	
17	RXD	Input from either the RS422 or RS232C serial link receiver from the P.I. card.
18	TXD	Output to the RS422 and RS232C serial link drivers from P.I. card.
19	/DTR	Data terminal ready--output indicating the P.I. card is ready to receive a byte on the serial link. Low when ready, high when not.
20	/CTS	Clear to send--input from serial link signalling when the receiver on the other end of the link is ready to receive a byte. When low, drive may transmit. When high, drive must not send anything out on TXD.
21	RST	Reset line from Control card to P.I. card, active high.
22	+5V	
23	+5V	
24	DCOM	
25	DCOM	
26	-24V	-24V raw power to P.I. card.

CONNECTOR 10PL — BETWEEN ANALOG PROCESS INTERFACE CARD AND OPTIONAL 3TB

PIN	3TB TERM #	NOMENCLATURE	DESCRIPTION
1	10	PCN	Optional negative input for process follower current (1-5, 4-20, 5-50, or 5-65mA.) or voltage (0-10V DC) reference.
2	11	PCP	Optional positive input for process follower reference.
3	12	MRI	Optional reference for manual system reference potentiometer to allow coordinated max speed scaling when PFRF is connected to Control card reference input via auto/manual selection.
4	13	PFRF	Optional process follower interface output (0-10V) which connects to a speed reference or other analog process input.
5	14	DTB	Optional square wave tach quadrature input (track B high). Operating Voltage: -0.5 to +0.8V is logic 0 and +2.4V to +15V is logic 1.
6	15	/DTB	Optional inverted square wave differential tach quadrature input (track B low).
7	16	DTA	Optional square wave tach main track input (track A high), Operating voltages: -0.5V to +0.8V is logic 0 and +2.4V to +15V is logic 1.
8	17	/DTA	Optional inverted square wave differential tach quadrature input (track A low).
9	18	PREF	Optional pulse reference frequency positive input, 0 to 6KHz Maximum.
10	19	PRFC	Optional high impedance common for pulse reference input.
11	20	SMET	Analog speed meter output, 10V DC. See P2 on the P.I. card (Table 1).
12	21	ACOM	Common for optional RS232C communications.
13	22	DCOM	Not for use external to drive.
14	23	R232	Optional RS232C receive line.
15	24	T232	Optional RS232C transmit line.
16	25	DM5	Spare connection tied to undedicated point DM5.

NOTE: */or* $\overline{}$ *= NOT: Example: /C or* $\overline{C}$ *= Not C*

TERMINAL BOARD 1TB LOCATED ON OPTIONAL PROCESS INTERFACE CARD

PIN	NOMENCLATURE	DESCRIPTION
1	TKN	Negative tachometer feedback signal. Maximum input is 310V DC or 344V AC.
3	TKP	Positive tachometer feedback signal.

CONNECTOR 11PL — BETWEEN MICROAPPLICATION CARD AND OPTIONAL 4TB

PIN	4TB TERM #	NOMENCLATURE	DESCRIPTION
1	30	EBR	Optional non-inverting differential or single-ended encoder B marker pulse output.
2	29	/EBR	Optional inverting differential encoder B marker pulse input.
3	85	MAP3	Microapplication card multiuse point 3. Connected to DFPL-2 (V_{GC}) for generator voltage feedback common signal. Connected to operational amplifier #1 via JP25 and JP31. See connector DFPL, this section, and Table 1. Not used if control equipped with 531X196DFAB__G version DFA card.
4	25	PSB	Strobe for optional 16 bit parallel input on 16PL. See Microapplication card JP7 (Table 1). Inputs must be set 50 micro-sec before strobe and strobe held for 100 micro-sec.
5	26	B03	Optional programmable 24V bit output #3.
6	27	B02	Optional programmable 24V bit output #2 .
7	28	B01	Optional programmable 24V bit output #1.
8	32	EBA	Optional non-inverting differential or single-ended encoder B Track A input.
9	31	/EBA	Optional inverting differential encoder B Track A input.
10	86	EBB	Optional non-inverting differential or single-ended encoder B Track B input.
11	87	/EBB	Optional inverting differential encoder B Track B input.
12	33	ENCR	Optional non-inverting differential or single-ended encoder A marker pulse input.
13	34	/ENCR	Optional inverting differential encoder A marker pulse input.
14	35	ENCB	Optional non-inverting differential or single-ended encoder A Track B input.
15	36	/ENCB	Optional inverting differential encoder A Track B input.
16	37	ENCA	Optional non-inverting differential or single-ended encoder A Track A input.
17	38	/ENCA	Optional inverting differential encoder A Track A input.
18	39	FC6	Bit 6 of optional fault code 24V output.
19	40	FC0	Bit 0 of optional fault code 24V output.
20	41	FC1	Bit 1 of optional fault code 24V output.
21	42	FC2	Bit 2 of optional fault code 24V output.
22	43	FC3	Bit 3 of optional fault code 24V output.
23	44	FC4	Bit 4 of optional fault code 24V output.
24	45	FC5	Bit 5 of optional fault code 24V output.

CONNECTOR 12PL — BETWEEN OPTIONAL PROCESS INTERFACE CARD AND OPTIONAL 3TB

PIN	3TB TERM #	NOMENCLATURE	DESCRIPTION
1	—	—	No connection.
2	—	—	No connection.
3	—	—	No connection.
4	—	—	No connection.
5	—	—	No connection.
6	—	—	No connection.
7	32	CTSA	Optional inverted RS422 clear-to-send input from serial link signaling when the Control card may transmit a byte, active high. (EIA standard nomenclature is CTSA.)
8	29	DTRA	Optional inverted RS422 data-terminal-ready output from Control card signaling when it is ready to receive a byte on the serial link, active high. (EIA standard nomenclature is DTRA.)
9	34	TXA	Optional inverted RS422 transmitted data signal from Control card to serial link. (EIA standard nomenclature is TXA.)
10	27	RXA	Optional inverted RS422 received data signal from serial link to Control card. (EIA standard nomenclature is RXA.)
11	36	RXB	Optional non-inverted RS422 received data signal from serial link to Control card. (EIA standard nomenclature is RXB.)
12	43	TXB	Optional non-inverted RS422 transmitted data signal from Control card to serial link. (EIA standard nomenclature is TXB.)
13	38	DTRB	Optional non-inverted RS422 data-terminal-ready output from Control card. (EIA standard nomenclature is DTRB.)
14	41	CTSB	Optional non-inverted RS422 clear-to-send input from serial link. (EIA standard nomenclature is CTSB.)
15	40	RST	Reset signal from Control card, active high, not for use external to drive.
16	39	+5V	Not for use external to drive.
17	42	+5V	Not for use external to drive.
18	37	DCOM	Not for use external to drive.
19	44	DCOM	Not for use external to drive.
20	45	-24V	Not for use external to drive.

CONNECTOR 13PL — BETWEEN MICROAPPLICATION CARD AND OPTIONAL 4TB

NOTE: *Refer to the Custom Instruction Book, software jumper description, for definitions of I/O through 13PL for particular application.*

PIN	4TB TERM #	NOMENCLATURE	DESCRIPTION
1	6	MODE 0	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
2	5	MODE 1	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
3	4	MODE 2	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
4	3	MODE 3	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
5	2	MODE 4	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
6	1	MODE 5	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
7	7	MODE 6	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
8	8	MODE 7	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
9	9	CI0	Provides MG Loop contactor (MD) indication to regulator (MD opened or closed).
10	10	CI1	DFM-RUN input, starts generator field control when connected to -15V. (Only if crane features are enabled.)
11	11	MAP1	Microapplication card multiuse point 1. Connected to DFPL-1 (V_{GX}) for generator voltage feedback signal. Connected to operational amplifier #1 via JP25 and JP31. See connector DFPL, this section and in Table 1. If control equipped with **531X196DFAB__G** version DFA card, connected to DFA card TB2 for motor voltage feedback signal.
12	12	MAP2	Microapplication card multiuse point 2. Connected to DFPL-4 (V_{MX}) for motor armature voltage feedback signal. Connected to operational amplifier #2 via JP27 and JP30. See connector DFPL, this section and in Table 1. If control equipped with **531X196DFAB__G** version DFA card, NOT USED.
13	13	AN2	General purpose Microapplication card analog input ±10V. May be scaled to voltage ranges of up to 48V via P5 — AN2 SCALE. Refer to elementaries and Custom Software Description for more information. NOT USED if control equipped with **531X196DFAA__G** version DFA card. If control equipped with **531X196DFAB__G** version DFA card, connected to DFA card TB1 for generator voltage feedback signal.
14	14	DA12	Optional analog output from 12-bit D-A converter. Range = -10V to +10V, 1 MA, with 450 ohms output impedance. Update rate is 30 to 40 times/second.

* *Functions of MODE0 - MODE7 are determined by Addresses 000-015 (ADR000 - ADR015) in the software description.*

CONNECTOR 13PL — BETWEEN MICROAPPLICATION CARD AND OPTIONAL 4TB (continued)

PIN	4TB TERM #	NOMENCLATURE	DESCRIPTION
15	15	SJ1	Optional analog input to summing junction #1. Operating range = ±10V; maximum input = ±15V.
16	16	SJ2	Optional analog input to summing junction #2. Operating range = ±10V; maximum input = ±15V.
17	17	PSREF	Optional analog position or process reference input (±10V).
18	18	LTA	Optional analog linear time adjust. Positive voltages increase timing, negative voltages decrease timing ±10 volts maximum. 10 volts is equal to 66 second adjust.
19	19	DA8	Optional analog output from 8-bit D-A converter. See DA12 for specifications. Normal update rate of 180 times/sec. Reduced to 30 to 40 times/sec if fast update selected for DA0.
20	20	FRA	Optional analog motor field reference adjust. Positive voltages up to +10V increase field current. Negative voltages down to -10V decrease field current. See Microapplication card JP32 and P2 (Table 1).
21	21	ASP1	Optional spare analog input #1 (programmable).
22	22	ASP2	Optional spare analog input #2 (programmable).
23	23	ASR	Optional auto speed reference input to drive, ±15V maximum positive for forward cell firing while motoring. ASR is the active speed reference in the Auto mode when analog auto reference is selected.
24	24	AN1	General purpose Microapplication card analog input (±10V). Connected to DFPL-10 (ACFB) for motor armature current feedback from the MG Loop shunt. 2 VDC input at rated armature current. See connector DFPL, this section, and P4-AN1 SCALE on the Microapplication card, Table 1. If control equipped with 531X196DFAB__G version DFA card, connected to DFA card TB3.

CONNECTOR 14PL — BETWEEN MICROAPPLICATION CARD AND OPTIONAL 4TB

PIN	4TB TERM #	NOMENCLATURE	DESCRIPTION
1	62	S0NO	Optional SLD #0 normally open contact.
2	63	S0NC	Optional SLD #0 normally closed contact.
3	64	S0C	Optional SLD #0 common contact.
4	65	S1NO	Optional SLD #1 normally open contact.
5	66	S1NC	Optional SLD #1 normally closed contact.
6	67	S1C	Optional SLD #1 common contact.
7	68	S2NO	Optional SLD #2 normally open contact.
8	69	S2NC	Optional SLD #2 normally closed contact.
9	70	S2C	Optional SLD #2 common contact.
10	71	S3NO	Optional SLD #3 normally open contact.
11	72	S3NC	Optional SLD #3 normally closed contact.
12	73	S3C	Optional SLD #3 common contact.
13	74	S4NO	Optional SLD #4 normally open contact.
14	75	S4NC	Optional SLD #4 normally closed contact.
15	76	S4C	Optional SLD #4 common contact.

CONNECTOR 16PL — LOCATED ON OPTIONAL APPLICATION CARD

PIN	NOMENCLATURE	DESCRIPTION
1	DCOM	Digital common.
2	+5V	Not used.
3	PI0	Optional parallel word input least significant binary bit input. For all inputs refer to Table 1, JP8 description for voltage level select. Input must be set for 50 *micro-sec prior to strobe.*
4	PI1	Optional parallel binary bit input 1.
5	PI2	Optional parallel binary bit input 2.
6	PI3	Optional parallel binary bit input 3 (or PI0--PI3 = BCD units digit).
7	PI4	Optional parallel binary bit input 4.
8	PI5	Optional parallel binary bit input 5.
9	PI6	Optional parallel binary bit input 6.
10	PI7	Optional parallel binary bit input 7 (or PI4--PI7 = BCD tens digit).
11	PI8	Optional parallel binary bit input 8.
12	PI9	Optional parallel binary bit input 9.
13	PI10	Optional parallel binary bit input 10.
14	PI11	Optional parallel binary bit input 11 (or PI8--PI11 = BCD hundreds digit).
15	PI12	Optional parallel binary bit input 12.
16	PI13	Optional parallel binary bit input 13.
17	PI14	Optional parallel binary bit input 14 (or PI12--PI14 = BCD thousands digit).
18	PI15	Optional sign bit input.
19	DCOM	Digital common.
20	DCOM	Digital common.

NOTE: *16PL connects the Microapplication card to a discrete external connection device to provide a 16 bit parallel input from outside sources. Inputs are compatible with ±24 volt sources, with a threshold of approximately 2.5 volts.*

CONNECTOR DFPL —
BETWEEN AUXILIARY CARD 531X196DFAA__G VERSION, 3TB and 4TB CARDS

PIN	TERM #	NOMENCLATURE DFPL (TERM)	DESCRIPTION
1	4TB11	V_{GX} (MAP1)	Resistance isolated generator voltage feedback signal. Feeds into Microapplication card operational amplifier #1, through analog input MAP1. See JP1 on Auxiliary I/O card and JP25, JP31 and P7 on Microapplication card, Table 1.
2	4TB85	V_{CC} (MAP3)	Resistance isolated generator loop voltage feedback common signal. Feeds into Microapplication card operational amplifier #1, through analog input MAP3. See JP2 on Auxiliary I/O card and JP24 and JP29 on Microapplication card, Table 1.
3	3TB61	V_{MC} (MUP1)	Resistance isolated motor armature voltage feedback common signal. Feeds into Microapplication card operational amplifier #2, through Control card analog input MUP1. See JP3 on Auxiliary I/O card and JP26 and P8 on the Microapplication card, Table 1.
4	4TB12	V_{MX} (MAP2)	Resistance isolated motor armature voltage feedback signal. Feeds into Microapplication card operational amplifier #2, through analog input MAP2. See JP4 on the Auxiliary I/O card and JP27 and JP30 on Microapplication card, Table 1.
5	3TB59	+24V (MUP3)	+24V unregulated power supply to Auxiliary I/O card.
6	3TB28	PCOM (PSPL)	Power Supply return for +24V.
7	3TB66	+15V	+15V regulated power supply input to Auxiliary I/O card.
8	3TB67	ACOM	Analog signal return for ±15V.
9	3TB64	-15V	-15V regulated power supply input to Auxiliary I/O card.
10	4TB24	ACFB (AN1)	Armature current feedback signal derived from generator loop shunt. Feeds into Microapplication card analog input, AN1, 1VDC at rated armature current. See P1 and P2 on Auxiliary I/O card and P4 on Microapplication card, Table 1.

CONNECTOR DFPL — BETWEEN AUXILIARY CARD 531X196DFAB__G VERSION, 3TB and 4TB CARDS

PIN	TERM #	NOMENCLATURE DFPL (TERM)	DESCRIPTION
1	4TB13	GFB (AN2)	Resistance isolated generator voltage feedback signal. See JP1 and JP2 on Auxiliary I/O card.
2	4TB11	MFB (MAP1)	Resistance isolated motor loop voltage feedback signal. See JP3 and JP4 on Auxiliary I/O card and JP31 on Microapplication card, Table 1.
3	4TB24	ACFB (AN1)	Armature current feedback signal derived from generator loop shunt. Feeds into Microapplication card analog input, AN1/. 2VDC at rated armature current. See P1 and P2 on Auxiliary I/O card and P4 on Microapplication card, Table 1.
4	NOT USED		
5	3TB59	+24V (MUP3)	+24V unregulated power supply to Auxiliary I/O card.
6	3TB28	PCOM (PSPL)	Power Supply return for +24V signal.
7	3TB66	+15V	+15V regulated power supply input to Auxiliary I/O card.
8	3TB67	ACOM	Analog signal return for ±15V signal.
9	3TB64	-15V	-15V regulated power supply input to Auxiliary I/O card.

ADDITIONAL CONNECTORS

CONNECTOR 1CPL — BETWEEN POWER SUPPLY CARD AND AC LINE CTs (CT1 AND CT3)

PIN	NOMENCLATURE	DESCRIPTION
1	J1	Input from line phase 1 CT.
2	J2	Input from line phase 1 CT.
3	J3	Input from line phase 3 CT.
4	J4	Input from line phase 3 CT.

CONNECTOR CNPL — BETWEEN POWER SUPPLY CARD AND MA CONTACTOR

PIN	NOMENCLATURE	DESCRIPTION
1	MAP	MA contactor control output.
2	X2	MA contactor control output.

CONNECTOR CPTPL — BETWEEN POWER SUPPLY CARD AND CONTROL POWER TRANSFORMER

PIN	NOMENCLATURE	DESCRIPTION
1	—	38 VAC, center tapped for controller power supplies.
2	—	38 VAC, center tapped for controller power supplies.
3	—	Center tap point for 38 VAC.
4	—	Not used.
5	—	115 VAC for controller cooling fan (if supplied).
6	—	115 VAC for controller cooling fan (if supplied).

CONNECTOR DRPL — BETWEEN POWER SUPPLY CARD AND SHUNT

PIN	NOMENCLATURE	DESCRIPTION
RED	IP	Generator field current feedback from shunt, 0-100mV. Positive when motoring with the forward power circuit, Red wire.
WH	IN	Generator field current feedback from shunt. Negative when drive is motoring with the forward power circuit, White wire.

CONNECTOR FAPL — BETWEEN POWER SUPPLY CARD AND CONTROLLER COOLING FAN (IF USED)

1	X1	115 VAC for controller cooling fan.
2	X2	115 VAC for controller cooling fan.

CONNECTOR FCPL — BETWEEN POWER SUPPLY CARD AND EXTERNAL MOTOR FIELD CONTROL ASSEMBLY

PIN	NOMENCLATURE	DESCRIPTION
1	ACOM	Analog common.
2	—	External motor field control feedback. Used only with external motor field control assembly, Figures 4-16 and 4-17.

CONNECTOR FSPL — BETWEEN POWER SUPPLY CARD AND FIELD POWER MODULE

PIN	NOMENCLATURE	DESCRIPTION
1	FLC	Gate pulse transformer output to field power module SCR, Red wire. Not used with external motor field control assembly, Figures 4-16 and 4-17.
2	FLG	Gate pulse transformer output to field power module SCR, White wire.

CONNECTOR OPTPL — LOCATED ON POWER SUPPLY CARD

PIN	NOMENCLATURE	DESCRIPTION
1	FX1	Fused 115 VAC from Control Power Transformer.
2	X2	115 VAC return from Control Power Transformer.

CONNECTOR PSPL — LOCATED ON POWER SUPPLY CARD

PIN	NOMENCLATURE	DESCRIPTION
1	+5V	+5V regulated digital supply (not for use external to drive).
2	DCOM	Digital signal return for +5V (not for use external to drive).

CONNECTOR COMPL —
OPTIONAL R232C SERIAL COMMUNICATIONS CONNECTOR ON OPTIONAL 3TB

PIN	NOMENCLATURE	DESCRIPTION
1	GND	Ground.
2	RXA or TXA	For Data Communications Equipment (DCE), dumb terminal mode, pin 2 is the RS232C receive line (RXA).
		For Data Terminal Equipment (DTE), modem or PDS system, pin 2 is the RS232C transmit line (TXA).
3	TXA or RXA	For DCE, pin 3 is the TXA line.
		For DTE, pin 3 is the RXA line.
4	—	Not used.
5	DTRA or CTSA	For DCE, pin 5 is the data-terminal-ready (DTRA) line.
		For DTE, pin 5 is the clear-to-send (CTSA) line.
6	DSR	Data set ready, no connection in DCE or DTE mode.
7	DCOM	Pin 7 is the communication system common.
8 thru 19	—	Not used.
20	CTSA or DTRA	For DCE, pin 20 is the clear-to-send (CTSA) line.
		For DTE, pin 20 is the data-terminal-ready (DTRA) line.
21 thru 25	—	Not used.

NOTE: *Refer to 3TB SW1 and SW2 description in Table 1, for setting DCE and DTE modes.*

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6 – DFM-300 ADJUSTMENTS

The following section details adjustments available in hardware and software. The hardware portion describes the location and function of jumpers and potentiometers on the cards. The software portion describes key software "parameter" adjustments available and provides a "Parameter Map" that relates the adjustments to electrical hardware equivalents.

HARDWARE JUMPER AND POT ADJUSTMENTS

Most of the jumper selections and pot adjustments have been factory set. When one of the cards has to be changed, make sure the replacement card switches, jumpers and pots are set to be the same as on the replaced card. For a full description of the jumper and pot adjustments, see Table 1.

NOTE: *If the Control card is replaced, the EEROM (U12) from the original card should be put into the replacement card. If the same failure symptoms still exist, the new EEROM, shipped with the replacement card, can be reinstalled but must be reprogrammed per the customer software jumper and adjustment values supplied in the Custom Software Description or parameter list in the drive door pocket.*

Hardware jumpers and pots are located on the Control card, Power Supply card, optional Process Interface card, Microapplication card, and optional Programmer card. Figures 6-1 thru 6-8 show the positions of pots, jumpers, test points and other key components.

Most of the pots and jumpers are used in the interface circuits of the microprocessor software control circuit. Some of the jumpers are only for manufacturer's use. The test data sheets supplied with each controller (in the drive door pocket) indicate the position of these adjustments as they were set in the factory. Adjustments not set in the factory can easily be made by the startup procedure in this book and by use of Table 1.

NOTE: *Only high impedance digital voltmeters or the optional "on-board" DVM (Test 3) should be used to make indicated measurements.*

NOTE: *Before changing any jumper positions, compare actual jumper positions with "as shipped" jumper positions through use of the Custom Software Description.*

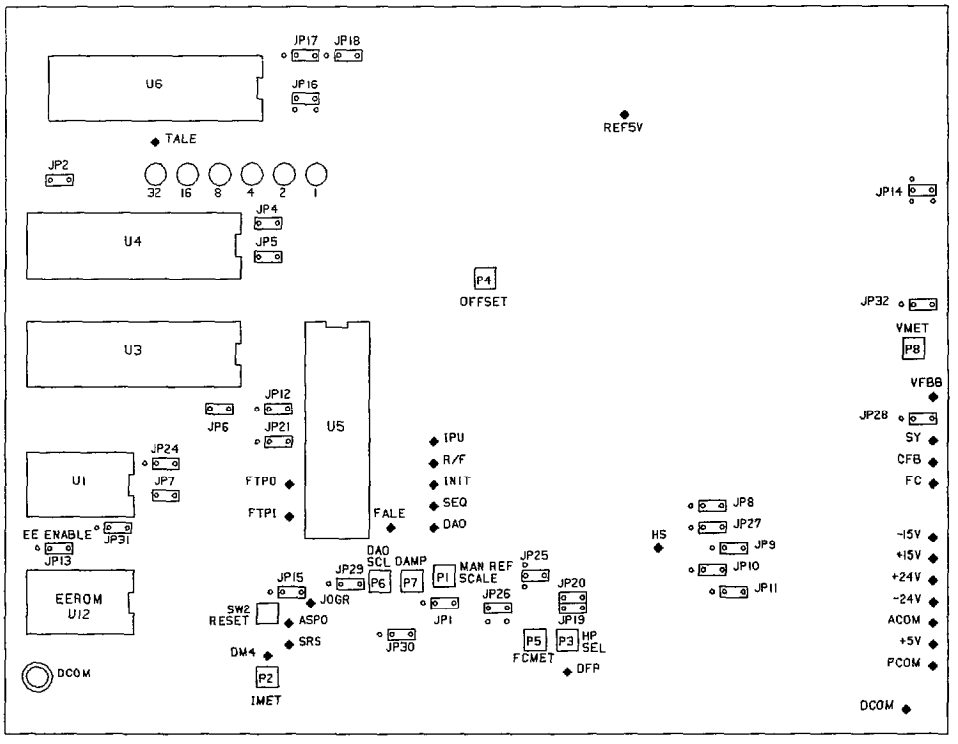


FIGURE 6-1
CONTROL CARD LAYOUT

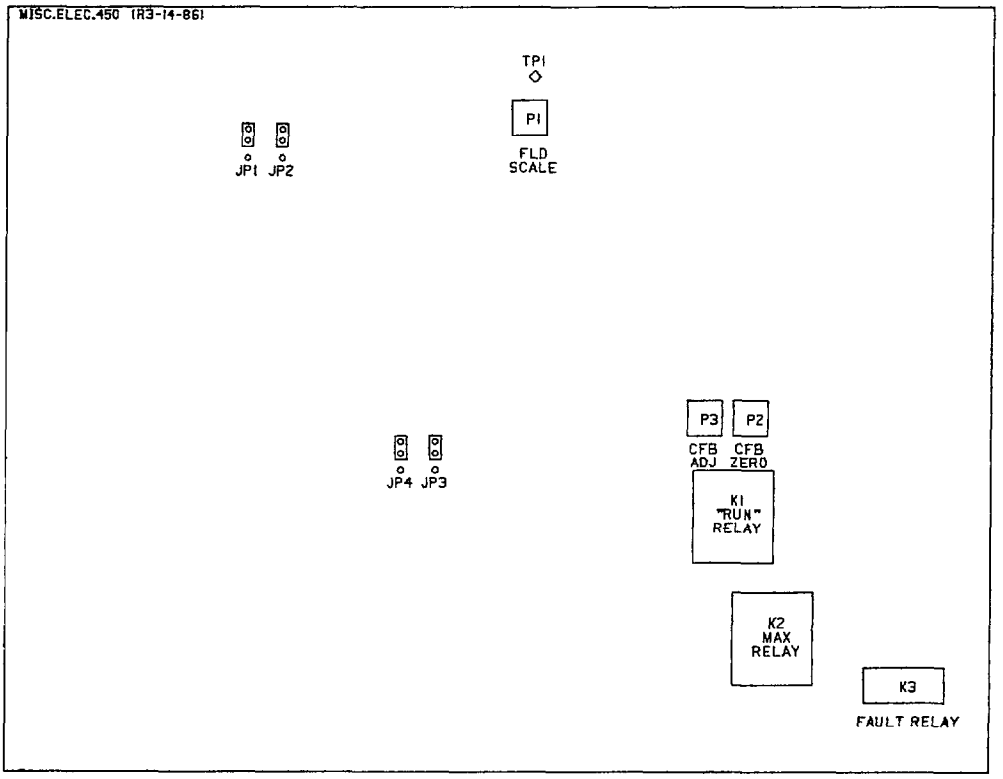


FIGURE 6-2
POWER SUPPLY CARD LAYOUT

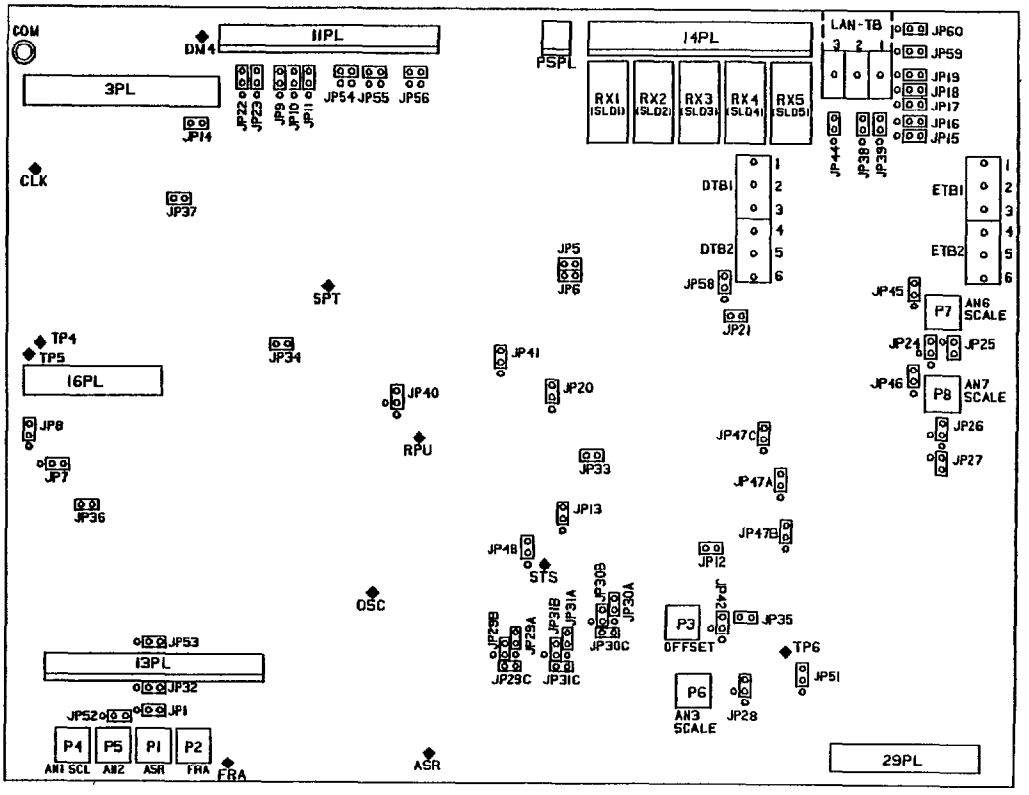


FIGURE 6-3
MICROAPPLICATION CARD LAYOUT

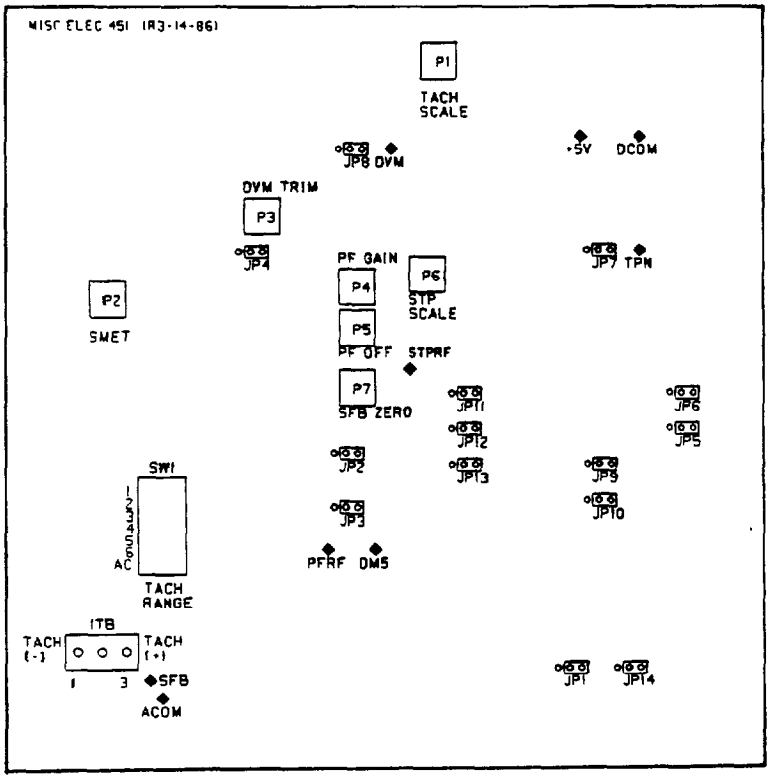


FIGURE 6-4
OPTIONAL PROCESS INTERFACE (531X133PRU SERIES) CARD LAYOUT

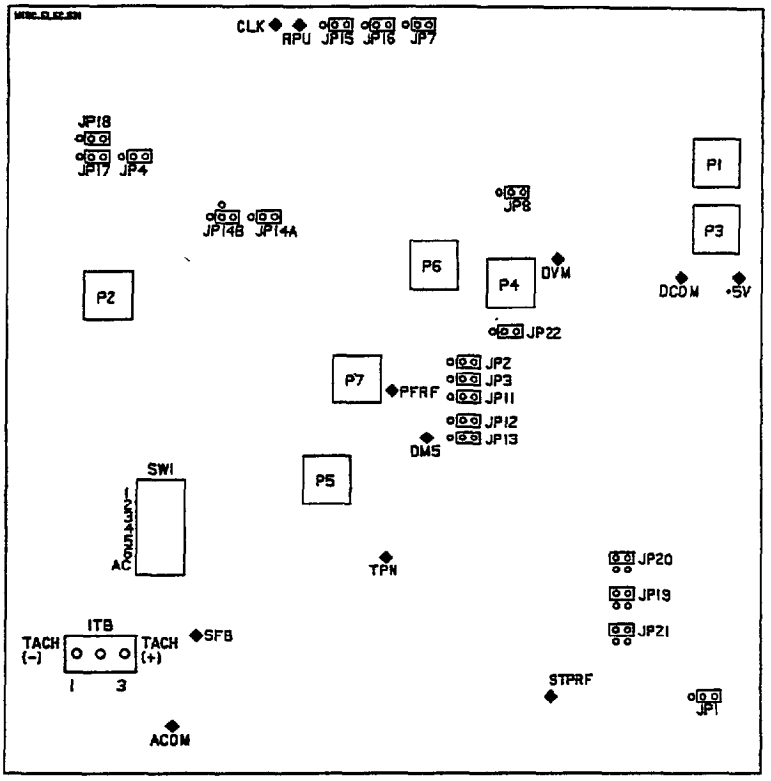


FIGURE 6-5
OPTIONAL ENCODER PROCESS INTERFACE (531X134EPR SERIES) CARD LAYOUT

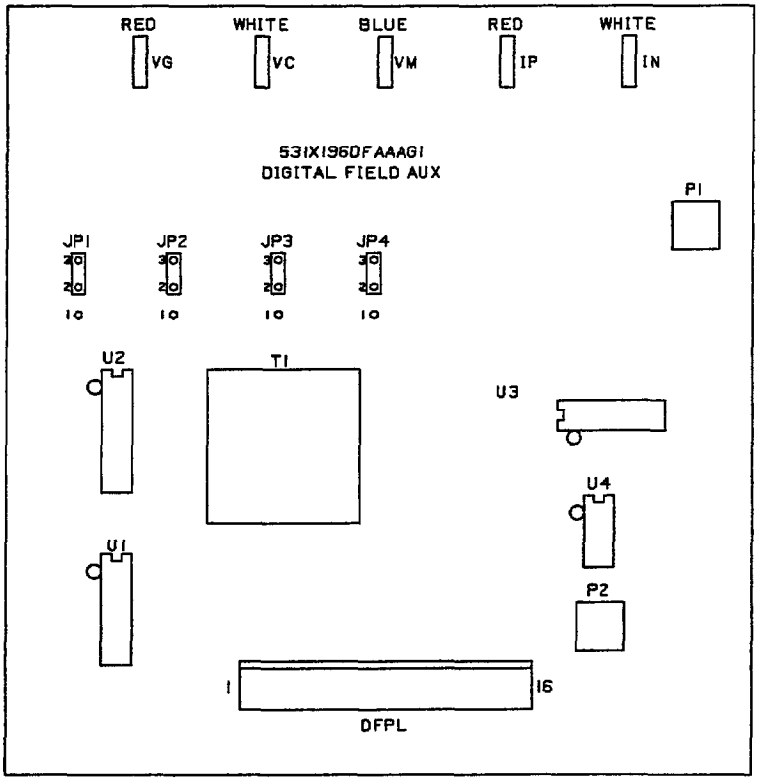


FIGURE 6-6
531X196DFAA__G VERSION DFA CARD LAYOUT

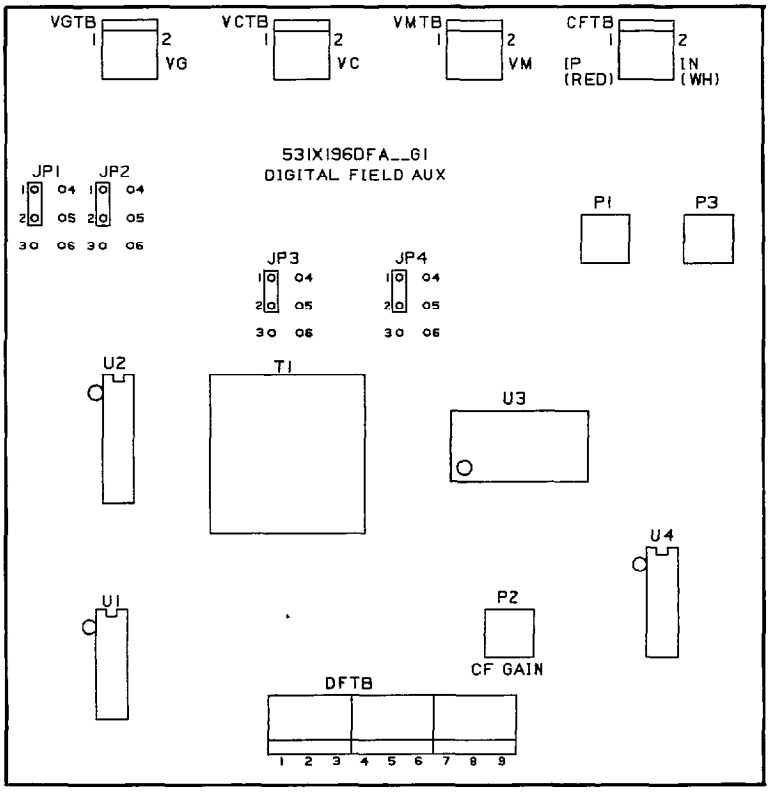


FIGURE 6-7
531X196DFAB__G VERSION DFA CARD LAYOUT

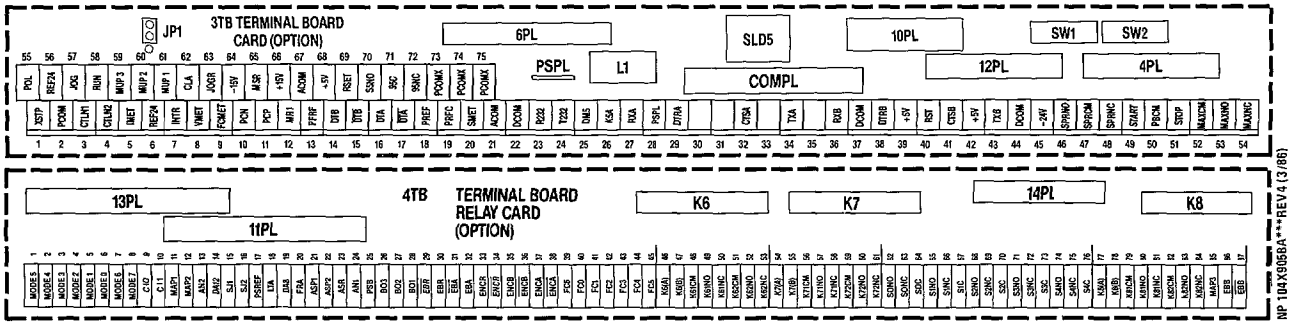


FIGURE 6-8
OPTIONAL TERMINAL BOARDS (3TB AND 4TB) LAYOUT

TABLE 1 – JUMPER AND POT ADJUSTMENTS

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Control card	Pot	P1	MAN REF	To adjust test point SRS to four (4) volts when top speed reference is at MSR, 3TB65 (6PL-5; connector 6PL, pin number 5). Used in conjunction with jumper JP1.
		P2	IMET	This pot is used to adjust the output of the analog generator field meter circuit. Adjust the output voltage to be from 1 to 8 volts at rated load. Maximum current load on this signal is 1000 ohms/volt.
		P3	HP SEL	Adjusted by the manufacturer. To adjust in the field, set to make it visually the same on replacement cards.
		P4	OFFSET	Adjusted by the manufacturer, no need to be adjusted in the field unless directed to do so by fault codes. Refer to adjustment procedure in "Use of the Optional Programmer Module" section if a Programmer is available.
		P5	FCMET	This pot is used to adjust the output of the analog motor field current voltmeter circuit. Adjust the output to be from 1 to 5 volts at rated maximum field current. Maximum current load on this signal is 1000 ohms/volt.
		P6	DA0 SCL	This pot is used to adjust the gain of the 12-bit digital-to-analog converter (DA0) located on the Control card. Full CCW sets the gain at 0.25, full CW sets the gain at 1.25. The output saturates at 12 VDC. (See JP29 for more detail.)
		P7	DAMP	Not used.
		P8	VMET	This pot is used to adjust the output of the analog generator field voltage meter circuit. Output is adjustable from 0 to 8 volts at rated voltage. Maximum current load on this signal is 10mA.
	Jumper	JP1	MAN REF	This jumper selects the maximum voltage range for the manual speed reference input. 1-2: 8 to 15 volts 2-3: 5 to 8 volts

TABLE 1 – JUMPER AND POT ADJUSTMENTS (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Control card	Jumper	JP2 JP4 JP5 JP6 JP7		JP2 - JP7 Set as follows: 1-2: Should be in this position for proper operation.
		JP8 JP9 JP10 JP11		These jumpers are set in accordance with the actual generator field current. Refer to the Custom Instruction Book for proper settings.
		JP12		1-2: Disables SCR test (normal operation).
		JP13		This jumper provides a hardware write protection for the software jumpers and adjustments. 1-2: Protected (should be in this position for maximum security and reliability of the drive). 2-3: Enables reprogramming. Do not perform a "hard" reset or turn power on or off in this position.
		JP14		This jumper selects proper line voltage. 1-2: 440 to 480 VAC 2-3: 380 to 415 VAC 2-5: 310 to 340 VAC 3-4: 220 to 240 VAC
		JP15		This jumper selects whether the generator field meter driver is to be bipolar or absoluted. 1-2: Absoluted (always positive) 2-3: Bipolar (zero center)
		JP16 JP17 JP18		JP16 - JP18 set as follows: Set by factory - refer to jumper listing in Custom Instruction Book in drive door pocket for proper setting.

TABLE 1 – JUMPER AND POT ADJUSTMENTS (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Control card	Jumper	JP19 JP20		JP19 & JP20 set the polarity of RUN, JOG, POL and XSTOP. 1-2 & 3-4 Inputs are active when pulled down between -24 to 0.4 VDC. Inputs are inactive when open or tied to +3.5 to +24 VDC. 1-3 & 2-4 Inputs are active when tied between +3.5 to +24 VDC. Inputs are inactive when open or held at -24 to 1.5 VDC.
		JP21		Allows interactive card test. 1-2: Disables card test (normal operation). 2-3: Initiates card test if drive is not running. To exit card test, move JP21 back to the 1-2 position and hard reset drive (push SW2).
		JP24 JP25 JP27 JP28 JP30 JP31		Set by factory - refer to jumper listing in Custom Instruction Book in drive door pocket for proper setting.
		JP26		This jumper directs MUP1 (DFPL-3) input. Connect to DM4 on Control card to feed signal to Microapplication card. NOT USED if control equipped with 531X196DFAB_G version DFA card.
		JP29		This jumper selects whether the DA0 output has a fixed gain or adjustable gain. 1-2: Fixed gain (see Table 3, RAM Variable Scaling). 2-3: Gain adjustable by P6-DA0 SCL.
	Switch	SW2	HARDWARE RESET	This switch causes a hardware reset ("hard" reset) of the controller when pressed. Used to clear some drive faults (refer to Table 6, Specific Fault Troubleshooting). The power supplies are toggled "OFF" and "ON" and the drive will return to the initial power up condition.

TABLE 1 – JUMPER AND POT ADJUSTMENTS (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
MFC/ Power Supply card	Pot	P1	FIELD SCALE	P1 is used to calibrate the field current feedback using a known field current. This pot is set and sealed by the manufacturer.
		P2	CFB* ZERO	P2 is used to adjust offset of the current isolator circuit so that test point CFB (on Control card) is zero.
		P3	CFB* ADJUST	**CAUTION** The drive must not be running when making this adjustment. P3 is used to adjust the gain of the current isolator circuit for CFB. Pots P2 and P3 are interdependent and should be adjusted alternately. **WARNING** **Electrical shock hazard. High voltage is present on the Power Supply card. Remove power from the drive before handling the plug and jumpers in this procedure.** P3 can be set by disconnecting the current feedback signal (DRPL) and moving jumpers JP3 and JP4 to the 2-3 position. This applies a calibration voltage (approx. 0.1V) between pin IP and IN of DRPL. P2 should be adjusted prior to moving JP3 and JP4. Adjust P3 until test point CFB on the Control card measures the value indicated on the drive calibration label. Setting P3, P2 should be rechecked. Return JP3 and JP4 to the 1-2 positions and reconnect DRPL. **CAUTION** If JP3 and JP4 are left in the 2-3 positions, erratic operation and card failures can occur.

** CFB is a signal directly proportional to generator field current.*

TABLE 1 – JUMPER AND POT ADJUSTMENTS (continued)

<table>
<tr><th>LOCATION</th><th>JUMPER OR POT</th><th>NUMBER</th><th>NOMEN-CLATURE</th><th>DESCRIPTION</th></tr>
<tr><td rowspan="3">MFC/ Power Supply card (continued)</td><td rowspan="3">Jumper</td><td>JP1
JP2</td><td></td><td>JP1 and JP2 are used to select motor field current feedback signal (FC) gain.

Version G1 MFC/Power Supply card (Figure 9).
JP1 JP2 GAIN FACTOR FIELD AMP RANGE
2-3 2-3 0.4 0.64 to 1.6 AMPS
1-2 2-3 1.0 1.6 to 4.0 AMPS
1-2 1-2 2.5 4.0 to 10.0 AMPS

Version G2 MFC/PSC (Figure 10) and optional 24ADC Remote Field Module (Figure 11).
JP1 JP2 GAIN FACTOR FIELD AMP RANGE
2-3 2-3 1.5 1.5 to 6.0 AMPS
1-2 2-3 3.0 6.0 to 12.0 AMPS
1-2 1-2 6.0 12.0 to 24.0 AMPS

Version G2 MFC/PSC (Figure 10) and optional 72ADC Remote Field Module.
JP1 JP2 GAIN FACTOR FIELD AMP RANGE
2-3 2-3 4.5 4.5 to 18.0 AMP
1-2 2-3 9.0 18.0 to 36.0 AMPS
1-2 1-2 18.0 36.0 to 72.0 AMPS</td></tr>
<tr><td>JP3
JP4</td><td></td><td>JP3 and JP4 are used for calibration of the generator current field feedback signal.
JP3 JP4 Function
1-2 1-2 Normal Operating Mode
2-3 2-3 CFB calibration
(See description at pot P3.)</td></tr>
<tr><td>JP5</td><td></td><td>K2 Coil Suppression Capacitor Jumper
1-2 Normal: Capacitor in.
2-3 Disable: Capacitor out.</td></tr>
<tr><td>Power Connection card</td><td>Jumper</td><td>P1
P2
P3
P4
P5
P6
P7
P8
P9
P10</td><td></td><td>Select generator field voltage feedback scaling resistors dependent on rated DC voltage output. Set by connecting wire jumpers with stab-on type connectors across the points indicated.
JUMPERS OUTPUT VOLTAGE
NONE 500V DC
P4-P8 and P5-P9 450V DC
P3-P7 and P6-P10 400V DC
P4-P7 and P5-P10 340V DC
P7-P8 and P9-P10 290V DC
P3-P4 and P5-P6 240V DC
P3-P8 and P6-P9 190V DC</td></tr>
</table>

TABLE 1 – JUMPER AND POT ADJUSTMENTS (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
3TB Terminal Board (optional)		JP1		JP1 selects whether optional SLD5 mounted on 3TB is controlled by the Control card or by an external contact closure. 1-2: Control card control of SLD5, programmed by ADR085, 086 and 087. 2-3: External control of SLD5, by connecting 3TB26 K5A to 3TB56, REF24.
	Switch	SW1 SW2		SW1 and SW2 select the mode of serial link communication through COMPL on 3TB (depress switch as shown by dot to set for up or down position). For use with Data Terminal Equipment (DTE): SW1-1 thru 3: Up SW1-4: Down SW2-1, 3, 4: Down SW2-2: Up SWITCH SET Up Down For use with Data Communication Equipment (DCE), all switches set in down position.
Micro-application card (optional)	Pot	P1	ASR SCALE	P1 is used in conjunction with JP1 to trim test point ASR to four (4) volts for top speed reference applied to 13PL-23 (4TB23).
		P2	FRA SCALE	P2 is used to scale the motor field reference adjust input at 13PL-20 (4TB20). One volt amplitude at test point FRA increases/decreases the FLDMX parameter setting by 204 counts.
		P4	AN1 SCALE	P4 scales the analog input AN1. Must be set full CW for the DFM-300 drive.
		P5	AN2 SCALE	P5 scales the analog input AN2. If control equipped with 531X196DFAB__G version DFA card, set in the fully clockwise position. NOT USED if control equipped with 531X196DFAA__G version DFA card.
		P7	AN6 SCALE	P7 is not used on the DFM-300 drive.
		P8	AN7 SCALE	P8 is not used on the DFM-300 drive.

TABLE 1 – JUMPER AND POT ADJUSTMENTS (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Micro-application card (continued)	Jumper	JP1		JP1 selects the voltage range for the auto speed reference input (ASR) at 4TB23 (13PL-23). 1-2: 8 to 15 volts 2-3: 5 to 8 volts
		JP2		JP2 selects whether the encoder Track A input at 11PL-16 (4TB37) is single ended or differential. 1-2: RS422 differential 2-3: TTL compatible single ended
		JP3		JP3 selects whether the encoder Track B input at 11PL-14 (4TB35) is single ended or differential. 1-2: RS422 differential 2-3: TTL compatible single ended
		JP4		JP4 selects whether the encoder marker track input at 11PL-12 (4TB33) is single ended or differential. 1-2: RS422 differential 2-3: TTL compatible single ended
		JP5 JP6		JP5 and JP6 select polarity of the eight logic inputs, MODE 0-MODE 7, at 13PL-1 thru 8 (4TB1 thru 8). 1-2 & 3-4: The control is active low. Inputs are active when pulled below 0.8 volts. 1-3 & 2-4: The control is active high. Inputs are active when pulled above 3.5 volts or driven by TTL high output.
		JP7		JP7 selects whether the sixteen bit parallel input is latched on the high or low level of the strobe. 1-2: Data latched on low to high transition of PSB. 2-3: Data latched on low level PSB.
		JP8		JP8 selects whether the sixteen bit parallel input and strobe signals are driven by an active high or low driver. 1-2: Lines internally pulled up to +5V. Must be driven by active pull-down to common driver. 2-3: Lines internally pulled down to DCOM. Must be driven by an active pull-up (above 3.5 volt) drive.

TABLE 1 – JUMPER AND POT ADJUSTMENTS (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Micro-application card (continued)	Jumper	JP14		JP14 set as follows: 1-2: Should be in this position for proper operation. Open: Only for manufacturer's use.
		JP24 JP25 JP26 JP27 JP28 JP29 JP30 JP31		JP24 thru JP31 define functions for the on-board general purpose amplifiers (GP AMP #1 and GP AMP #2). Refer to drive elementaries, operating notes and Custom Software Description for more information. Used on DFM-300 drives equipped with 531X196DFAA__G version DFA card to route motor armature voltage and current feedback signals. JP24 thru JP30 are NOT USED if control equipped with 531X196DFAB__G version DFA card. JP31 must set in 2-5 position to connect MAP1 to AN4 if 531X196DFAB__G version card.
		JP32		JP32 selects the scaling range of P2-FRA SCALE. 1-2: Normal range 2-3: Extended range
		JP36 JP37		JP36 & JP37 set as follows: 1-2: Should be in this position for proper operation. Open: Only for manufacturer's use.
		JP40 JP41		JP40 & JP41 select whether the encoder input is used as a feedback or a pulse reference: **JP40 JP41** 1-2 1-2 Normal encoder operation 2-3 1-2 Enables encoder track A as a pulse reference frequency input at 3TB37 (11PL-16, ENCA), 40k Hz maximum. 2-4 — Not used.
		JP45		JP45 selects whether P7-AN6 SCALE is in or out of the GMP AMP #1 circuit. 1-2: Connects P7 into amplifier circuit. 2-3: Disconnects P7 from amplifier circuit. Selection 2-3 is used on DFM-300 drives.

TABLE 1 – JUMPER AND POT ADJUSTMENTS (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Micro-application card (continued)		JP46		JP46 selects whether P8-AN7 SCALE is in or out of the GMP AMP #1 circuit: 1-2: *Connects P8 into amplifier circuit.* 2-3: Disconnects P8 from amplifier circuit. Selection 2-3 is used on DFM-300 drives.
		JP52		1-2: Enables 13PL-9 as CIO input.
		JP53		1-2: Enables 13PL-11 as CI1 input.
		JP54		Jumpers set max input voltage to optical isolators from ENCR (11PL-12) 1-2 10 volts 2-3 5 volts 3-4 1.5 volts OPEN 15 volts
		JP55		Jumpers set max input voltage to optical isolators from ENCB (11PL-14) 1-2 10 volts 2-3 5 volts 3-4 1.5 volts OPEN 15 volts
		JP56		Jumpers set max input voltage to optical isolators from ENCA (11PL-16) 1-2 10 volts 2-3 5 volts 3-4 1.5 volts OPEN 15 volts
		JP57		NOT USED - SET 1-2
		JP58		**NOTE:** *JP58, JP59, JP60 must all be set either 1-2 or 2-3.* 1-2 (Def) Standard LAN selected 2-3 Optically isolated LAN selected
		JP59		1-2 (Def) Standard LAN selected 2-3 Optically isolated LAN selected
		JP60		1-2 (Def) Standard LAN selected 2-3 Optically isolated LAN selected

TABLE 1 – JUMPER AND POT ADJUSTMENTS (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Micro-application card (continued)	Resistors*			Four scaling resistors are located on the Microapplication card amplifier terminal blocks as follows: **GENERATOR OUTPUT VOLTAGE** DTB2 - ETB2 DTB3 - ETB3 **MOTOR ARMATURE TERMINAL VOLTAGE** DTB5 - ETB5 DTB6 - ETB6 These resistors **MUST** be added to new replacement cards. Consult the Custom Instruction Book for values.
Digital Field Auxiliary (DFA) card **531X196DFAA__G version card**	Pot	P1	CF ZERO	Adjusts armature current feedback to AN1 (I_A) for zero when there is no armature current (MD contactor is open).
		P2	CF GAIN	Adjusts armature current feedback gain to AN1 (I_A) for full scale 2.0 volts when armature shunt output is 100 millivolts.
	Jumper	JP1 JP2		These jumpers select input scaling for high impedance isolator as follows: **ARMATURE VOLTAGE** — **JP1** — **JP2** 240VDC — 2-3 — 2-3 500VDC — 1-2 — 1-2
Digital Field Auxiliary (DFA) card **531X196DFAB__G version card**	Pot	P1	CF ZERO	Adjusts armature current feedback to ANI (I_A) for zero when there is no armature current (MD contactor is open).
		P2	CF GAIN	Adjusts armature current feedback gain to ANI (I_A) for full scale 2.0 volts when armature shunt output is 100 millivolts.
		P3	AMET	Provides meter calibration connection with a gain of 1 to 6.5 from DFTB4.

** Resistors used only if control equipped with 531X196DFAA__G version DFA card.*
NOT USED if control equipped with 531X196DFAB__G version DFA card.

TABLE 1 – JUMPER AND POT ADJUSTMENTS (continued)

<table>
<tr><th>LOCATION</th><th>JUMPER OR POT</th><th>NUMBER</th><th>NOMEN-CLATURE</th><th>DESCRIPTION</th></tr>
<tr><td>Digital Field Auxiliary (DFA) card 531X196DFAB__G version card (continued)</td><td>Jumper</td><td>JP1
JP2
JP3
JP4</td><td></td><td>These jumpers select input scaling for high impedance isolator (8V output) as follows:
<table>
<tr><th>ARMATURE VOLTAGE</th><th>JP1</th><th>JP2</th><th>JP3</th><th>JP4</th></tr>
<tr><td>702VDC</td><td>1-2</td><td>1-2</td><td>1-2</td><td>1-2</td></tr>
<tr><td>575VDC</td><td>2-5</td><td>2-5</td><td>2-5</td><td>2-5</td></tr>
<tr><td>498VDC</td><td>4-5</td><td>4-5</td><td>4-5</td><td>4-5</td></tr>
<tr><td>371VDC</td><td>1-4</td><td>1-4</td><td>1-4</td><td>1-4</td></tr>
<tr><td>251VDC</td><td>5-6</td><td>5-6</td><td>5-6</td><td>5-6</td></tr>
<tr><td>124VDC</td><td>2-3</td><td>2-3</td><td>2-3</td><td>2-3</td></tr>
</table></td></tr>
<tr><td>Programmer card (optional)</td><td>Jumper</td><td>JP1, JP3
JP4, JP5</td><td></td><td>When used with 18PL on Control card select as follows:
<table>
<tr><th>JP1</th><th>JP3</th><th>JP4</th><th>JP5</th></tr>
<tr><td>1-2</td><td>1-2</td><td>1-2</td><td>2-3</td></tr>
</table></td></tr>
<tr><td>Process Interface card (optional)</td><td>Switch</td><td>SW1</td><td>Tach Range Select</td><td>These switches provide coarse scaling of the tach circuit as follows:
DIP switch positions are selected for the legend of the side depressed. To select position 3 ON, depress the switch at position 3 next to the ON legend.
<table>
<tr><th>SWI POSITION NO. "CLOSED" OR "ON"</th><th>TACH VOLT AT MAX SPEED DC TACH</th><th>AC TACH</th></tr>
<tr><td>All "Open" or "Off"</td><td>10 - 12.5V</td><td>11.1 - 13.9V</td></tr>
<tr><td>1</td><td>12.5 - 15.5V</td><td>13.9 - 17.2V</td></tr>
<tr><td>2</td><td>15.5 - 19V</td><td>17.2 - 21.1V</td></tr>
<tr><td>1,2</td><td>19 - 22V</td><td>21.1 - 24.4V</td></tr>
<tr><td>3</td><td>22 - 25V</td><td>24.4 - 27.8V</td></tr>
<tr><td>1,3</td><td>25 - 30V</td><td>27.8 - 33.3V</td></tr>
<tr><td>1,2,3</td><td>30 - 37.5V</td><td>33.3 - 41.7V</td></tr>
<tr><td>4</td><td>37.5 - 46V</td><td>41.7 - 51.1V</td></tr>
<tr><td>1,2,4</td><td>46 - 56V</td><td>51.1 - 62.2V</td></tr>
<tr><td>1,2,3,4</td><td>56 - 70V</td><td>62.2 - 77.8V</td></tr>
<tr><td>5</td><td>70 - 87V</td><td>77.8 - 96.7V</td></tr>
<tr><td>2,3,5</td><td>87 - 105V</td><td>96.7 - 117V</td></tr>
<tr><td>1,2,4,5</td><td>105 - 120V</td><td>117 - 133V</td></tr>
<tr><td>1,2,3,4,5</td><td>120 - 144V</td><td>133 - 160V</td></tr>
<tr><td>6</td><td>144 - 180V</td><td>160 - 200V</td></tr>
<tr><td>1,2,4,6</td><td>180 - 205V</td><td>200 - 228V</td></tr>
<tr><td>5,6</td><td>205 - 250V</td><td>228 - 278V</td></tr>
<tr><td>1,2,3,4,5,6</td><td>250 - 310V</td><td>278 - 344V</td></tr>
</table>
NOTE: SW1 position 7 must be off (or open) for DC tachs, on (or closed) for AC tachs. Maximum voltages less than 10 volts are not to be used.</td></tr>
</table>

TABLE 1 – JUMPER AND POT ADJUSTMENTS (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Process Interface card (continued)	Pot	P1	TACH SCALE	Used to scale the tach circuit to obtain 4.0 volts at test point SFB at maximum speed.
		P2	SMET	Used to adjust the analog output signal at 3TB20 of the speed metering circuit for 10 volts at maximum motor speed. Maximum current load on this signal is 2mA.
		P3	DVM TRIM	With DVM test point connected to +5V and optional Programmer set to DVM operation, adjust P3 until the Programmer displays 5.00.
		P4 P5	PF GAIN PF OFF	These pots allow trimming the gain and offset of the circuit that provides up to 10 volts from 1 to 5, 4 to 20, 5 to 50, 5 to 65 mA transducer inputs. To adjust, start with PF gain at the full CCW position, and perform the following procedure: – Apply the current reference corresponding to minimum speed. – Adjust PF OFF for 0 volts at test point PFRF. – Adjust the current reference corresponding to maximum speed. – Adjust PF GAIN for 10.0 volts at PFRF. Repeat as necessary. **NOTE:** *To prevent reversing when the current reference is less than the minimum current, the drive should be set up in a non-reversing mode.*
		P6	STP SCALE	Set point scale, used to adjust scaling of definable input at test point STPRF. Refer to elementaries and engineering notes if used.
		P7	SFB ZERO	This pot adjusts test point SFB to be zero with zero tach feedback. Set speed reference to zero (jumper MSR or ASR to PCOMX if used) and adjust P7 for zero motor speed when a run is commanded.

TABLE 1 – JUMPER AND POT ADJUSTMENTS (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Process Interface card (continued)	Jumper	JP1		JP1 selects circuitry for the serial link communication mode. 1-2: Enables RS422 Reception. 2-3: Enables RS232C Reception.
		JP2 JP3		JP2 and JP3 select the process follower mode per the following table: **JP2 JP3** 1-2 1-2 5-65 mA current loop follower. 1-2 2-3 5-50 mA current loop follower. 2-3 1-2 4-20 mA current loop follower. 2-3 2-3 1-5 mA current loop follower or 10V voltage follower.
		JP4		JP4 selects the speed meter (SMET) circuit mode. 1-2: Unipolar speed meter driver mode. 2-3: Bipolar meter drive mode (DC tach only).
		JP5 (531X133PRU Series cards only)		JP5 selects digital tach quadrature mode. 1-2: Quadruples frequency of digital tach input (only when JP6 is in the doubler mode). 2-3: Disables quadrature mode.
		JP5* (531X134EPR Series cards only)		JP5 selects marker pulse mode. 1-2: Differential RS422 compatible, with 0.33K ohm termination. 2-3: Single-ended input at PCP.
		JP6 (531X133PRU Series cards only)		JP6 selects digital tach doubler mode. 1-2: Doubles frequency of digital tach input 3TB16. 2-3: Disables double mode.
		JP7 (531X133PRU Series cards only)		JP7 selects type of digital tach. 1-2: AN tach linerization mode or reluctance digital tach. 2-3: Square wave digital tach.

** See JP21 if JP5 is not on card.*

TABLE 1 – JUMPER AND POT ADJUSTMENTS (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Process Interface card (continued)	Jumper	JP7 (531X134EPR Series cards only)		JP7 enables marker pulse output. 1-2: Not used. 2-3: Enables marker pulse output to TUP.
		JP8		JP8 selects analog tach filter mode. 1-2: Enables 4MS filter in analog tach feedback; required for AC tach operation. 2-3: Disables 4MS filter; required for speed regulator performance above 20 radians (DC or Digital Tach only).
		JP9 JP10		JP9 and JP10 select the digital tach mode. 1-2: Differential RS422 compatible with terminating resistor. 2-3: Single-ended input at 3TB14 and 3TB16. Removed: Differential operation with no terminating resistor.
		JP11 JP12 JP13 JP14*		**JP11 JP12 JP13 JP14** 1-2 1-2 1-2 1-2 RS422 Serial Link – Standard 2-3 2-3 2-3 2-3 RS232C Serial Link – Standard (DTR low) 2-3 2-3 1-2 2-3 RS232C Serial Link (DTR hi) 1-2 2-3 2-3 2-3 RS422 Xmit/RS232C Handshake (Series Six™ Programmable controller compatible) **NOTE:** *For Series Six™ interfaces using the CCM card, set JP1 and JP11 to 1-2 and JP12-JP14 to 2-3. 12PL-7 should be tied to +5V.*
		JP15 (531X134EPR Series cards only)		**Encoder Counter Direction Control** 1-2: Normal control by encoder direction of rotation. 2-3: Up only.

** See JP14A and JP14B if JP14 is not on card.*

TABLE 1 – JUMPER AND POT ADJUSTMENTS (continued)

<table>
<tr><th>LOCATION</th><th>JUMPER OR POT</th><th>NUMBER</th><th>NOMEN-CLATURE</th><th>DESCRIPTION</th></tr>
<tr><td rowspan="6">Process Interface card (continued)</td><td rowspan="6">Jumper</td><td>JP16
(531X134EPR Series cards only)</td><td></td><td>Encoder A Quad B Enable
1-2: Normal A quad B encoder operation.
2-3: Count on leading edge of track A only.</td></tr>
<tr><td>JP17
(531X134EPR Series cards only)</td><td></td><td>LUP Marker Pulse Select
1-2: Marker pulse not connected to 9PL-14 (normal operation).
2-3: Marker pulse circuit output passed to 9PL-14 for use by LUP.</td></tr>
<tr><td>JP18
(531X134EPR Series cards only)</td><td></td><td>LUP DM5 Select
1-2: DM5 not connected to 9PL-13 (normal operation).
2-3: DM5 connected to 9PL-13 for use by LUP.</td></tr>
<tr><td>JP19
JP20
JP21
(531X134EPR Series cards only)</td><td></td><td>JP19, JP20 and JP21 functionally replace JP9, JP10 and JP5 respectively and should be set as follows:
<table>
<tr><th>MAX INPUT VOLTAGE</th><th>JP19 DTA-/DTA (3TB63, 62)</th><th>JP20 DTB-/DTB 3TB65,64)</th><th>JP21 DTM-/DTM (3TB60, 58)</th></tr>
<tr><td>1.5v</td><td>3-4</td><td>3-4</td><td>3-4</td></tr>
<tr><td>5V</td><td>2-3</td><td>2-3</td><td>2-3</td></tr>
<tr><td>10V</td><td>1-2</td><td>1-2</td><td>1-2</td></tr>
<tr><td>15V</td><td>1-4</td><td>1-4</td><td>1-4</td></tr>
</table></td></tr>
<tr><td>JP22
(531X134EPR Series cards only)</td><td></td><td>Set JP22 in the 1-2 position always.</td></tr>
<tr><td>JP14A
JP14B
(531X134EPR Series cards only)</td><td colspan="2">JP14A and 14B functionally replace JP14 and should be set as follows:
<table>
<tr><th></th><th>JP14A 1-2</th><th>JP14A 2-3</th><th>JP14B 1-2</th><th>JP14B 2-3</th><th>JP14B 2-4</th><th></th></tr>
<tr><td>JP14 (1-2)</td><td>Y</td><td></td><td>Y</td><td></td><td></td><td>(For Differential Inputs)</td></tr>
<tr><td>JP14 (2-3)</td><td>Y</td><td></td><td></td><td>Y</td><td></td><td>(For Single-Ended Input)</td></tr>
<tr><td>JP14 (2-3)</td><td></td><td>Y</td><td></td><td></td><td>Y</td><td>(/CTSB @ -15V; Disabled)</td></tr>
<tr><td>OR</td><td>Y</td><td></td><td></td><td>Y</td><td></td><td>(CTSB @ -15V; Disabled)</td></tr>
<tr><td>JP14 (2-3)</td><td></td><td>Y</td><td>Y</td><td></td><td></td><td>(/CTSB Open; Disabled)</td></tr>
<tr><td>OR</td><td></td><td>Y</td><td></td><td></td><td>Y</td><td>(/CTSB Open; Disabled)</td></tr>
</table></td></tr>
</table>

TABLE 2 – TEST POINTS

As a further aid to troubleshooting the DFM-300 drive controller, a series of test points have been provided to allow for specific signal observation and/or measurements. Test points not defined in the following listing are for manufacturer's use only and should not have instrumentation connected to them.

CONTROL CARD TEST POINTS

TEST POINT NAME	DESCRIPTION	NOMINAL VALUE
ACOM	Common for analog circuit power supplies.	
CFB	Generator field current feedback signal – positive when forward SCRs are conducting. Maximum external loading is 5 mA.	Calibration label voltage level at rated drive current (approx. 1.0 VDC).
DA0	Scaled output from 12 bit digital-to-analog converter (DAC).	
DCOM	Common for digital circuit power supply.	
DFP	Delayed firing power for the SCR gate circuits. Maximum external loading is 25 mA. Enabled only when generator field SCR firing is allowed.	+24V when enabled or OV when disabled.
MUP1 (DM1)	Multiuse point 1, connects to 6PL-9 and DFPL-3 (V_{MC}). **NOTE:** *NOT USED if control equipped with 531X196DFAB__G version DFA card.*	
MUP2 (DM2)	Multiuse point 2, connects to 6PL-10.	
MUP3 (DM3)	Multiuse point 3, connects to 6PL-11 and +24V.	+24V
FC	Motor field current feedback from Power Supply card. Maximum external loading is 1 mA. $FC = \frac{\text{MAX RATED MOTOR FIELD AMPS X 4VDC}}{\text{MAX FIELD CURRENT SET BY JP1 and JP2}}$ **NOTE:** *See Table 1, Power Supply card for maximum field current set by JP1 and JP2.*	
JOGR	Jog reference input at 6PL-7.	
-15V	-15 VDC power supply	-14.25 to -15.75 VDC
+15V	+15 VDC power supply	+14.25 to +15.75 VDC
+5V	+5 VDC power supply	+4.95 to +5.05 VDC
R/F	High when reverse SCRs being fired. Low when forward SCRs being fired.	High = 2 to 5 V Low = 0 to 2 V
SRS	Scaled Speed reference signal	4 volts at top speed reference.
VFBB or VFB	Generator field voltage feedback signal. Positive when forward motoring. Maximum external loading is 1 mA. Not for external use. **NOTE:** *VFBB is affected by P8-VMET.*	4 volts at rated generator field voltage.

TABLE 2 – TEST POINTS (continued)

POWER SUPPLY CARD TEST POINTS

TEST POINT NAME	DESCRIPTION	NOMINAL VALUE
TP1	Motor field current feedback signal. Maximum loading is 1 mA.	

OPTIONAL PROCESS INTERFACE CARD TEST POINTS

TEST POINT NAME	DESCRIPTION	NOMINAL VALUE
ACOM	Common of analog circuit power supplies.	
DCOM	Common of digital, circuit power supply.	
+5V	+5 VDC power supply.	+4.95 to +5.05 VDC
DVM	Input probe connection for digital voltmeter. Requires optional Programmer.	
SFB	Analog speed feedback signal. No value when digital tach or no tach is being used in drive system.	4 volts at top speed
PFRF	Output of process follower interface circuitry.	10 volts at top speed
DM5	Dummy point 5, connects to 10PL-16.	
TPN	Digital or AN tach pulse output.	

MICROAPPLICATION CARD TEST POINTS

TEST POINT NAME	DESCRIPTION	NOMINAL VALUE
ASR	Auto System Reference signal.	4 volts at top speed reference
DA8	Output from the 8 bit D-A converter.	±10 volts = maximum signal
DA12	Output from the 12 bit D-A converter.	±10 volts = maximum signal
FRA	Motor field reference adjust signal.	±10 volts = full adjust
RPU	Encoder interface marker track, one pulse per revolution.	

SOFTWARE ADJUSTMENTS

Table 3 on the following pages is a listing of "standard" DFM-300 software adjustments (parameters) common in most applications. Included is the "DFM-300 Parameter Map" (equivalent analog representation of parameters), the "Key Parameter Adjustments", and the "RAM Variable Scaling" table, which defines the scaling of RAM addresses available for interrogation of the drive control "circuit". The parameters can be examined and adjusted with the optional Programmer (refer to "Use of Optional Programmer Module") or via the serial link.

CUSTOM SOFTWARE DESCRIPTION

A Custom Software Description is shipped with every drive. The Custom Software Description describes the setting and/or function of various programmable jumpers and adjustments (hardware and software) within the drives that are identified with the specific model number of that drive. Included in that book is a listing of specific programmed options, their description and a set of software adjustment definitions pertinent to the specific drive.

PARAMETER MAP (FIGURE 6-9)

The Parameter Map is a diagnostic aid designed to reference specific software adjustments and correlate them to their specific regulating functions. For example, ADR015 - BSLPG is represented as a potentiometer feedback across the base speed regulating amplifier, giving an effective linear gain adjustment of input to output.

Software adjustments are shown next to their symbol with the mnemonic and the parameter address in a box beside it. Hardware adjustments are represented by a symbol, the adjustment name and the card it is located on (e.g.: FCMET is a potentiometer, P5 on the Control card). RAM addresses that can be used to examine ouputs of certain stages of regulation through TEST 11, TEST 21, TEST 22 or D-A converters, are shown as a three- or four-digit number in a circle, connected to the regulator section it is associated with (e.g.: Speed loop output can be examined by using RAM address 053 in Test 11, 21, 22 or the D-A converters).

The optional Programmer display scaling adjustments are shown in the lower left section of the map as well as their relation to feedback.

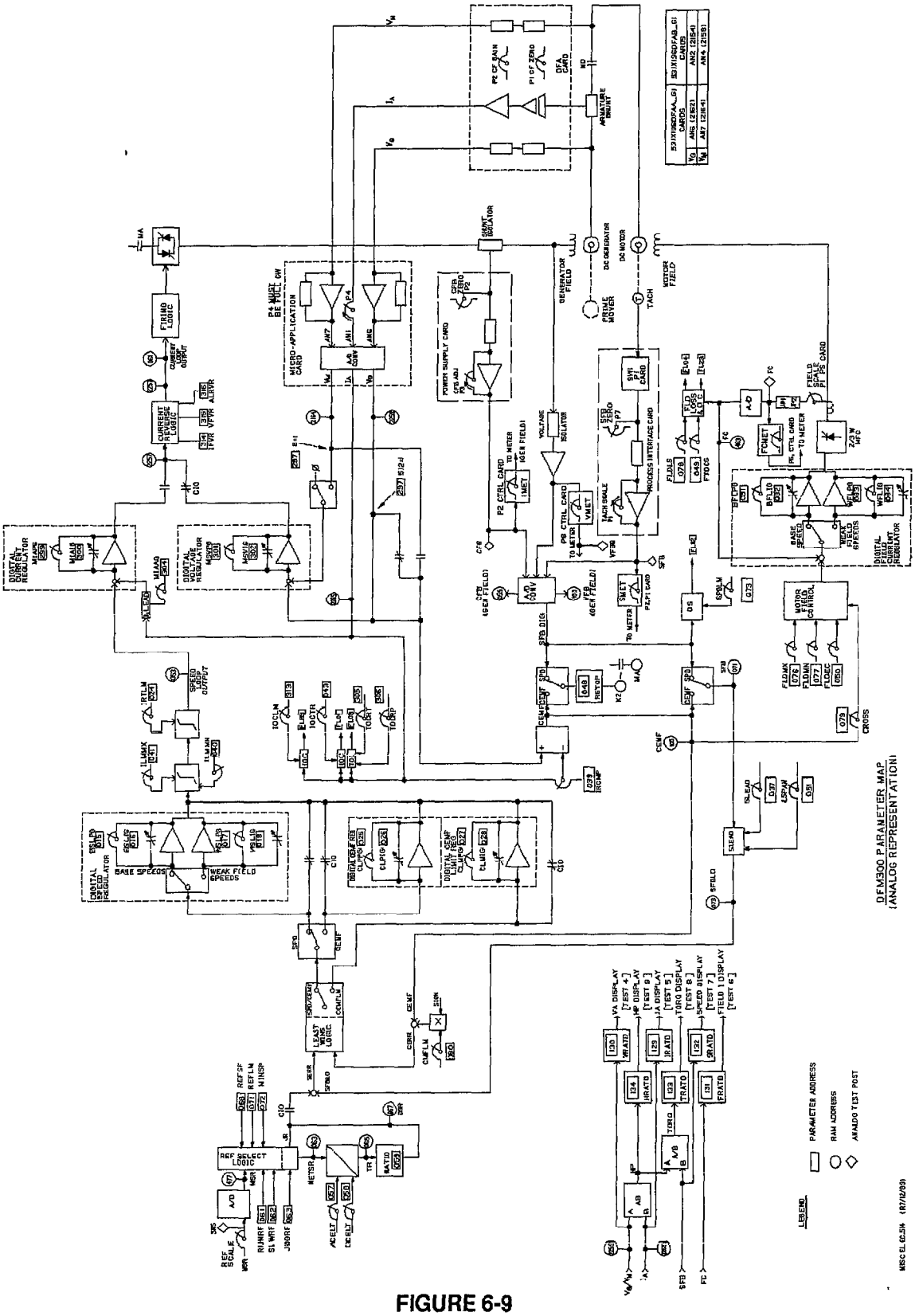


FIGURE 6-9
DFM-300 PARAMETER MAP
(ANALOG REPRESENTATION)

TABLE 3 – KEY PARAMETER ADJUSTMENTS

The following parameter adjustments listing Programmer mnemonic, address number, function name, and description, will prove helpful in setup and troubleshooting of the DFM-300 drive. This is a partial list and should be used in conjunction with the "Parameter Map". For a more complete list of parameters contained in a specific drive, refer to the Custom Software Description furnished with that drive.

The following is a description of column headers:

NAME Mnemonic displayed by Programmer when interrogating parameter.

ADR Parameter address.

UNITS Unit designation of displayed number (example - RPM).

MIN Minimum allowable value.

MAX Maximum allowable value.

STEP Amount by which value can be incremented or decremented.

DEFAULT Standard value if no special setting is required.

NOTE: *Parameters listed in Table 3 are grouped functionally per the parameter map (Figure 6-9).*

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT

MOTOR FIELD

FLDMX **076** **None** **0** **2047** **1** **800**

Full Field Current - sets the motor field current below base speed. Also is the voltage reference in the open loop mode of field operation. In the current regulated mode, 1638 corresponds to 4 volts at test point FC. The field current scaling jumpers on the Power Supply card should be set to give the largest feedback at FC that does not exceed roughly 4.5 volts. In the open loop mode, 0 corresponds to fully retarded firing and 2047 corresponds to fully advanced firing.

$$FLDMX = \frac{2048 \times I_F\ (max)}{5 \times \text{Gain Factor}}$$

The gain factor is determined by the field current scaling jumpers on the Power Supply card (refer to Table 1).

FLDMN **077** **None** **0** **2047** **1** **200**

Minimum Field Current - sets the minimum motor field current level the drive will regulate. 1639 equal 4 volts at FC.

$$FLDMN = \frac{0.6 \times 2048 \times I_F\ (min)}{5 \times \text{Gain Factor}}$$

NOTE: *This equation varies for material handling applications.*

TABLE 3 – KEY PARAMETER ADJUSTMENTS (continued)

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT

FLDLS 078 None 0 2047 1 100

Field Loss Trip Set Point - sets the level of motor field current below which the drive will trip. 1638 equals 4 volts at FC.

$$FLDLS = \frac{2048 \times I_F \text{ (min)}}{12.5 \times \text{Gain Factor}}$$

NOTE: *A setting of 60,000 disables the field loss trip function. This equation varies for material handling applications.*

CROSS 079 None 0 2047 1 1850

Field Crossover - sets the CEMF level at which the motor field will begin to weaken. A setting of 1638 corresponds to rated motor voltage. Determine % of Rated Armature volts desired at Crossover. Multiply that % by 1638 to obtain CROSS parameter setting.

FLDEC 050 None 0 255 1 0

Field Economy - sets level of motor field current when drive is stopped. Determine % of FLDMX desired for field economy. Then FLDEC will equal that percent x 255.

Example: 70% of maximum field current desired for field economy.
0.70 x 255 = 178.5
Setting of FLDEC = 179.

NOTE: *A setting of zero defaults FLDEC to 70% field current also.*

BFLPG 031 None 0 255 1 40

Base Field Loop Proportional Gain - sets the proportional gain of the motor field current regulator below base speed.

BFLIG 032 None 0 255 1 20

Base Field Loop Integral Gain - sets the integral gain of the motor field regulator below base speed.

WFLPG 033 None 0 255 1 40

Weak Field Loop Proportional Gain - sets the proportional gain of the motor field current regulator above base speed.

WFLIG 034 None 0 255 1 20

Weak Field Loop Integral Gain - sets the integral gain of the motor field current regulator above base speed.

FTOCG 049 None 0 255 1 82

Field TOC Gain - sets the time constant of the motor field timed overcurrent integrator. Set to 0 to inhibit this feature. For field forcing of FF per unit over a setting of FLDMX, the drive will trip in 6990/(FLDMX * FF * FTOCG) minutes. The default allows 0.1 P.U. forcing over a nominal FLDMX of 850 for 1 minute.

TABLE 3 – KEY PARAMETER ADJUSTMENTS (continued)

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT

LINEAR TIME

ACELT 057 **Decisec** 0 9999 1 20

Acceleration Time - sets the rate at which motor speed will increase in response to command speed changes. For linear time, enter time in tenths of a second for accel from 0 to top speed. ACELT = Time (in sec.) x 10. Some drives may have an optional extended linear time range. Refer to the Custom Software Description for information.

DCELT 058 **Decisec** 0 9999 1 20

Deceleration Time - sets the rate at which motor speed will decrease in response to command speed changes. For linear time, enter time in tenths of a second for decel from top speed to 0. DCELT = Time (in sec.) x 10. Some drives may have an optional extended linear time range. Refer to the Custom Software Description for information.

REFERENCES AND SCALING

RATIO 059 **% X 100** 0 19999 1 10000

Ratio - sets the ratio (draw) between the actual speed and the speed reference (NETSR). A ratio greater than 100% (RATIO set at 10,000) causes the drive to run faster than the speed reference; below 100%, slower.

REFSF 068 **None** 0 6553 1 6553

Reference Scale Factor - sets the factor by which references entered from the serial link and Programmer will be multiplied and by which speeds displayed on the Programmer and sent to the serial link will be divided. Enter the value of speed reference which corresponds to top speed (4 volts at SFB). ***Important*** - *affects settings of RUNRF, SLWRF, JOGRF, R1RF, R2RF, SLWRF, parallel input scaling, and speed reference display.* With Tach, REFSF = Top Speed RPM; for CEMF regulator, REFSF = Base Speed RPM at rated armature volts.

RUNRF 061 **RPM** 0 6553 1 0

"Run" Key Reference - sets the preset reference for RUN commands from the serial link. Also may be used as the preset reference for manual start commands. Also serves as the top speed reference for motor operated rheostat (MRH) mode. May be a speed, torque or position depending on jumper setting in ADR006-RUNJP. Normally RUNFR = Top Speed (RPM).

SLWRF 062 **RPM** 0 6553 1 0

"Slow" Key Reference - sets the preset reference for SLOW commands from the serial link. SLWRF = Slow Speed (RPM).

JOGRF 063 **RPM** 0 6553 1 **10% of REFSF**

"Jog" Key Reference - sets the preset reference for Jog commands from the serial link. Also may be used as the preset reference for manual jog commands. JOGRF = Jog Speed (RPM).

TABLE 3 – KEY PARAMETER ADJUSTMENTS (continued)

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT
RPMSF	070	RPM	0	6553	1	6553
		RPM Scale Factor - sets the speed in RPM which corresponds to 4 volts at SFB. **NOTE:** *Change parameter only while the drive is not running, and toggle the reset switch afterwards to make the change fully effective.* ***Important*** *- affects setting of REFLM, MINSP, SPDLM, tapered current limit speed setting and digital tach scaling. Same as setting for REFSF.*				
REFLM	071	RPM	0	6656	1	6560
		Maximum Speed - sets the level at which speed reference will be clamped if the input reference magnitude exceeds the level. Maximum value should not exceed 1,015 x RPMSF. Typically, REFLM = RUNRF + 100.				
MINSP	072	RPM	0	6553	1	0
		Minimum Speed - sets the level at which speed reference will be clamped if the input reference magnitude is less than the level. MINSP = desired Minimum Speed (RPM). Maximum value should not exceed RPMSF.				
SPDLM	073	RPM	0	7936	1	7208
		Speed Limit Trip Set Point - sets the speed at which the drive will trip. SPDLM = RPM at drive overspeed trip. Should not exceed 1.2 x RPMSF.				
SPEED REGULATOR						
BSLPG	015	None	0	255	1	7
		Base Speed Loop Proportional Gain - sets the proportional gain of the outer speed regulator below base speed when tachometer feedback is used.				
BSLIG	016	None	0	255	1	8
		Base Speed Loop Integral Gain - sets the integral gain of the outer speed regulator below base speed when tachometer feedback is used.				
WSLPG	017	None	0	255	1	7
		Weak Field Speed Loop Proportional Gain - sets the proportional gain of the outer speed regulator above base speed.				
WSLIG	018	None	0	255	1	8
		Weak Field Speed Loop Integral Gain - sets the integral gain of the outer speed regulator above base speed.				
SLEAD	037	None	0	255	1	10
		Speed Feedback Lead - sets the lead time constant on tachometer feedback.				

TABLE 3 – KEY PARAMETER ADJUSTMENTS (continued)

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT
LSPAN	051	None	0	15	1	0
RSTOP	048	None	0	255	1	100
SPDPM	318	None	0	6553	1	100

Speed Feedback Lead/Lag Span - sets the span of the tachometer lead/lag network. A setting of 0 is treated as a two-to-one span.

Regenerative Stop Speed - sets the speed at which the contactor opens during a regenerative stop. One unit = 0.04% speed (for Speed Reg.) One unit = 0.12% volts (for CEMF Reg.).

Maximum Speed Permissive - sets maximum speed at which MD contactor may close into rotating motor. Top rated motor speed (RPMSF) set by 6553.

CEMF REGULATOR

CMFLM	080	None	0	2047	1	1800
CLPPG	025	None	0	255	1	8
CLPIG	026	None	0	255	1	10
CLMPG	027	None	0	255	1	8
CLMIG	028	None	0	255	1	10
IRCMP	039	None	0	255	1	0

CEMF Limit - sets the CEMF level to which the drive will clamp. A setting of 1638 corresponds to rated motor voltage.

$$\text{CMFLM} = \frac{1638 \text{ X (\% motor Rated Voltage desired for CEMF limit)}}{100}$$

(Tachless) CEMF Loop Proportional Gain - sets the proportional gain of the outer CEMF regulator when CEMF is used as a feedback instead of a tachometer.

(Tachless) CEMF Loop Integral Gain - sets the gain of the outer CEMF regulator in no tach mode.

CEMF Limit Proportional Gain - sets the proportional gain of the spillover CEMF limit regulator.

CEMF Limit Integral Gain - sets the integral gain of the spillover CEMF limit regulator.

IR Compensation - sets positive or negative IR compensation for CEMF calculation.

TABLE 3 – KEY PARAMETER ADJUSTMENTS (continued)

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT

ARMATURE CURRENT REGULATOR

ILMMX **041** **% Rated** **10** **510** **2** **150**

Maximum Current Limit - sets the maximum DC motor armature current level allowed by the drive. One unit = 1% current.

ILMMN **040** **% Rated** **10** **510** **2** **10**

Minimum Current Limit - sets the minimum allowable motor current limit level; must be set **below** the desired **maximum** current limit setting. Also acts as the third current limit setting when tapered current limit is used. One unit = 1% current.

IOCTR **043** **% Rated** **10** **510** **2** **400**

Instantaneous Overcurrent Trip Setpoint - sets the generator field current level at which the drive will trip. One unit = 1% current.

TOCRT **305** **% Rated** **10** **510** **2** **120**

Timed Overcurrent Run Threshold - sets the armature current level above 1% speed, at which the TOC integrator begins to time up. One unit = 1% current. Refer to the Custom Software Description if multi-motor function is used.

TOCRP **306** **None** **0** **255** **1** **8**

Timed Overcurrent Run Gain - sets the time constant of the TOC integrator. Refer to the Custom Software Description if multi-motor function is used.

$$TOCRP = \frac{335.536}{(\text{Trip Time})\,(X\text{-}TOCRT)}$$

where X = % of armature current corresponding to desired trip time.

TOCST **307** **% Rated** **10** **65535** **1** **30**

Timed Overcurrent Sense Threshold - sets the current level below 1% speed, at which the TOC integrator begins to time up. One unit = 1% current. Refer to the Custom Software if multi-motor function is used.

TOCSP **308** **None** **0** **255** **1** **17**

Timed Overcurrent Response - sets the time constant of the TOC integrator. Refer to the Custom Software Description if multi-motor function is used.

$$TOCSP = \frac{335.536}{(\text{Trip Time})\,(X\text{-}TOCST)}$$

where X = % of armature current corresponding to desired trip time.

IOCLM **313** **% Rated** **0** **500** **1** **200**

Instantaneous Overcurrent Limit - sets armature current level where drive will trip. One unit = 1% current.

TABLE 3 – KEY PARAMETER ADJUSTMENTS (continued)

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT
MIAPG	299	None	0	65535	1	80
Current Loop Proportional Gain - sets the proportional gain of the inner armature current regulator.						
MIAIG	300	None	0	65535	1	80
Current Loop Integral Gain - sets the integral gain of the inner armature current regulator.						
MIAAG	304	None	0	65535	1	256
Current Loop Lead Gain - sets lead gain for armature current regulator.						
IRTLM	024	PU/S	1	255	1	50
Current Rate Limit - sets the maximum rate of change of current in response to commanded changes. Exact relation is current rate = 0.9 (IRTLM) per unit per second.						

ARMATURE VOLTAGE REGULATOR

MSVPG	301	None	0	65535	1	80
Voltage Loop Proportional Gain - sets the proportional gain of the inner voltage regulator which is used when the armature contactor (MD) is open.						
MSVIG	302	None	0	65535	1	80
Voltage Loop Integral Gain - sets the integral gain of the inner voltage regulator which is used when the armature contactor (MD) is open.						
VGLIM	310	None	0	2047	1	1800
Generator Voltage Maximum Limit Fault Trip - sets maximum generator output voltage as a ratio of 2047.						

GENERATOR FIELD

MIFPG	303	None	0	65535	1	65535
Generator Field Current Gain - this adjustment controls the gain of the generator field current regulator.						
KGMIN	311	None	0	3267	1	0
Generator Ratio - this adjustment sets the fault point of detecting prime mover underspeed or drop in speed. With field current greater than 20% and the generator voltage equal to 1638 at rated voltage:						

$$KGMIN = \frac{V_G}{I_F}$$ as a minimum acceptable ratio. in %

TABLE 3 – KEY PARAMETER ADJUSTMENTS (continued)

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT
IFLIM	312	None	0	65535	1	650

Generator Field Current Limit - this adjustment sets the maximum limit of generator field current output. Generally, this adjustment is to be left at the factory setting.

IFRVR	314	None	0	255	1	4

Generator Field Current Reversing - this adjustment sets the maximum generator field (percent current) at which reversal is allowed to occur. Do not change from factory setting unless instructed to do so by the factory.

VFRVR	315	None	0	65535	1	255

Generator Field Voltage Reversing - *this adjustment sets the maximum generator field* voltage at which reversal is allowed to occur. Do not change from factory setting unless instructed to do so by the factory.

ALRVR	316	None	0	127	1	30

Firing Angle Reverse Limit - this adjustment sets the maximum firing angle permissable at the time of reversal in order for reversal to occur. Do not change from factory setting unless instructed to do so by the factory.

ALPLM	317	None	0	127	1	0

Firing Angle Advance Limit - *this adjustment sets the maximum firing angle allowed.* A setting of zero represents no limit on the firing angle whereas a setting of 127 will limit the firing angle to retard limit. Do not change from factory setting unless instructed to do so by the factory.

TOCTH	044	% Rated	10	510	2	120

Timed Overcurrent Sense Threshold - *sets the generator field current level at which* the TOC integrator begins to time up. One unit = 1% current. Refer to the Custom Software Description if multi-motor function is used.

TOCGN	045	None	0	255	1	24

Timed Overcurrent Response - sets the time constant of the TOC integrator for the generator field. Refer to the Custom Software Description if multi-motor function is used.

$$\text{TOCGN} = \frac{66.666}{(\text{Trip Time})\,(\text{X-TOCTH})}$$

where X = % of armature current corresponding to desired trip time.

NOTE: *The default values for TOCTH and TOCGN allow the drive to operate at 150% current for 90 seconds.*

TABLE 3 – KEY PARAMETER ADJUSTMENTS (continued)

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT

OUTPUT SIGNALS

DAC0 **090** **None** **0** **65535** **1** **0**

D-A #0 Output Control - controls what internal variable is sent to the 12 bit D-A converter (DAC0) on the Control card. Enter the desired RAM variable number into ADR090 to program DAC0. Refer to "RAM Variable Scaling", Table 4, at the end of this listing for relative value and RAM addresses.

DAC8 **096** **None** **0** **65535** **1** **0**

D-A #1 Output Control - controls what internal variable is sent to the 8 bit D-A converter (DAC8) on the Microapplication card. See DAC0 for programming information.

DAC12 **097** **None** **0** **65535** **1** **0**

D-A #2 Output Control - controls what internal variable is sent to the 12 bit D-A converter (DAC12) on the Microapplication card. See DAC0 for programming information.

BIT #1 **098** **None** **0** **65535** **1** **0**

Bit #1 Output Control - controls what internal bit is sent to bit output #1 (B01) on the Microapplication card. Possible outputs include:

RAM ADDR	VARIABLE NAME	RAM ADDR	VARIABLE NAME
061	Direction of Firing	148	Reference Enable
137	Fault	152	Precondition
140	Current Limit Flag	156	Zero Current Flag
141	CEMF Limit Flag		

In addition, the sign of all of the variables given in DAC8 output may be used as outputs.

BIT #2 **099** **None** **0** **65535** **1** **0**

Bit #2 Output Control - controls what internal bit is sent to bit output #2 (B02) on the Microapplication card. See Bit #1 for programming information.

BIT #3 **100** **None** **0** **65535** **1** **0**

Bit #3 Output Control - controls what internal bit is sent to bit output #3 (B03) on the Microapplication card. See Bit #1 for programming information.

TABLE 3 – KEY PARAMETER ADJUSTMENTS (continued)

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT

DIAGNOSTICS

DIAG1	122	None	0	65535	1	0

Diagnostic Variable #1 - selects 1 of 2 variables which can be displayed on the serial link or Programmer using "display diagnostics". See DAC0 output control for programming information.

DIAG2	123	None	0	65535	1	0

Diagnostic Variable #2 - selects second variable which can be displayed on the serial link or Programmer using "display diagnostics". See DAC0 output control for programming information.

SCHRL	127	None	1	26	1	18

Left Scaled Unit Character - determines which character is displayed as the left character of the scaled speed status message. 1 represents "A"; 2, "B; 26, "Z".

SCHRR	128	None	1	26	1	13

Right Scaled Unit Character - see ADR127-SCHRL for programming information.

IRATD	129	None	0	3276	1	100

Displayed Arm Current Scaling - controls the factor by which per unit armature current is multiplied for display on the Programmer.

VRATD	130	None	0	2621	1	100

Displayed Arm Volts Scaling - controls the factor by which per unit armature voltage is multiplied for display on the Programmer.

FRATD	131	None	0	43690	1	1000

Displayed Field Current Scaling - controls the factor by which per unit motor field current is multiplied for display on the Programmer.

SRATD	132	None	0	6200	1	100

Displayed Speed Scaling - controls the factor by which per unit speed is multiplied for display on the Programmer.

TRATD	133	None	0	3276	1	100

Displayed Torque Scaling - controls the factor by which per unit torque is multiplied for display on the Programmer.

HRATD	134	None	0	3276	1	100

Displayed HP Scaling - controls the factor by which per unit horsepower is multiplied for display on the Programmer.

TABLE 4
RAM VARIABLE SCALING FOR PROGRAMMER AND DAC OUTPUTS

RAM ADDR	VARIABLE NAME	VALUE AT 100% SIGNAL DECIMAL	HEX(1)	VOLTS ON DAC(2) AT 100% SIGNAL
053	SLOPO-Speed Loop Output	100	0064	2.0
055 2275	CFB - Current Feedback (Generator Field)	100	0064	2.0
061	ILOPO - Current Loop Output	220	00DC	4.3
063	NETSR - Net Speed Reference	6553	1999	8.0
065	TR - Timed Reference	6553	1999	8.0
067	DSR - Draw Speed Reference	6553	1999	8.0
069	JR - Jog Reference	6553	1999	8.0
071	SFB - Speed Feedback	6553	1999	8.0
073	SLEAD - Speed Feedback Lead	6553	1999	8.0
103 2277	VFB - Voltage Feedback (Generator Field)	1638	0666	8.0
105 2255	CEMF - Counter Electromotive Force	1638	0666	8.0
463	FC - Motor Field Current (rated current - not scaled)	2047	07FF	10.0
477	MSR - Manual Speed Reference	6553	1999	8.0
486	ASR - Automatic Speed Reference	6553	1999	8.0
501	TOCT - Timed Overcurrent Timer (trips at 1024)	1024	0400	0.3
2265 2152	I_A - Armature Circuit Current Feedback	100	0064	2.0
2164	V_M - Motor Terminal Voltage (531X196DFAA__G version cards)	1638	0666	8.0
2158	V_M- Motor Terminal Voltage (531X196DFAB__G version cards)	1638	0666	8.0
2259	V_M or V_G - (Depends on value in EE297) (V_G unless 512 bit set in EE297)	1638	0666	8.0
2162	V_G - Generator Voltage (531X196DFAA__G version cards)	1638	0006	8.0
2154	V_G - Generator Voltage (531X196DFAB__G version cards)	1638	0006	8.0
2779	MD Permissive OFF ON	0 128	0000 0080	

NOTE: *(1) Negative numbers are represented in Two's complement.*

(2) D/A converters DAC8 and DAC12 have an op-amp with a gain of two, this factor is included in the listed values.

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7 – STARTUP PROCEDURE

This startup procedure is meant as a general guide for the initial checkout and startup of a DFM-300 drive. It references some general hardware and software checks relative to most basic drive applications. Refer to the sections on "Hardware Jumpers and Pot Adjustments" and "Key Parameter Adjustments" and the accompanying "Parameter Map" (Figure 6-9) for more complete information on adjustments available on the DFM-300 drive. Refer to the Custom Software Description Elementaries for information on wiring and adjustments specific to your drive application.

NOTE: *Prior to beginning actual drive startup, read this entire manual and be aware of all pertinent information. If an optional Programmer was supplied with the drive, review the "Use of Optional Programmer Module" and "Diagnostic Mode for Drive Setup" sections. Special diagnostic functions are available which will greatly aid drive startup.*

BEFORE APPLYING POWER

(Review warning and caution notices at the beginning of this instruction book before proceeding.)

Every DFM-300 DC SCR drive controller has been factory tested and is ready to operate provided the power and control connections have been properly made and the following steps are performed:

1. Verify that all incoming wiring is in agreement with the elementary drawings supplied with the drive, and conforms to approved wiring practices (wire size, separation of power from control, etc.).
2. Verify that incoming 3 phase voltage is in accordance with nameplate requirements (+10%, -5%), unless otherwise specified.
3. Confirm that the following jumpers (JP) and switch positions (SW) are in accordance with the jumper selection chart supplied with the drive in the drive door pocket and Custom Software Description. (Refer to the card outlines, Figures 6-1 thru 6-8, for the location of jumpers). Particular note should be made of the following:

Control Card:

JP1 - Reference Voltage Range For Maximum Reference of	Position of JP1
8 - 15V	1-2
5 - 8V	2-3

JP15 - Loadmeter Select For Percent Load Output	Position of JP15
Absoluted (Zero Left Scale)	2-3
Bipolar (Zero Center Scale)	1-2

JP26	Pin 2 is jumpered to DM4.

Power Supply Card:

JP1 & JP2 – Field Current Range
(version G1 Power Supply card, Figure 4-14).

For Motor Max Field Amps	Position of JP1	JP2	Gain* Factor (GF)
0.64 - 1.6A	2-3	2-3	0.4
1.6 - 4.0A	1-2	2-3	1.0
4.0 -10.0A	1-2	1-2	2.5

JP1 & JP2 – Field Current Range
(version G2 Power Supply card, Figure 4-15).

For Motor Field Max Amps		Position of		Gain Factor (GF)*	
24A	72A	JP1	JP2	24A	72A
6.0A	18.0A	2-3	2-3	1.5	4.5
12.A	36.0A	1-2	2-3	3.0	9.0
24.0A	72.0A	1-2	1-2	6.0	18.0

** Gain Factor is required in "Apply AC Power" section of startup procedure, Item 7.*

JP3 and JP4 - Generator Field Feedback Signal

For Operating Mode	Position of JP3	JP4
Normal	1-2	1-2

Process Interface Card (if used)

SW1, Tach Scale (Refer to SW1 setting chart in "Hardware Jumpers and Pot Adjustments" section, Table 1.

Microapplication Card

Armature Feedback Amplifier Jumpers*			
Jumper	Position	Jumper	Position
JP24	2-3	JP29	2-5
JP25	1-4	JP30	2-4
JP26	2-3	JP31**	2-4
JP27	1-4	JP45	2-3
JP28	2-4	JP46	2-3

** Jumpers 24 thru 30 are NOT USED if control is equipped with 531X196DFAB__G version DFA card.*

*** Jumper 31 must be set in 2-5 position if control is equipped with 531X196DFAB__G version DFA card.*

Armature Voltage Feedback Resistors* Verify Resistor Values Connected	
From Terminal	To Terminal
DTB - 2	ETB - 2
DTB - 3	ETB - 3
DTB - 5	ETB - 5
DTB - 6	ETB - 6

** These resistors are not present if control is equipped with 531X196DFAB__G version DFA card.*

Armature Current Feedback Pot	
P4	FULL CLOCKWISE

Additional Jumper Settings:		
Jumper	Function	Normal Position
JP52	CI0 to 13PL-9	1-2
JP53	CI1 to 13PL-11	1-2

DFA (Digital Field Auxiliary) Card

Armature Voltage Isolation Jumpers JP1 and JP2 (531X196DFAA__G version)		
Armature Voltage	Jumper JP1	Position JP2
500 V	1 - 2	1 - 2
240 V	2 - 3	2 - 3

Armature Voltage Isolation Jumpers JP1, JP2, JP3 and JP4 (531X196DFAB__G version)				
Armature	Jumper Position			
Voltage	JP1	JP2	JP3	JP4
702 VDC	1-2	1-2	1-2	1-2
575 VDC	2-5	2-5	2-5	2-5
498 VDC	4-5	4-5	4-5	4-5
371 VDC	1-4	1-4	1-4	1-4
251 VDC	5-6	5-6	5-6	5-6
124 VDC	2-3	2-3	2-3	2-3

4. Verify that all connections are tight.
5. Verify that the DC Motor, Generator and Prime Mover have been properly installed and that their shafts are free to rotate.

WARNING

Burn and Shock Hazard: Repeated application and removal of power is required during the following start-up and tuning procedure. Exercise extreme caution to assure that power is removed whenever making any connections or otherwise contacting potentially live circuits.

CAUTION

When changing drive parameter values it will be necessary to enable programming of the EEROM by placing JP13 on the Control card in the 2-3 position. Do not perform a "Hard Reset" or turn the power to the drive on or off while the EEROM is enabled. Replace JP13 in the programming disable. EEROM protected. 1-2 position to prevent accidental reprogramming.

BASIC PROCEDURE

The suggested procedure consists of the following steps:

1. Check all system connections and verify correctness per the system elementary diagrams.
2. Load and verify EEROM Parameters
3. Check Generator Voltage Feedback Calibration
4. Check Generator Armature Current Feedback Calibration
5. Check Generator Field Connection Polarity
6. Verify Generator Open Loop Voltage Regulator
7. Motor Field Regulator Operation and Tune-Up
   a. Tune motor field regulator
   b. Set motor field protection and crossover features
8. Voltage Regulator Operation and Tune-Up
   a. Verify generator output voltage polarity
   b. Tune voltage regulator
   c. Motor shaft rotation check
   d. Tachometer polarity verification (if required)
   e. Verify armature current feedback polarity
9. Current Regulator Operation and Tune-Up
   a. Tune current regulator
   b. Set-up motor/generator armature protection features
10. Speed Regulator Operation and Tune-Up
    a. Tune speed regulator
    b. Set motor speed protection features
11. System Operational Checks
    a. Emergency Stop interlocking
    b. System sequencing check-out
    c. Reference system verification
    d. Instrumentation calibration check-out
    e. Gear-in speed calibration

REQUIRED TEST EQUIPMENT

The following test equipment is required to set up the DFM system.

1. Variable Power Supply adjustable from 2 to approximately 50 VDC with 0.5 amps capability
2. Voltmeter capable of reading 1 to 750 VDC at least 10 megohm input impedance.
3. Programmer (hand-held or local)

CHECK CONNECTIONS

Check all system connections and verify correctness per the system elementary diagrams.

1. Determine correctness of motor and generator nameplate data. Note changes for future software revision.
2. Check all motor/generator armature and field connections. Pay attention to the correct wiring of series or parallel field connections per the elementary diagrams.
3. Check mechanical operation of the DC loop contactor and the auxiliary contact connection to the DFM input.
4. Determine generator prime mover rotation as per the elementary diagrams.

CAUTION

Do not operate the DFM with the generator field energized without the prime mover rotating or damage to this field due to overcurrent and overvoltage can occur. External interlocking should prevent this from happening if the wiring is correct as per the elementary diagram.

5. Check connection and polarity of voltage and current feedback to the DFA isolation card as per the elementary diagrams.

APPLY AC POWER

1. Check for proper direction of rotation of all three phase blower motors on both the power unit(s) and the DC motor(s).
2. Verify that the Diagnostic LEDs are blinking one at a time from right to left. If the optional Programmer is supplied, confirm that it displays the word OPERATE.

   NOTE: *If at any time the LEDs remain locked, indicating a fault message, the Programmer displays a fault code, or either display is blank, refer to the "Troubleshooting" section for corrective procedures.*
3. Check the motor field for proper voltage. Since the drive may be in field economy until a "START" is called for, check the setting in ADR050-FLDEC in the Custom Software Description to determine the proper field voltage.
4. Place the EEROM write device on the Control card in the "ENABLE" position to allow programming.

   NOTE: *The EEROM write protect device is JP13 on the Control card. To enable programming move JP13 to the 2-3 position.*

CAUTION

Do not perform a "Hard Reset" or turn power "Off" or "On" with the EEROM write device in the "Enable" position.

5. Access the Parameter mode by pressing [SET] [DRV#] [7] [7] and [ENTER] on the keypad (where the number or function is encased in brackets [ ] indicates the Programmer key to be pressed).

   The word PARAMETER should be displayed. The **BLACK** keypad functions are now usable. To access a parameter address, press all three numbers on the keypad corresponding to the three-digit number of the desired address. The name of the address and its present value will be displayed.

   To access a new parameter, press [CLEAR/MODE] and the three digits corresponding to the next parameter desired.
6. In order to allow programming of new parameter values, first verify that the parameter address #003, DGNJP (ADR003-DGNJP) contains an odd number (1, 3, 9, etc.) by pressing [0] [0] [3]. If this address contains an **even** number, add a "1" to this number and enter the new value.

   Example: ADR003 contains the number 8. Add 1 to obtain 9. Press [9] and [ENTER].

   Press [CLEAR/MODE] and the three digits for the next desired parameter.

   NOTE: *Confirm that the drive parameters agree with the "as shipped" parameter list inside the drive door or in the Custom Software Description prior to making any adjustments.* ***Document any changes.***
7. Verify maximum motor field current (ADR076-FLDMX) to the motor nameplate value per the following formula (must be in Parameter mode):

   $$\text{MAX FLD SETTING} = \frac{\text{MAX N.P. FLD. CURRENT} \times 2048}{\text{GAIN FACTOR} \times 5}$$

   NOTE: *Obtain the gain factor (GF) from* ***"Power Supply Card"****, under* ***"BEFORE APPLYING POWER"****.*

   Example:
   Maximum Nameplate Field Current = 4.5A, and with G1 Power Supply Card, GF = 2.5

   a. SETTING = $\frac{4.5 \times 2048 = 737}{2.5 \times 5}$

   b. Call up ADR076 - FLDMX by pressing:
      [0] [7] [6]

   c. Enter setting by pressing:
      [7] [3] [7] [ENTER]

   d. Press [CLEAR/MODE]

8. Similarly verify minimum field by calling up ADR077-FLDMN. (Minimum field is defined at 60% of the lowest running field current.)

   NOTE: *Field current is at the field economy setting ADR050-FLDEC until a "START" is called for (START, RUN, JOG, etc.).*

   MIN Field Setting:

   $$\text{MIN FLD SETTING} = \frac{\text{MIN N.P. FLD. CURRENT} \times 0.6 \times 2048}{\text{GAIN FACTOR} \times 5}$$

   Example: Minimum Nameplate Field Current = 2.0 amps; other parameters same as above:

   a. $\text{SETTING} = \frac{2.0 \times 0.6 \times 2048 = 196.6}{2.5 \times 5}$

   b. Press [1] [9] [7] [ENTER]

   c. Press [CLEAR/MODE]

9. Verify acceleration and deceleration time by use of ADR057-ACELT and ADR058-DCELT. Setting can be any number between 0 and 9999 (999.9 sec). Some drives may have an optional extended linear time or optional "S-curve" capability. Refer to the Custom Software Description for information.

   Example: 22.5 SECS, SETTING = 225

10. Call up ADR070-RPMSF and verify the desired motor top speed in RPM (ADR068-REFSF should contain the same value).

    Example: Motor top speed is to be 1500 RPM. Enter 1500 in both ADR068 and ADR070.

11. Verify Minimum Speed and Jog Speed (if available) through ADR072-MINSP and ADR063-JOGRF respectively. For example, Jog or Minimum Speed of 10% of top speed.

    SETTING = 0.10 x ADR070 SETTING

    NOTE: *JOG reference may be supplied by an analog input at 3TB64, JOGR (6PL-7).*

12. Armature Voltage Settings

    The values for ADR079-CROSS or ADR080-CMFLM can be obtained from the following:

    A setting of 1638 in ADR079-CROSS or ADR080-CMFLM is equal to rated armature volts.

    $$\text{DESIRED PARAMETER VALUE} = \frac{\text{Required Volts} \times 1638}{\text{Rated Volts}}$$

    Typically, ADR080-CMFLM is set approximately 108% for non-regenerative, or 105% for regenerative drives.

    Example: Rated Armature Volts = 500 VDC
    Desired CEMF Limit = 520 VDC

    $$\text{ADRO-CMFLM} = \frac{520 \times 1638}{500}$$

    ADR080-CMFLM = 1704

13. Current Feedback (Generator Field)

    a. The current feedback circuit is factory adjusted to give 1 VDC at CFB for the rated current of the shunt in the generator field circuit.

    b. All generator field current parameter adjustments can be set in percent (100 = 100% shunt rated current).

    c. Check P2-CFB ZERO on the Power Supply card and adjust so that test point "CFB" is as close to zero volts as possible with no current flowing in the generator field circuit (see Table 1 for CFB adjustments).

14. Exit the PARAMETER mode by pressing [CLEAR/MODE] [CLEAR/MODE] [CLEAR/MODE]. "OPERATE" should appear. Place JP13 on Control card in the "PROTECT" 1-2 position.

15. Verify voltage feedback calibration with a simulated feedback signal, as follows:

CAUTION

The prime mover should be stopped during this procedure and the drive disabled from operating in any mode by disconnecting 3TB-CNTL1.

   a. With POWER OFF remove the voltage feedback leads V_G and V_C from the motor/generator armature circuit and connect a variable voltage power supply to act as feedback to the drive.

   b. APPLY POWER to the drive, but DO NOT ALLOW THE DRIVE TO RUN. Using the programmer in the PARAMETER MODE, as before, load ADR298-DFMJ1 with 4096 and ADR014-SPCJP with 132. Place JP13 on the Control card in the PROTECT position and perform a HARD RESET by pushing SW2.

c. Select DIAGNOSTIC MODE and using TEST 11 monitor RAM location 2259 to verify voltage feedback calibration. A value of 664(H) in RAM 2259 represents 1.0 PU (per unit) feedback. Negative feedback is represented as a Two's Complement such that 1PU is F99A(H). With 100 VDC approximately the following values should be obtained.

RATED MOTOR VOLTAGE	+100 VDC FEEDBACK	-100 VDC FEEDBACK
500 VDC	147(H)	FEB9(H)
250 VDC	28F(H)	FD71(H)

If the actual values obtained vary significantly from those given, recheck the resistors between DTB and ETB terminal boards on the Microapplication card, jumpers on the Microapplication card and jumpers on the DFA card.

d. With POWER OFF, reconnect the voltage feedback leads to the armature circuit as before.

e. REPEAT the above test for motor voltage feedback using V_M and Vc leads and RAM 2164.

f. RETURN ADR014-SPCJP and ADR298-DFMJ1 to the original values shown on the Custom Software description. With POWER OFF reconnect 3TB-CTLN1 as before.

16. Verify current feedback calibration with a simulated feedback signal, as follows:

CAUTION

The prime mover should be stopped during this procedure and the drive disabled from operating in any mode by disconnecting 3TB-CNTL1.

a. With POWER OFF remove the current feedback leads 1P and 1N from the motor/generator armature circuit shunt and connect them together.

b. APPLY POWER to the drive, but DO NOT ALLOW THE DRIVE TO RUN. Using the programmer in the PARAMETER MODE, as before, load ADR298-DFMJ1 with 4096 and ADR014-SPCJP with 132. Place JP13 on the Control card in the PROTECT position and perform a HARD RESET by using SW2.

c. Select DIAGNOSTIC MODE and using TEST 11 monitor RAM location 2265 and adjust P1-CF ZERO on the DFA card to obtain a value of approximately 0(H) in RAM 2265, typically between FFFE(H) and 0002(H).

d. With POWER OFF disconnect the current feedback leads 1P and 1N and connect a variable millivolt power supply to them as feedback to the drive.

e. APPLY POWER to the drive, but DO NOT ALLOW THE DRIVE TO RUN. Monitor RAM location 2265 and apply 500 millivolts (5PU) to 1P and 1N in the motoring positive polarity. Adjust P2-CF GAIN on the DFA card to obtain a value of approximately 1FF(H) in RAM 2265 (note that FE00(H) would indicate reversed polarity).

f. REPEAT the above test for reversed polarity feedback to obtain approximately FE00(H) in RAM 2265 with 500 millivolts on 1P to 1N.

g. RECHECK for both zero input and 100 millivolt (1PU) input. The value in RAM 2265 should be approximately 66(H) for +100 millivolt and FF9A(H) for -100 millivolt inputs. REPEAT the preceding calibration as required.

h. With POWER OFF, reconnect the current feedback leads to the armature circuit shunt as before.

RETURN ADR014SPCJP and ADR298-DFMJ1 to the original values shown on the Custom Software description. With POWER OFF reconnect 3TB-CNTL1 as before.

17. Verify the polarity of the generator field by applying a test voltage and noting the output polarity as follows.

CAUTION

Power must be disconnected from the drive and the armature circuit (MD) contactor open during this procedure.

a. With POWER OFF disconnect the generator field leads from the output of the DFM-300 drive controller and connect an appropriate, temporary DC supply (i.e. 12 VDC @ 3ADC) through a switch to feed the generator field.

b. Start the prime mover and close the switch to excite the generator field with the 12 VDC supply. Using a DC Voltmeter check both the field and armature voltages of the generator and note the polarities at the generator's terminals.

c. Open the generator field supply switch and STOP the prime mover. AFTER the generator has come to a complete STOP and with POWER OFF disconnect the temporary field supply from the generator and mark the generator field terminals (leads) with the polarity associated with positive generator output voltage.

d. With POWER OFF, reconnect the DFM-300 drive controller to the generator field circuit with the positive field lead going to terminal DA1 and the negative field lead to DA2.

18. Verify the actual EEROM contents versus the values supplied in the Custom Software Description provided with the drive. Make sure that any changes which have been made are correct and reasonable. NOTE all changes on the listing for future reference.

19. Assure that the connections and polarities are correct for the armature current (1P and 1N) and voltage feedback (V_G and Vc) leads.

20. Ensure that all STOP circuits are functional and open the MA contactor in the drive input.

CAUTION

"STOP" from an operator station or other 115 VAC start/stop circuit wired to 3TB49, 50 and 51 will not function if the DFM-300 drive is started from the programmer keypad. Likewise, the "RUN" input at 3TB8 will not function.

21. Verify armature feedback polarities by operating the DFM-300 controller into the generator with the armature circuit disconnected from the drive motor, as follows:

CAUTION

The main DC armature circuit contactor (MD) should be prevented from closing at this time to prevent inadvertant motor operation.

a. With POWER OFF, check that the auxiliary contact from the MD contactor is connected to 4TB-9 (CI0) and that the contact is open. No other circuit should be wired to 4TB-9 (CI0). Disable the MD contactor from closing by opening and isolating the contactor coil circuit.

b. Connect a voltmeter to the generator armature, noting polarity and using the proper VDC scale for full armature voltage.

c. APPLY POWER to the DFM-300 drive controller but DO NOT ALLOW THE DRIVE TO "RUN". Using Diagnostic TEST 11 monitor RAM address 2162 (if equipped with 531X196DFAA__G version DFA card) or 2154 (if equipped with 531X196DFAB__G version DFA card), V_G, which should read approximately 0(H) at this time.

d. START the prime mover and note the magnitude and polarity of any "residual" armature voltage. Compare the reading on the voltmeter with the value displayed for RAM, remembering that the RAM value is shown as a HEXidecimal number with negative values in Two's Complement (i.e. F# # #).

CAUTION

The next steps will apply power to the generator field using the DFM-300 controller. Test all means of safely removing power in case emergency shutdown is required.

e. Set the speed reference input for minimum output per the Custom Instructions and elementaries provided with the drive. Place the DFM-300 drive controller in the RUN mode and observe the voltmeter monitoring the generator armature circuit. Any residual voltage should drop to within +/-0.01PU voltage.

CAUTION

If voltage is not within this range, immediately stop the DFM-300 controller. Stop the prime mover and disconnect power. Recheck feedback polarities and EEROM parameters. Correct as indicated before repeating this procedure.

f. The DFM-300 drive should now be regulating the generator field, and in turn the armature voltage to zero. The SCR firing angle should be low, providing less than 10 VDC to the generator field in most cases.

g. STOP the DFM-300, stop the prime mover and disconnect power.

h. Check polarities for all armature and motor feedback and instrumentation circuits relative to the polarity established in the preceding tests.

i. Reconnect the MD contactor coil circuit as before this test.

22. Check the direction of motor shaft rotation and verify the polarity of tachometer (speed) feedback as follows:

    a. With POWER OFF connect the motor armature and all associated circuitry. Unload the motor shaft and be sure it is free to turn.

    b. Start the system with the following steps in the ORDER GIVEN:

    - START the prime mover
    - APPLY POWER to the DFM-300 drive input
    - INITIATE the RUN command sequence to the drive

CAUTION

If at any point the drive "Runs Away" and motor speed increases uncontrollably, stop the drive immediately and recheck connections and polarities.

SLOWLY increase the speed reference input until the motor begins to rotate. If the motor fails to rotate or rotates in the wrong direction check:

- Drive Diagnostic Display Fault Codes (LEDs or Programmer)
- MD Contactor circuit
- Generator connections/polarities
- Motor connections/polarities
- Tachometer connections/polarities

NOTE: *With proper rotation the positive tach lead will connect to the terminal marked "Tach (-)" on the process interface card and the negative tach lead will connect to the "Tach (+)" Terminal.*

Check all meters and feedback signals for proper indication and polarity.

Stop the drive system as follows:

- STOP the DFM-300 controller
- WAIT for the MA contactor to open
- STOP the prime mover
- DISCONNECT the drive input power

NOTE: *If optional Programmer and Process Interface cards are included in the drive, an on-board digital voltmeter (DVM) and electronic wire check are available to verify wiring and make the following adjustments. Refer to the "Use of Optional Programmer Module" section for instructions.*

23. With the drive **stopped,** increase the speed reference to 100% speed. Adjust P1-REF SCALE on the Control card, to obtain 4 VDC at test point SRS (Manual Speed Reference only. Jumper JP1 on the Control card selects the manual speed reference input voltage range). Measure with a volt-ohmmeter (or the DVM diagnostic test as mentioned above) at the appropriate test point and PCOMX. If Auto System Reference (ASR) is used on the Microapplication card, adjust P1-ASR SCALE for 4 VDC at test point ASR, also located on the Microapplication card. Set the speed reference to 0% speed.

    NOTE: *If no tach or digital tach is used. SFB will be "0" VDC in the next step.*

24. Start the drive and gradually increase speed reference to maximum. Check for 4 VDC at test points SRS and SFB on the Control and Process Interface cards. (If ASR is used, check test point ASR on the Microapplication card.) Adjust P1-TACH SCALE (requires optional Process Interface card), if necessary, to reach the correct top speed (only on drives with an AC or DC tach). Confirm rated armature voltage.

    NOTE: *If experiencing difficulty in making the adjustments in Step 24, it may be necessary, due to the tolerances of the tach, to verify the actual tach output voltage/RPM. Check the tachometer's actual speed and output voltage with a hand tach and voltmeter to verify the proper SW1 setting on the Process Interface card.*

WARNING

Burn and Shock Hazard: Exercise caution as high voltage may be present.

25. Confirm good motor commutation (absence of sparking between brushes and commutator) under load.
26. Slowly decrease the speed. Verify stability through the full speed range. If the drive is **NOT** stable, refer to the Stability Adjustment procedure in the Troubleshooting section.
27. After the drive is stable throughout the full speed range, confirm that the armature current does not exceed the nameplate rating.

Before Continuing, it is recommended that the following be done:

a. The list of "as shipped" parameter settings provided with the controller be checked against the actual parameters in the drive.

b. All incoming signals be checked for proper value and correct connection. Refer to the "Use of Optional Programmer Module" section Test 20-Electronic Wire Check.

Similarly, other parameters as sold and shipped with the drive can be accessed and changed as necessary. (A complete list of pertinent parameters is contained in the Custom Software Description shipped with the drive.) A partial listing of "KEY" Parameters, their descriptions, and a Parameter Map are provided in this instruction book.

NOTE: *Final responsibility for parameter settings resides with the customer. Document all changes to parameter settings in the space provided in the Custom Software Description.*

28. Further accidental reprogramming should be prevented by subtracting a 1 from the value presently in ADR003-DGNJP (refer to Step 3 of this section), and entering the new value; then place the EEROM write device in the "PROTECT" position.
29. Exit the parameter mode by pressing [CLEAR/MODE] [CLEAR/MODE]. OPERATE should appear. The **RED** keypad functions will now control the drive.

    The above list of adjustments should result in properly running drive with desired readouts at any available diagnostic test on the Programmer.

    The drive is now ready for operation.

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8 - TROUBLESHOOTING

This section contains information on detecting and correcting problems which may be encountered when working on a DFM-300 drive controller. The controller contains a built-in fault detection system which indicates when a fault occurs and the nature of the fault. Tables 5 and 6 are intended to aid in the correction of problems both indicated by the drive fault system and those not indicated by the fault display. Following Table 6 is a "Stability" Troubleshooting section.

Table 5, the BugBuster™ Troubleshooting Chart" is for aid in troubleshooting problems which have no associated fault or error indication or can not be remedied through the standard fault system. Table 5 includes a "Quick Reference Guide", an outline of the possible problem descriptions included in the chart.

Table 6, "Specific Fault Troubleshooting" defines drive fault codes. Fault codes are displayed on the Diagnostic LEDs and Programmer module, and broadcast on the optional serial link.

Table 7, "Specific Error Troubleshooting" defines error codes that are only related to Programmer or serial link errors, and therefore, only displayed by the Programmer module or broadcast on the serial link.

TABLE 5
BUGBUSTER TROUBLESHOOTING CHART
QUICK REFERENCE GUIDE

PROBLEM DESCRIPTION

DRIVE FAULTS, LED INDICATION

1. All six diagnostic LEDs off
2. All six diagnostic LEDs on
3. Fault messages displayed (repetitive)
4. Fault messages displayed (non-repetitive, random)

PROGRAMMER PROBLEMS

5. Programmer display off
6. Error messages displayed
7. Programmer ineffective – no errors displayed
8. Parameter changes ineffective
9. Diagnostics do not perform

FUSE BLOWING, SCR PROBLEMS, MOV FAILURES, CARD FAILURES

10. Fuse blowing, AC or DC power fuses
11. Motor Field fuses blowing
12. Power supply fuses blowing
13. MOV failures
14. Repetitive card failures
15. Random SCR misfirings
16. Line voltage problems

OPERATIONAL PROBLEMS

17. 24 Volt logic not operating properly
18. Drive/Motor does not run
19. Motor Field current or voltage wrong value
20. Drive runs in wrong direction
21. Drive does not reverse
22. Drive speed is incorrect
23. Spindle orient not positioning (option)
24. Serial link does not function properly (option)
25. Erratic operation
26. Stability problem
27. Signal level detector chatter
28. Too much coast time on deceleration

MOTOR PROBLEMS

29. Motor sparking
30. Motor runs hot
31. Motor audible noise
32. Motor vibration

GENERATOR PROBLEMS

33. Generator sparking
34. Generator runs hot
35. Generator audible noise
36. Generator vibration

NOISE, WIRING, POWER SUPPLIES, ETC.

37. EEROM memory rewrites
38. Noise or random nuisance faults
39. Wiring procedures
40. Checking power supplies

TABLE 5 – BUGBUSTER TROUBLESHOOTING CHART

PROBLEM	CHECK/ADJUST/REPLACE
1. All Diagnostic LEDs OFF	1. Check Programmer display for drive fault (if used). 2. Check for 3-phase AC input. 3. Check for open AC line fuses (FU1-3) or breaker. 4. Check for open fuses on the Power Supply card (CFU1-8). 5. 2PL disconnected, open, loose or intermittent. 6. Power supplies loaded down or failed – refer to "Checking Power Supplies", this section. 7. If Hand-held Programmer is available, connect and attempt to enter the PARAMETER mode. Refer to Error Code List if any detected. A hard reset should clear this lock-up condition. 8. Replace Control card software (U1, U5 and U6). 9. Replace Control card.
2. All Diagnostic LEDs ON	1. Drive faulted – check fault code list. 2. U5 and U6 microprocessor interchanged on the Control card. 3. RSET-3TB69 connected to +15V. 4. Extended memory U1 installed – check JP24 per test data sheet in drive door pocket. 5. Replace Control card software (U1, U5 and U6).
3. Fault Messages Displayed (Repetitive)	1. See Specific Fault Troubleshooting, Table 6, for troubleshooting procedures related to the specific drive fault number.
4. Fault Messages Displayed (Non-Repetitive, Random)	1. Specific problem description on "Noise or Random Nuisance Faults". 2. See "Wiring Procedure".
5. Programmer Display OFF	1. Is plug connected correctly? • Local Programmer – 10 conductors must be in **right** side of plug when connected. • Hand-held Programmer – red stripe must appear on the left-hand side of the cable leaving the Programmer and on the left-hand side of the cable when inserted in 18PL. 2. Check power supplies – refer to "Checking Power Supplies", this section. 3. Replace Programmer card.
6. Error Messages	1. See Error Message Listing in Table 7.

TABLE 5 – BUGBUSTER TROUBLESHOOTING CHART (continued)

	PROBLEM	CHECK/ADJUST/REPLACE
7.	Programmer Ineffective No Errors Displayed	1. Is plug inserted properly? • Local Programmer – 10 conductors must be in **right** side of plug when connected. • Hand-held Programmer – red stripe must appear on the left-hand side of the cable leaving the Programmer and on the left-hand side of the plug when inserted in 18PL. 2. Check jumpers JP3 and JP4 on the Programmer card for proper position. 3. Check Programmer jumpers JP1 and JP2 – must be in 1-2 position. 4. Is the drive in Auto-mode – this mode inhibits Diagnostic Run, SCR test and certain programmed Operate mode references, i.e., RUNRF, JOGRF, etc. 5. Replace Programmer.
8.	Parameter Changes Ineffective	1. Check JP13 write protect on the Control card, must be in 2-3 or "ENABLE" position for programming. 2. Check setting in ADR003-DGNJP – must be an odd number to allow reprogramming. 3. If the drive uses multi-motor operation for different gear ranges, spindle orient, etc., verify that the proper motor parameters are being adjusted. Check by setting the IOC trip setpoint of the motor parameters being adjusted to 10. Starting the drive will cause IOC fault if the proper motor parameters have been adjusted. 4. Check "Programmer Ineffective – No Errors Displayed" section. 5. Replace Control card software (U1, U5 and U6).
9.	Diagnostics Do Not Perform	1. Verify that the particular function is available – refer to the Custom Software Description, software jumper descriptions. 2. Check JP13 on the Control card, must be in the 2-3 position for some Diagnostic tests to work. Refer to "Use of Optional Programmer Module". 3. Check the drive mode, Auto or Manual. The drive must be in Manual mode to run Diagnostic tests. 4. Refer to "Programmer Ineffective – No Errors Displayed", this section.

TABLE 5 – BUGBUSTER TROUBLESHOOTING CHART (continued)

PROBLEM	CHECK/ADJUST/REPLACE
10. Fuse Blowing AC or DC Power Fuses	1. Compare the fuse size with the Renewal Parts List. 2. Check for shorted SCRs with an ohmmeter from each MA contactor pole to both sides of the DC bus supplying the generator field. 3. Check the phase rotation and connection of the AC sync wires at the incoming fuse, breaker and Power Supply card. 4. Check the phase rotation and connection of the AC power wires at the incoming fuses, breaker, MA contactor and SCRs. 5. Check the setting of ADR080-CMFLM. 6. Check for out of tolerance or sagging AC line voltage – refer to "Line Voltage Problems" and "Noise or Random Nuisance Faults" this section. 7. Check the field power module for a short. 8. Using a megger, check for a ground in motor field, generator field or blower circuit. 9. Adjust the field control parameters to provide a lower armature voltage at top speed – lower ADR079-CROSS on field weakening drives or ADR076-FLDMX on base speed drives. 10. Refer to "Random SCR Misfirings" this section.
11. Field Fuses Blowing	1. Check for a ground in the motor field or conduit wiring. 2. Check the field power module for a short or failure. 3. Verify that all wires on the Power Supply card are connected to the correct stab-on. 4. Verify that FSPL is plugged into the proper position with the correct polarity. 5. Check for shorted MOVs on the Power Supply card. 6. Check for shorted Control Power Transformer (CPT) primary.
12. Power Supply Fuses Blowing	1. Verify that the fuses are correct. 2. Check for any shorted or over-loaded power supplies (Refer to "Checking Power Supplies", this section). 3. When using Hand-held Programmer, verify that it is properly connected to 18PL with red wire on the left-hand side of the cable. 4. If the +5 volt DC supplies a digital tach, verify that a cooling fan is installed below the card assembly on the controller and operating properly. 5. Replace the Power Supply card.
13. MOV Failures	1. Refer to "Line Voltage Problems", this section. 2. Generator fault causes AC fuses to open. Check for grounded field or wiring.
14. Repetitive Card Failures	1. Check all connections at 2TB, 3TB, 4TB & 21TB. 2. Verify that all ribbon cables and plugs are connected properly (i.e., not offset by one row or pin). 3. Check the stab-on connection points on the Power Supply and Power Connection cards. 4. Verify that all voltages (incoming AC and internal DC) are correct. 5. Check for moisture. 6. Check for damage due to airborne chemical contamination.

TABLE 5 – BUGBUSTER TROUBLESHOOTING CHART (continued)

PROBLEM	CHECK/ADJUST/REPLACE
15. Random SCR Misfirings	1. Check for inoperative SCRs. 2. Check for loose connections on the fuses, breaker, MA contactor, SCRs or any armature motor field or generator field connections. 3. Check for loose connections on SCR gate leads, Power Connection card or 5PL. 4. Check for notching on the AC line – this could be caused by other drives on the same incoming AC line. 5. Refer to "Line Voltage Problems", this section.
16. Line Voltage Problems	1. Verify that the line voltage is within drive specifications, -5% to +10%. 2. Verify that the AC line voltage is balanced line-to-line and line-to-ground. 3. Check for AC line notches or spikes. 4. Verify that the line frequency is within ±1 Hz of rated. 5. When AC line is supplied by a diesel driven alternator, check for changes in line frequency during drive acceleration. 6. Check for more than one and one-half cycles of ringing following commutation notches in the AC line caused by the drive. Amplitude should be less than or equal to 200 volts. 7. Check for an ungrounded secondary on the power system transformer directly in front of the drive (if used). (The transformers are generally a delta/wye type with the neutral of the "wye" grounded through a high resistance or directly to ground.)
17. 24 Volt Logic Not Operating Properly	1. Check the ±24 volt power supplies. 2. Check logic polarity control Jumpers per the Custom Software Description (ADR012-OPTJP), JP19 and 20 on the Control card, JP5 and JP6 on the Microapplication card. 3. Check control logic connections at 3TB and 4TB. 4. 24 volt logic being switched too quickly to 4TB or Microapplication card – there must be at least a 70 millisecond delay between two logic inputs changing state. 5. Check for proper polarity and common circuit if 24 volt supply is from a source other than drive. 6. Check that the drive is in the proper motor mode (0, 1, 2 or 3) for the particular motor or function being operated.

TABLE 5 – BUGBUSTER TROUBLESHOOTING CHART (continued)

PROBLEM	CHECK/ADJUST/REPLACE
18. Drive/Motor Does Not Run	1. No connections to REF 24, ±15V or common should be made at 4TB. Move any such connections to 3TB. 2. FU1, 2, 3 or 4 open (or optional circuit breaker open). 3. CFU1, 2, 3, 4 or 5 open and/or MOVs shorted on Power Supply card. 4. Check all hardware and software jumpers per the Custom Software Description, particularly ADR298-DFMJ1, ADR012-OPTJP, JP19 and JP20 on the Control card and JP5 and JP6 on the Microapplication card. 5. Check for the speed reference at 3TB or 4TB (MSR, JOGR or ASR). 6. Check control logic voltage and state at 3TB and 4TB (RUN, CI0, CI1, JOG, SLOW, MODE 0 thru MODE 7, etc.) as well as power supply voltage levels. 7. Check that the drive is in the proper operate mode (AUTO, MANUAL or ORIENT). 8. Drive may be stalled in current limit. 9. Check MD contactor operation and DFM RUN and MD logic at 4TB.
19. Field Current or Voltage Wrong Value	1. Is the drive in field economy? 2. Check for proper motor field connections. 3. Check JP1 and JP2 on the Power Supply card for proper maximum field settings. 4. Check the P1-FIELD SCALE pot seal – factory adjusted for field current. 5. Check ADR076-FLDMX and ADR077-FLDMN per the Custom Software Description. 6. Check 4TB20 for connection to FRA if external field forcing is used. 7. Check for any external resistors or diodes in the field circuit. 8. Check the field power module for shorted elements. 9. Check the field current (Test point FC). Replace the Power Supply card if failed. 10. Replace the Control card.
20. Drive Runs in the Wrong Direction	1. Check for logic signal on 3TB15 (POL). 2. Armature connections reversed (correct armature and tach leads). 3. Motor field connections reversed (correct field and tach leads). 4. Generator field connections reversed. 5. Check feedback polarity to DFA card from armature circuit. 6. Check all software jumpers (ADR000 to ADR014, ADR297 and ADR298) per the Custom Software Description. 7. Check the reference polarity. 8. Check logic polarity control jumpers per the Custom Software Description (ADR012-OPTJP), JP19 and JP20 on the Control card, JP5 and JP6 on the Microapplication card. 9. Check Custom Software Description to verify that the reverse function is enabled when operating from the serial link.

TABLE 5 – BUGBUSTER TROUBLESHOOTING CHART (continued)

PROBLEM	CHECK/ADJUST/REPLACE
21. Drive Does Not Reverse	1. Verify that the drive is designed for reversing. 2. Check settings of the soft jumpers (ADR000 to ADR014, ADR297 and ADR298) Custom Software Description. 3. Check the reference polarity. 4. Check control logic to POL (3TB55) if used for reversing. 5. Check the logic polarity control jumpers per the Custom Software Description (ADR012-OPTJP), JP19 and JP20 on the Control card, JP5 and JP6 on the Microapplication card. 6. Check the Custom Software Description to verify that the function is enabled when operating from the serial link.
22. Drive Speed is Incorrect	1. Verify that the motor design is suitable for the application. 2. Is the proper reference (MSR, ASR, RUNRF, R1RF, R2RF, SLWRF, JOGRF) enabled? Refer to Custom Software Description. 3. Check JP1 on the Control card for proper scaling. 4. Adjust P1-REF SCALE on the Control card. 5. ADR068-REFSF set incorrectly? 6. Check JP1 on the Microapplication card for proper ASR scaling. 7. Adjust P1-ASR SCALE on the Microapplication Card (if used). 8. Adjust P7-SFB ZERO on Process Interface card if analog AC or DC tach is used. 9. Confirm the setting of SW1-TACH SCALE on the Process Interface card if an analog AC or DC tach is used. 10. Check output voltage at the drive if a pulse tach is used. May be low due to excessive cable length. 11. RPMSF or REFSF may be misadjusted, check per the Custom Software Description. 12. Verify the proper settings in ADR053-DTCHR and ADR074-CPPR per the Custom Software Description. 13. Check for noisy tach signal. Refer to "Wiring Procedures", this section. 14. Set ADR001-FDBJP per the Custom Software Description. 15. Confirm that JP5-10 on the Process Interface card are in accordance with application requirements. 16. Check the setting of ADR041-ILMMX if the drive is suspected of being in current limit. 17. Check the setting of ADR080-CMFLM if the drive is suspected of being in CEMF limit. 18. Check ADR002-REGJP and ADR079-CROSS per the Custom Software Description. 19. Check ADR076-FLDMX and ADR077-FLDMN per the Custom Software Description and by the calculations in the key parameter adjustment list. 20. Check ADR059-RATIO per the Custom Software Description, should be set at "10000" for no draw. 21. Check that P4 on the Microapplication card is full clockwise.

TABLE 5 – BUGBUSTER TROUBLESHOOTING CHART (continued)

PROBLEM	CHECK/ADJUST/REPLACE
23. Spindle Orient Not Positioning	1. Confirm that position loop is enabled by the control logic. 2. Check that the position loop has been enabled by the hardware and software jumpers. 3. Verify that the proper motor parameters are selected by changing the IOC trip parameter for that motor to 10 and getting an IOC when position commanded. 4. If spindle continues to turn or orients in only one direction, adjust P7-SFB ZERO on the Process Interface card. 5. If an encoder is used, verify that only one marker pulse per revolution is present and not noisy. 6. Confirm that the motor field is weakened when in position.
24. Serial Link Does Not Function Properly	1. Verify that JP11, JP12 and JP13 on the Process Interface card are set correctly. 2. Check the baud rate of the drive (ADR126-BAUDR) compared to serial link device (hard reset must be performed after BAUDR is changed). 3. Check serial link connections at 3TB22, 23 and 24 (DCOM, R232 and 3TB22). 4. If RS422 is used, verify that the handshaking lines are properly connected. Verify that sending device is responding to the drive's DTR signal (low level) when the device is allowed to transmit to the drive. 5. Replace the Process Interface card.
25. Erratic Operation	1. Check power supplies for magnitude and ripple (refer to "Checking Power Supplies"). 2. Verify that 1CPL on the MFC/Power Supply card is inserted correctly (not backwards or offset). 3. Check connections on all ribbon cables. 4. Tighten all connections on 2TB, 3TB, 4TB, 21TB and all power connections. 5. Confirm correct hardware jumper setting per the Custom Software Description. 6. Check all parameters per the software test data sheet in the drive door or Custom Software Description. 7. Check for noisy ground connection – refer to "Wiring Procedures". 8. Examine CFB and VFB with an oscilloscope for stability – refer to "Stability". 9. If a digital tach is used, check the integrity of the signal. 10. Check resistors and connections at DTB and ETB on Microapplication card.

TABLE 5 – BUGBUSTER TROUBLESHOOTING CHART (continued)

PROBLEM	CHECK/ADJUST/REPLACE
26. Stability Problems	1. Examine CFB with an oscilloscope to determine if any SCRs have failed. 2. Check settings of all parameters per the software test data sheet inside the drive door pocket. 3. Check the setting of the SW1-TACH SCALE switch on the Process Interface card if an analog AC or DC tach is used. 4. Check the settings of JP7 and JP8 on the Process Interface card if an analog tach with tach linearity is used (should be 1-2 position for both). 5. Verify that all contacts on the MA contactor are closing. 6. Check and adjust for any offset on CFB, P2-CFB ZERO, P3-CFB ADJUST (on the Power Supply card), JP3 and JP4 on the Power Supply card or the P4-OFFSET (on the Control card). 7. Check P4 on the Microapplication card. Should be full clockwise. 8. Check scaling resistors at ETB and DTB on Microapplication card. 9. Check setting of P2 and P1 on DFA card. 10. Refer to the "Troubleshooting Stability Problems" section in this manual. 11. Make sure armature current reads correctly at RAM2152 (Diag Test 11).
27. Signal Level Detector Chatter	1. Is SLD looking at a "noisy" signal (e.g., CFB, tach feedback, etc.)? 2. Is enough hysteresis programmed into the appropriate parameter address?
28. Too Much Coast Time On Deceleration	1. Lower the value in ADR048-RSTOP. 2. Check settings in Custom Software Description and confirm.
29. Motor Sparking	1. Check motor commutator for contamination. 2. Check setting in ADR024-IRTLM, may be too high. 3. Check the field current – refer to "Motor Field Current or Voltage Wrong Value". 4. Check the motor field current stability – refer to "Stability Troubleshooting" section. 5. Check the commutator condition. 6. Confirm that all brushes are seated properly with proper brush tension. 7. Check the brush grade. 8. Check the brush neutral adjustment. 9. Check for sagging AC line voltage – refer to "Line Voltage Problems", this section. 10. If tapered current limit is required for the application, verify that ADR081-ADR084 reflect this.

TABLE 5 – BUGBUSTER TROUBLESHOOTING CHART (continued)

PROBLEM	CHECK/ADJUST/REPLACE
30. Motor Runs Hot	1. Motor may be overloaded, check for mechanical problems. 2. Check the motor winding temperature. If the motor has Class F insulation and the winding temperature is less that 130°C, this is normal. 3. Check the rotation of the motor blowers. 4. Check the motor field current with respect to the motor nameplate rating. Adjust ADR076-FLDMX if necessary. 5. Check the motor field current when motor is not running. If the field current is full nameplate rating when the motor or blower is not running, check ADR050-FLDEC and ADR297. 6. If motor runs hot at very low speeds, check motor application guide for full-torque speed range.
31. Motor Audible Noise	1. Loose pole bolts – tighten per manufacturer's torque specifications. 2. Check for motor vibration.
32. Motor Vibration	1. Coast Stop Drive – if the vibration is electrically induced the vibration will disappear. If it does not disappear, check for mechanical causes. 2. Refer to "Stability" procedures. 3. Motor not mounted level. 4. Motor base vibrating.
33. Generator Sparking	1. Check generator commutator for contamination. 2. Check the setting in ADR024-ITLM, may be too high. 3. Check the generator field current. 4. Check the commutator condition. 5. Confirm that all brushes are seated properly with proper brush tension. 6. Check the brush grade. 7. Check the brush neutral adjustment. 8. Check for sagging AC line voltage – refer to "Line Voltage Problems", this section. 9. If tapered current limit is required for the application, verify that ADR081-ADR084 reflect this.
34. Generator Runs Hot	1. Motor may be overloaded, check for mechanical problems.. 2. Check the generator winding temperature. If Class F insulation and the winding temperature is less than 130°C, this is normal. 3. Check the rotation of generator blowers (if any). 4. Check the generator field current with respect to the nameplate rating. 5. Confirm that all SCRs are firing properly by monitoring current feedback.
35. Generator Audible Noise	1. Loose pole bolts – tighten per manufacturer's torque specifications. 2. SCRs misfiring – check current feedback. 3. Cross-firing SCRs – increase deadband. 4. Unstable regulator – check "Troubleshooting Stability Problems" section.

TABLE 5 – BUGBUSTER TROUBLESHOOTING CHART (continued)

PROBLEM	CHECK/ADJUST/REPLACE
36. Generator Vibration	1. Coast Stop Drive – if the vibration is electrically induced the vibration will disappear. If it does not disappear, check for mechanical causes. 2. Refer to "Stability" procedures. 3. Generator not mounted level. 4. Generator base vibrating.
37. EEROM Memory	1. JP13 on the Control card must be in the 1-2 position during "Power Up" or hard reset. 2. Check programmer operating procedure and verify keying sequence and programmer modes used.
38. Noise or Random Nuisance Faults	1. Verify that the 5-volt DC supply is not used externally. 2. Check if ACOM, DCOM or PCOM are used externally – PCOMX is the only point that should be used for common connections. 3. Verify that PCOM passes through a ferrite core (not required if PCOMX is on 3TB73-75). 4. Check for suppression on all relays, contactors and starters. 5. Place JP16 on the Control card in the 2-3 position on all drives not using a frequency reference. 6. Check all power supplies – refer to "Checking Power Supplies", this section. 7. When an encoder is used, verify that all of the encoder wires except the shield have been passed through a ferrite core. 8. Refer to "Wiring Procedures", this section.
39. Wiring Procedures	1. Check the wire sizes on the armature, field and AC power connections. 2. Verify that the control wires (24 volt DC, 115 volt AC) do not run in the same pick-up. 3. Verify that all of the signal wires (i.e., tach, encoder, reference, load cell or RS-232C/422) are twisted pairs, shielded and shields grounded at the drive. 4. Verify that the signal wires are not grounded at more than one end or other than the drive end only. 5. Verify that the signal wires do not run in the same conduit or parallel closer than four inches to the power wires. 6. Verify that there is only one system ground. 7. Verify that the ground bus is large enough to prevent voltage drops or noise pick up.
40. Checking Power Supplies	1. Check the +15 test point on the Control card for +15VDC, ±5%. Verify that there is less than or equal to 100 millivolts of ripple with an oscilloscope. 2. Check the -15V test point on the Control card for -15 volts VDC, ±5%. Verify that there is less than or equal to 100 millivolts of ripple with an oscilloscope. 3. Check the +5 test point on the Control card for +5VDC, ±1%. Verify that there is less than or equal to 100 millivolts of ripple with an oscilloscope. 4. When the 5V power supply issued to power an encoder or digital tach, check for power supply heating if no fan is supplied.

SPECIFIC FAULT TROUBLESHOOTING

FAULT/ERROR CODES

The DFM-300 drive controller displays faults with a series of Diagnostic LEDs (interpreted as a binary number) or with a fault message (fault number and name) on the optional Programmer.

NOTE: *In the normal condition the drive's LEDs will flash from right to left:*

one at a time in the standby mode

two at a time in the RUN or JOG modes

The following table lists predefined fault codes in order by the assigned fault number. To determine the fault number from the diagnostic LEDs, add the number value of the LEDs printed on the card above or below the LEDs (also add to this total the value of 64 if the LEDs are blinking). For example:

TYPES OF FAULTS

The faults listed in Table 6 may be divided into four categories:

L - The fault is latched, i.e., once it occurs it will not clear itself even if the cause of the fault no longer exists. The controller must be reset to clear it.

N - The fault is non-latched, i.e., it will clear itself if the cause of the fault no longer exists.

T - The fault trips the drive, i.e., it causes the controller to remove power from the motor. The controller must be reset. All trip faults are latched.

A - The fault is annunciated only, i.e., if the drive is running it will not cause the drive to shut down. However, an annunciated fault will prevent the drive from starting if it is not running. Annunciated faults may or may not be latched.

There are several ways to reset the controller. A software reset may be executed from the Programmer card, or by serial command from a device connected to the serial link. A hardware reset may be executed by toggling the reset switch on the Control card, by momentarily connecting 6PL-1 to 6PL-4 (3TB69 to 3TB66), or by momentarily interrupting incoming power to the drive. Most latched faults may be cleared by a software reset, however, some faults should be cleared by a hardware reset. These are indicated in Table 6 by the letter "H". If a software reset does not seem to restore the drive to normal operation, a hardware reset should be attempted.

FAULT CODE INDICATION

LED Indication	Binary Number						Decimal Number	Fault Number
	32	16	8	4	2	1		
Steady display	0	1	1	1	0	0	16+8+4 = 28	FL28
Blinking display	0	1	0	1	0	0	16+4+64 = 84	FL84

NOTE: 0 = LED OFF 1 = LED ON 1 = BLINKING LED

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
000001	FL01	IOC	Instantaneous DC overcurrent (L,T) • Tighten armature connections. • Armature grounded. • Examine motor commutator for damage or evidence of flashover. • Verify proper brush seating and tension (refer to motor manual). • Mechanical Problems – binding shaft, unreleased brake, vibration, gearbox, or load problems. • IOC parameter set too low. Set at least 128 units above current limit settings. • Current feedback isolator failure – check DFPL. • P4 on Microapplication card misadjusted. Set full clockwise. • P1 or P2 on DFA card misadjusted or DFA card failed. • Check input at SJ2 (4TB16 or 13PL-16). • ADR011-SJAJP set incorrectly. • Current regulator mistuned – adjust MIAPG and MIAAG (ADR299 and ADR300).
000010	FL02	OVRSP	Overspeed (L,T) • ADR073-SPDLM set too close to ADR068-REFSF and ADR071-REFLM – set a minimum of 5% greater than REFSF and REFLM. • Speed overshoot occurs – speed regulator response too high, reduce gain. (Refer to Table 7, Troubleshooting "Stability" Problems.) • Conflicting settings of ADR001-FDBJP, ADR002-REGJP and ADR013-POSJP for digital tach and standard Process Interface card (531X133PRUA-G1) or Encoder Process Interface card (531X134EPRA-G1) – refer to Custom Software Description (Encoder Process Interface card cannot accept encoder and digital tach feedback at same time). • Noise on digital tach input. • Loss of field current.
000011	FL03	SPARE	

KEY: L - LATCHED N - NON-LATCHED H - REQUIRES HARD RESET
T - TRIPS DRIVE A - ANNUNCIATED ONLY

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
000100	FL04	FLOSS	Loss of motor field current (L,T) • CFU1, 2 or 3 on Power Supply card open. • 1PL disconnected or damaged. • Field grounded. • Field lead connections made improperly. • Field regulator saturation; add "16" to ADR012-OPTJP. • Open field circuit components – resistors, contactors or DC fuses. • ADR078-FLDLS set too high – set at least 25% below ADR077-FLDMN and 50% below ADR076-FLDMX. • FC gain jumpers set wrong – JP1 and JP2 on Power supply card. • External field supply – set ADR078-FLDLS at "60000" to inhibit field loss. • Field regulator gain too low – adjust BFLPG, BFLIG WFLPG, WFLIG (ADR031, 032, 033, 034).
000101	FL05	TPOL	Tach polarity incorrect (L,T) • Reversed tach connections. • Switch SW1-7 on Process Interface card set improperly, should be open for DC tach, closed for AC tach. • Armature or field reversing contactors – ADR001-FDBJP must contain a 2, check Custom Software Description to see if this feature is supplied.
000110	FL06	JMPER	Software jumper illegally changed while running (L,A) • Drive must be stopped to change certain parameters (i.e., ADR000 to ADR014).
000111	FL07	RESET	Reset while running (L,T) • Drive powered up with RUN or JOG closed — check 6PL-12 and 6PL-13 (3TB58 and 57). • Momentary loss of incoming power while running. • Incorrect logic polarity, check ADR012-OPTJP.
001000	FL08	TOC	Timed generator field overcurrent (L,T) • CFU2 on Power Supply card open. • ADR044-TOCTH and ADR045-TOCGN set too low – check against the parameter list shipped with drive. • Current feedback isolator failure or 1PL disconnected or damaged. • P3-CFB ADJ on the Power Supply card misadjusted. • Generator field winding failed.

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
001001	FL09	LNKE	Serial Link trip (drive trip via serial link) (L,T) • Drive received external trip signal – check for external fault. • 12PL or 18PL incorrectly connected or damaged.
001010	FL10	LNKFL	Serial Link fault (communications failure) (L,T) • Programmer card jumpers incorrect – refer to Table 1. • Baud rate set incorrectly – check ADR126-BAUDR. • 12PL or 18PL incorrectly connected or damaged. • Intermittent connection between 3TB32 (CTSA) and 12PL-7 or 3TB41 (CTSB) and 12PL-14. • FL36 appears momentarily previous to FL10 latching - refer to FL36.
001011	FL11	DFPON	Delayed firing power on when it should be off (L,T) • Q2 failed on – check test point DFP. • Replace Control card.
001100	FL12	DFPOF	Delayed firing power off when it should be on (L,T) • Q2, Q3, R33 or U14 failed on Control card – replace Control card.
001101	FL13	EEXXX	Checksum error or (L, T, H) parameter out of range (L, A, H) where XXX equals a 3-digit number, indicating an out-of-range parameter address or altered checksum. • Spontaneous parameter alteration – check all parameters and verify EEROM is not write "ENABLED" during power up or power down, or "hard" reset. • Value other than zero in ADR255-UNDFD - replace EEROM.
001110	FL14	ILOOP	Current loop regulator saturated (L,T) • Power Supply card failed – replace. • Control card failed – replace. • Prime mover not running. • Scaling resistors on Microapplication card DTB and ETB missing or incorrect. • Insure that the MD Contactor is not opening before the interlock at CIO on 4TB. • Check starting logic and verify CIO opens before 3TB run.
001111	FL15	VFBLS	Loss of armature voltage feedback (L,T) • This Fault Code not used with DFM-300 drive. If FL15 occurs, check that ADR004 is greater than 16384.

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
010000	FL16	NOTRP	Trip from N.O. on Application card (L,T) • External input at 4TB8 (13PL-8). • Check jumpers on Microapplication card per Custom Software Description. • Replace Microapplication card.
010001	FL17	XSTOP	Coast stop trip (L,T) • Check for input at XSTOP on 3TB1 (6PL-16). • Check ADR004-STPJP – a 4 causes coast stop when XSTOP is selected.
010010	FL18	STALL	Generator field overcurrent (L,T) • ADR046-STLTH and ADR047-STLGN set wrong – should have value of 500 in ADR046 and 10 in ADR047. • Generator field failed. • Loose generator field connection.
010011	FL19	OVERV	Excessive armature voltage (L,T) • ADR080-CMFLM set too low – compare to parameter list. • VFB circuit failure – check test point VFB.
010100	FL20	TLOSS	Loss to tach feedback (L,T) • Tach leads disconnected. • Tach feedback circuit failure – 7PL disconnected or damaged. • Process Interface card jumpers or switches incorrect. Refer to Table 1. • P7-SFB ZERO on Process Interface card misadjusted – set test point SFB for zero volts at zero speed. • AC or digital tach connected but CEMF regulator being used, remove JP7 from Process Interface card. • ADR053-DTCHR set too high when AC Tach being used – lower value to 10. • ADR002-REFJP set incorrectly for type of Process Interface card used – PI card (531X133PRUA-G1) or Encoder PI card (531X134EPRA-G1); refer to Custom Software Description. • JP17 on Control card in 1-2 position. Move to 2-3 position. • Process Interface card failed – replace.
010101	FL21	NCTRO	Trip from N.C. interlock on Application card (L,T) • Loss of external input on 4TB7 (13PL-7). • Microapplication card absent. • If preceded by FL07 and soft reset, check logic polarity ADR012-OPTJP.

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
010110	FL22	PHLOS	AC phase loss (L,T) • JP14 on Control card in wrong position – see Table 1. • Fuse open in AC line – FU1-3. • 1PL disconnected or damaged. • Low or unbalanced AC line voltage – line-to-line or line-to-neutral more than 5% low. • Loose AC power connections. • CFU1 open. • Verify that ADR187-FUSLS contains a zero – change if necessary. • ±24-volt or ±15-volt power supplies failed – check CFU6 and 7 on Power Supply card. • ±15 volt power supplies outside limits (±5%) – replace Power Supply card.
010111	FL23	OVERF	Motor Field overcurrent (L,T) • Field regulator unstable – refer to Stability procedure in Troubleshooting section. • Field forcing at FRA (4TB20, 13PL-20) – check input scaling, see Table 1. • ADR076-FLDMX or ADR049-FTOCG incorrect – check parameter list. • Field SCR shorted – replace Motor Field Module. • Voltage regulated field – check for miswired or grounded field.
011000	FL24	U3 FL	Faulty RAM on U3 8155 (L,T) • Replace Control card.
011001	FL25	COMTR	SCR commutation failure (L,T) • Low AC line voltage (less than 90% rated). • 1CPL disconnected. • 1CPL reversed – phase 1 CT should be wired to bottom of 1CPL at Power Supply card. • P3-HP SEL and JP8-11 gains set too high on Control card. • 1PL disconnected or damaged. • Gate plug(s) incorrectly connected. • SCR(s) failed.

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
011010	FL26	ACIOC	Circulating AC line fault (L,T) • Low or noisy AC line. • Incorrect setting of P3-HP SEL and JP8-11 on Control card. • Noise on SCR gate leads. • 1PL disconnected or damaged. • SCR(s) failed.
011011	FL27	LYSNC	Loss of AC line signal (L,T) • Fuse open in AC line – FU1-3. • 1PL disconnected or damaged. • Low or unbalanced AC line voltage – line-to-line or line-to-neutral more than 5% low. • Loose AC power connections. • CFU1 open. • 15 volt power supplies outside limits (±5%) – replace Power Supply card. • AC line frequency below 45Hz or above 63Hz (70Hz if called for by ADR001-FDBJP). • Check 50/60Hz square wave at test point SY on Control card - replace Control if no signal.
011100	FL28	RMODE	Illegal selection of RUN mode (N,A) • Do not command DIAGNOSTIC STATIC while running or vice versa. • Check ADR197, ADR014, ADR001.
011101	FL29	CNLON	Control on input opened (L,T) • Open circuit between 3TB3 and 3TB4. • 6PL incorrectly connected or damaged. • Noise between 3TB3 and 3TB4.
011110	FL30	NOAPP	Microapplication card failure (L,T) • Microapplication card absent. • 3PL incorrectly connected or damaged. • Check jumpers on Microapplication card per Custom Software Description. • Replace Microapplication card.
011111	FL31	NOSEQ	Phase sequence indeterminate (L,T) • CFU1 on Power Supply card open. • Excessively noisy or notched AC line. • Line unbalance, phase to ground, by more than 15%. • 15 volt supplies outside limits (±5%) – replace Power Supply card. • 1PL incorrectly installed or damaged.

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
100000	FL32	TOCAN	Timed Generator Field overcurrent annunciation (N,A) • CFU2 on Power Supply card open. • ADR044-TOCTH and ADR045-TOCGN set too low – check against the parameter list shipped with drive. • P3-CFB ADJ on the Power Supply card misadjusted. • Generator field winding failed.
100001	FL33	PHSEQ	Phase sequence reversed (N,A) • Reverse any two AC input lines, recheck rotation of any three phase motors. • Excessively noisy or notched AC line. • Line unbalance, phase to ground by more than 15%. • 15 volt supplies outside limits (±5%), noisy or excessive ripple, replace Power Supply card. • 1PL incorrectly installed or damaged. • ADR012-OPTJP not properly set for phase sequence insensitivity option.
100010	FL34	PLLSY	AC line PLL out of sync (N,A) • Excessively noisy or notched AC line. • CFU1 on Power Supply card open. • Line unbalance, phase to ground, by more than 15%. • 15 volt power supplies outside limits (±5%) – replace Power Supply card. • On variable frequency AC lines – ADR029-PLLGN too low.
100011	FL35	O SPD	Motor rotating when RUN commanded (N,A) • RUN command initiated during motor rotation. • ADR048-RSTOP set too low. • Test point SFB contains an offset at zero speed – adjust P7-SFB ZERO on the Process Interface card. • ADR004-STPJP not set properly for "flying restarts".

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
100100	FL36	LNKER	Undecipherable messages detected on serial link by drive or other device (N,A) • Excessive noise on 12PL-7 thru 14. • 12PL incorrectly connected or damaged. • 5V DC power supply noisy. • Baud rate incorrectly set – ADR126-BAUDR normally set at 4800. • Check Process Interface card jumpers JP1, JP11, JP12, JP13 and JP14. • Serial link message protocol incorrect, message too long, checksum byte incorrect or handshaking incorrect – refer to "Serial Link User's Guide". • Replace Process Interface card. • Replace control card.
100101	FL37	IHUNG	Armature current not zero when fully retarded (N,A) • Not used on DFM-300.
100110	FL38	ELOCK	Attempt to change software adjustments when soft locked (N,A) • ADR003-DGNJP must be odd to program EEROM – add one to value in DGNJP.
100111	FL39	EFAIL	Software adjustment not stored in EEROM (N,A) • JP13 on Control card in hard lock position. • EEROM failed – replace.
101000	FL40	MXSPD	Speed reference greater than REFLM speed adjustment (N,A) • Manual Reference jumper JP1 on Control card improperly set – see Table 1. • P1-REF SCALE on the Control card misadjusted – see Table 1. • ADR071-REFLM set below ADR070-RPMSF. • P1-ASR SCALE, on the Microapplication card misadjusted (if used) – see Table 1. • P4-PF SCALE or P5-PF OFF, misadjusted on Process Interface card (if used) see Table 1. • ADR075-RPM/H set improperly for frequency reference (if used). • Top speed reference for parallel input (if used) is set higher than ADR068-REFSF and ADR070-RPMSF. • Preset references (RUNRF, R1RF, R2RF, SLWRF) set too high (if used).

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
101001	FL41	KEYOF	Keypad function disabled (N,A) • Programmer key depressed for a function that is not enabled.
101010	FL42	RLOGC	Run logic error (N,A) • JOG or RUN, called for with XSTOP open. • Keypad STOP called for during an external run cycle. • Incorrect logic polarity – check ADR012-OPTJP, JP19 and JP20 on the Control card and JP5 and JP6 on the Microapplication card.
101011	FL43	U11FL	A-D converter failure (L,T) • Check for noise on ACOM, move wiring to PCOMX. • Control card failed – replace.
101100	FL44	U4LAT	U4 to U5 communication error (L,T) • Check PROM revision compatibility. • Check for noise on power supplies or analog inputs. • Replace Control card.
101101	FL45	SLOOP	Speed tracking fault (L,T) • Drive entered current limit or CEMF limit when tracking enabled. • ADR036-SERLM set too low – should be between 1 and 7. • Speed regulator unstable – see "Stability Troubleshooting" section.
101110	FL46	IOFST	Current feedback offset fault (L,T) • Current feedback signal at test point AN1 offset by 35%. • P2-CFB ZERO on MFC/Power Supply card misadjusted.
101111	FL47	OADJ>	A-D converter offset too high (N,A) • P4-OFFSET on the Control card misadjusted – turn CW. • Check for noise on ACOM.
110000	FL48	OADJ<	A-D converter offset too high (N,A) • P4-OFFSET on the Control card misadjusted – turn CCW. • Check for noise on ACOM.
110001	FL49	NO U4	U4 to U6 communications error (L,T) • Check for PROM revision compatibility. • Check for noise on power supplies – or analog inputs. • Replace Control card.

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
110010	FL50	U4BAD	U4 RAM, timer or ALU is faulty (L,T) • Verify settings of ADR011, 014, 056, 089, 187, 209 and 210 per Custom Software Description. • Incorrect software installed for application check revision (Diagnostic Test 10). • U4 failed – replace Control card.
110011	FL51	U1ROM	U1 ROM is faulty (L,T) • Replace U1 on the Control card. • Replace Control card.
110100	FL52	NCLSD	Annunciate N.C. interlock on Microapplication card (N,A) • Check external input on 4TB7 (13PL-7). • Microapplication card absent but selected by soft jumper – ADR010-APPJP should be zero if no Microapplication card is present.
110101	FL53	SCLER	SCALE ERROR (L,T) • Occurs if position Feedback is not enabled from the Process Interface card and the product of ADR070-RPMSF and ADR074-TCPPR is greater than 2,490,367 – check setting in ADR070 and 074.
110110	FL54	NO U5	U5 to U4 communication error (L,T) • Check for PROM revision compatibility. • Replace Control card or U5.
110111	FL55	U5BAD	U5 RAM, timer or ALU is faulty (L,T) • U5 failed – replace. • Replace Control card.
111000	FL56	U5ROM	U5 ROM is failed (L,T) • U5 failed – replace.
111001	FL57	NOPEN	Annunciate N.O. interlocks on Microapplication card (N,A) • External input on 4TB8 (13PL-8). • Microapplication card absent but selected by soft jumper – ADR010-APPJP should be zero if no Microapplication card is present.
111010	FL58	U5COM	U4 not communicating correctly with U5 (L,T) • Check for PROM revision compatibility. • Digital tach pulse rate exceeds 41,000 Hertz. • Noise on tach input. • Replace Control card.

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
111011	FL59	NO U6	U6 does not communicate properly with U4 (L,T) • Check for PROM revision compatibility. • Replace Control card.
111100	FL60	U6BAD	U6 RAM, timer or ALU is faulty (L,T) • U6 failed – replace. • Control card failed – replace.
111101	FL61	U6ROM	U6 ROM is faulty (L,T) • U6 failed – replace. • Control card failed – replace.
111110	FL62	SPARE	
111111	FL63	U6COM	U4 not communicating correctly with U6 (L,T) • Check for PROM revision compatibility. • Digital tach pulse rate exceeds 41,000 Hertz when using Process Interface card (531X133APRUA-G1) or 410,000 Hertz when using Encoder Process Interface card (531X134EPRA-G1). • ADR002-REGJP not set for Encoder Process Interface card (531X134EPRAG1) and digital tach pulse rate above 41,000 Hertz used – refer to Custom software description. • Check for noise on tach input signal. • U1 extended memory – move JP24 to the 2-3 position. • U5 and U6 (TUP) in wrong sockets. • Replace Control card.
000000	FL64	SPARE	
000001	FL65	A1FOP	SCR A1F is not operating. • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • Gate circuit 1F disconnected or open. • 5PL disconnected or damaged. • Failed SCR – check and replace if necessary.* • Replace Power Connection card. • Replace Control card.

** (see door label for module location)*

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
000010	FL66	A2FOP	SCR A2F is not operating (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • Gate circuit 2F disconnected or open. • 5PL disconnected or damaged. • Failed SCR – check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
000011	FL67	A3FOP	SCR A3F is not operating (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • Gate circuit 3F disconnected or open. • 5PL disconnected or damaged. • Failed SCR – check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
000100	FL68	A4FOP	SCR A4F is not operating (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • Gate circuit 4F disconnected or open. • 5PL disconnected or damaged. • Failed SCR – check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
000101	FL69	A5FOP	SCR A5F is not operating (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • Gate circuit 5F disconnected or open. • 5PL disconnected or damaged. • Failed SCR – check and replace if necessary.* • Replace Power Connection card. • Replace Control card.

** (see door label for module location)*

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
000110	FL70	A6FOP	SCR A6F is not operating (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • Gate circuit 6F disconnected or open. • 5PL disconnected or damaged. • Failed SCR – check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
000111	FL71	FWDOP	All forward SCRs are not operating (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • 5PL disconnected or damaged. • Replace Power Connection card. • Replace Control card.
001000	FL72	ARMOP	Armature circuit open (L,T) • DC fuse open. • Generator field circuit open. • 5PL disconnected or damaged. • Main contactor not picking up – check pilot relay operation. • Loss of current feedback – check 1PL-20 and DRPL. • Generator field not connected. • Offset voltage at CFB with zero armature current – adjust P2-CFB ZERO on the Power Supply card. • Power Supply card failed – replace. • Control card failed – replace.
001001	FL73	A1ROP	SCR A1R is not operating (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • Gate circuit 1R disconnected or open. • 5PL disconnected or damaged. • Failed SCR – check and replace if necessary.* • Replace Power Connection card. • Replace Control card.

** (see door label for module location)*

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
001010	FL74	A2ROP	SCR A2R is not operating (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • Gate circuit 2R disconnected or open. • 5PL disconnected or damaged. • Failed SCR – check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
001011	FL75	A3ROP	SCR A3R is not operating (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • Gate circuit 3R disconnected or open. • 5PL disconnected or damaged. • Failed SCR – check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
001100	FL76	A4ROP	SCR A4F is not operating (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • Gate circuit 4R disconnected or open. • 5PL disconnected or damaged. • Failed SCR – check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
001101	FL77	A5ROP	SCR A5R is not operating (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • Gate circuit 5R disconnected or open. • 5PL disconnected or damaged. • Failed SCR – check and replace if necessary.* • Replace Power Connection card. • Replace Control card.

** (see door label for module location)*

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
001110	FL78	A6ROP	SCR A6R is not operating (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • Gate circuit 6R disconnected or open. • 5PL disconnected or damaged. • Failed SCR – check and replace if necessary.* • Replace Power Connection card. • Replace Control card.
001111	FL79	REVOP	All reverse SCRs are not operating (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • 5PL disconnected or damaged. • ADR002-REGJP not set correctly. • Replace Power Connection card. • Replace Control card.
010000	FL80	SCTFL	SCR short circuit test failed (L,T) • High current test may be enabled (ADR012, 256 bit set to 1) – run low current test (set ADR012, 256 bit to zero). • If torque proving enabled, add 8192 to ADR012-OPTJP. • Check for shorted SCRs – replace.*
010001	FL81	A1FSH	SCR A1F or A4R is shorted (L,T) • High current test may be enabled (ADR012, 256 bit set to 1) – run low current test (set ADR012, 256 bit to zero). • If torque proving enabled, add 8192 to ADR012-OPTJP. • Check for shorted SCRs – replace.*
010010	FL82	A2FSH	SCR A2F or A5R is shorted (L,T) • High current test may be enabled (ADR012, 256 bit set to 1) – run low current test (set ADR012, 256 bit to zero). • If torque proving enabled, add 8192 to ADR012-OPTJP. • Check for shorted SCRs – replace.*
010011	FL83	A3FSH	SCR A3F or A6R is shorted (L,T) • High current test may be enabled (ADR012, 256 bit set to 1) – run low current test (set ADR012, 256 bit to zero). • If torque proving enabled, add 8192 to ADR012-OPTJP. • Check for shorted SCRs – replace.*

** (see door label for module location)*

TABLE 6 - SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
010100	FL84	A4FSH	SCR A4F or A1R is shorted (L,T) · High current test may be enabled (ADR012, 256 bit set to 1) - run low current test (set ADR012, 256 bit to zero). · If torque proving enabled, add 8192 to ADR012-OPTJP. · Check for shorted SCRs - replace.*
010101	FL85	A5FSH	SCR A5F or A2R is shorted (L,T) · High current test may be enabled (ADR012, 256 bit set to 1) - run low current test (set ADR012, 256 bit to zero). · If torque proving enabled, add 8192 to ADR012-OPTJP. · Check for shorted SCRs - replace.*
010110	FL86	A6FSH	SCR A6F or A3R is shorted (L,T) · High current test may be enabled (ADR012, 256 bit set to 1) - run low current test (set ADR012, 256 bit to zero). · If torque proving enabled, add 8192 to ADR012-OPTJP. · Check for shorted SCRs - replace.*
010111	FL87	NOUAP	Microapplication card failure (L,T) · Microapplication card absent. · 3PL incorrectly connected or damaged. · U30 or U33 on Microapplication card failed - replace. · Replace Microapplication card.
011000	FL88	UAPWD	MUP watchdog misoperated (L,T) · U30 on Microapplication card failed or jumpers set incorrectly - replace card if necessary. · Refer to BugBuster, "Noise or Random Nuisance Faults" item. . Check JP47-This should NOT be connected. Discard if necessary.
011001	FL89	MUPTO	RUP to MUP communication failure (L,T) · U30 on Microapplication card failed - replace. · Replace Microapplication card. · Noise
011010	FL90	NOMSG	RUP to MUP watchdog detected no message received within time period expected (L,T) · Serial Link or Local Area Network transmission lines disconnected. · 12PL damaged or disconnected.

* (see door label for module location)

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
011011	FL91	SPARE	
011100	FL92	SPARE	
011101	FL93	SPARE	
011110	FL94	SPARE	
011111	FL95	WATCH	FUP watchdog detected RUP has misoperated (L,T) • Same as FL58.
100000	FL96	PASSED*	Annunciates when SCR test has passed.
100001	FL97	OCTFL*	SCR open circuit test failed (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • DC fuse open. • Generator field circuit open. • 5PL disconnected or damaged. • Main contactor not picking up – check pilot relay operation. • Loss of current feedback – check 1PL-20 and DRPL. • Offset voltage at CFB with zero armature current – adjust P2-CFB ZERO on the Power Supply card. • Power Connection card failed – replace. • Power Supply card failed – replace. • Control card failed – replace.
100010	FL98	OCTMF*	Multiple open SCRs (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • Check all gate plugs. • 5PL disconnected or damaged. • Check SCRs for failure – replace. • Replace Power Connection card. • Replace Control card.

KEY: L - LATCHED N - NON-LATCHED H - REQUIRES HARD RESET
T - TRIPS DRIVE A - ANNUNCIATED ONLY

** Varying inductances in generator field may not allow drive to pass these tests even if SCRs are functioning properly. In this instance, an alternate method of performing these tests is to disconnect the generator field, and substitute a 25 ohm, 100 watt resistor.*

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
100011	FL99	ALLOP*	All SCRs tested open (L,T) • Low current test may be enabled (ADR012, 256 bit set to zero) – run high current test (set ADR012, 256 bit to 1). • DC fuse open. • Generator field circuit open. • 5PL disconnected or damaged. • Main contactor not picking up – check pilot relay operation. • Loss of current feedback – check 1PL-20 and DRPL. • Generator field not connected. • Offset voltage at CFB with zero armature current – adjust P2-CFB ZERO on the Power Supply card. • Power Supply card failed – replace. • Control card failed – replace.
110010	FLB4	MAPNO	Microapplication Software mismatch • Check MUP and RUP software revision levels. • Check EEROM content versus Custom Software Description ADR014-SPCJP, ADR251-UAPJY.. • Voltage loop enabled.
110111	FLB9	IOC	Instantaneous Overcurrent (L, T) Measured armature current is greater than allowed by ADR313-IOCLM. • Check armature circuit and motor armature for short circuits. • Verify setting in ADR313-IOCLM. • Check motor for overload.
111000	FLC0	REGTO	Regulator Timeout (L, T) • Verify parameters in EEROM per the Custom Software Description. • Consult the factory.

KEY: L - LATCHED N - NON-LATCHED H - REQUIRES HARD RESET
T - TRIPS DRIVE A - ANNUNCIATED ONLY

** Varying inductances in generator field may not allow drive to pass these tests even if SCRs are functioning properly. In this instance, an alternate method of performing these tests is to disconnect the generator field, and substitute a 25 ohm, 100 watt resistor.*

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
111001	FLC1	UNDKG	Under KG (N, T) Prime mover speed below programmed limit set by ADR311-KGMIN. • Check speed of prime mover. • Verify setting in ADR311-KGMIN.
111010	FLC2	RVRTO	Reversing Timeout (N, T) • Check ADR314-IFRVR, ADR315-VFRVR and ADR316-ALRVR per Custom Software Description. • Increase AD057-ACCEL and ADR058-DECEL.
111011	FLC3	TOC	Timed Overcurrent Annunciated (N, A) Armature current has exceeded threshold level set by ADR305-TOCRT for the time-current factor set by ADR036-TOCRP while operating at 1% or greater full scale motor speed. • Verify settings of ADR305-TOCRT and ADR306-TOCRP. • Check armature circuit for shorts or faults. • Check motor for faults or overload.
111100	FLC4	STALL	Stalled TOC Trip (L, T) Armature current has exceeded threshold level set by ADR307-TOCST for the time-current factor set by ADR308-TOCSP while operating at less than 1% full scale motor speed. • Verify settings of ADR307-TOCST and ADR308-TOCSP. • Check armature circuit for shorts or faults. • Check motor for faults or overload.
111101	FLC5	TOC	Timed Overcurrent Trip ((L,T) Armature current has exceeded threshold level set by ADR305-TOCRT for the time-current factor set by ADR306-TOCRP while operating at 1% or greater full scale motor speed. • Verify settings of ADR305-TOCRT and ADR306-TOCRP. • Check armature circuit for shorts or faults. • Check motor for faults or overload.

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
111110	FLC6	OVRVG	Generator Overvoltage (L, T) Generator voltage has exceeded level set by ADR310-VGLIM. • Verify setting of ADR310-VGLIM. • Check current and voltage regulator tuning parameters. ADR025-CLPPG, ADR026-CLPIG, ADR027-CLMPG, ADRO28-CLMIG, ADR299-MIAPG, ADR300-MIAIG, ADR301-MSVPG, and ADR302-MSVIG.
111111	FLC7		Application Faults Series of application related faults with last three digits in Programmer fault NAME field indicating specific fault as follows:
		001	Torque Proving Fault (L, T) Insufficient armature current detected. • Check armature current feedback. • Check MD contactor and pilot devices. • Check P1 and P2 settings on DFA card. • Check P4 setting on Microapplication card, should be full clockwise. • Check inner loop current regulator tuning parameters, ADR299-MIAPG and ADR300-MIAIG. • Verify ADR314-IFRVR, ADR315-VFRVR and ADR316-ALRVR per Custom Software Description. • Verify ADR298-DJMJ1 per Custom Software Description.
		002	MD Contactor Open While Brake Released (L, T) • Check MD contactor and pilot devices. • Check interlock contact circuit from MD to DFM-300 controller.

TABLE 6 – SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
		003	RSTOP Fault (L, T) Time for motor to stop exceeds programmed value. • Verify ADR325-RSTPT per Custom Software Description. • Check motor speed feedback circuit. • Check motor for stopping time. • Check if motor driven externally. • Check if testpoint SFB contains an offset – if so adjust P7-SFB ZERO on the Process Interface card. • Check if A/D offset misadjusted – adjust P4-OFFSET on the Control card. • Check jumpers or switches on the Process Interface card. • Pulse rate set incorrectly for digital tach. • Check EEROM parameters per Custom Software Description.
		004	Load Cell Timeout (L, T) Load Limited Speed regulator error. • Verify parameters in EEROM per the Custom Software Description. • Consult the factory. • If LAN used, reduce LAN traffic.
		005	Software Error (L, T) • Consult the factory.
		006	EEROM Capacity Error (L, T) The application requires a "2816" 16K EEROM (2048 x 8 Bit) when only a "2084" 4K EEROM (512 x 8 Bit) is provided on the Control card. • Verify EEROM parameters per Custom Software Description. • Check EEROM on Control card and, if not correct capacity, replace. • Replace Control card.
		007	Armature Loop Open (L, T) • Check armature current feedback. • Check MD contactor and pilot devices. Check P1 and P2 settings on DFA card. • Check P4 setting on Microapplication card, should be full clockwise.

TABLE 6 - SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
		008	Armature Current Offset (L, T) Armature current feedback (IA) offset more than 32%. · Check P1 setting on DFA card. . Check MD CONTACTOR LOGIC. Make sure CIO is NOT open if MD is closed.
		009	Microapplication card not present. (see FL30).
		010	Software Error (L, T) · Verify ADR014-SPCJP per Custom Software Description.

TABLE 7 – SPECIFIC ERROR TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
—	ER01	BUART	Faulty UART on the 8031 U.C. • This is a normal response to Test 50. • Check JP3 and JP4 on Programmer card. • Programmer card failed, replace.
—	ER02	BKPAD	The keypad has shorted leads. • Replace ribbon cable. • A key pressed during power up or reset. • Programmer card failed, replace.
—	ER03	BDMEM	The random access memory is faulty. • Programmer card failed, replace.
—	ER04	BPRGM	The program memory checksum is incorrect. • Programmer card failed, replace.
—	ER05	BTIMR	The timers in 8031 U.C. are faulty. • Programmer card failed, replace.
—	ER06	NADDR	An address is not specified. • "ENTER" key pressed without specifying a parameter address.
—	ER07	2 KEYS	Two keys were pressed together. • Press keys – one at a time. • Keypad failed, replace. • Programmer card failed, replace.
—	ER08	ILBRD	Illegal broadcast. • Programmer card failed, replace.
—	ER11	NOACK	No acknowledgement received. • Device connected to the serial link not responding to the acknowledged requests from the Programmer card. • Wires to the serial link disconnected. • Driver or receiver chip failed on Programmer, Process Interface card, or external device. • Processors locked up, use hard reset. • If Hand-held Programmer, may be failed U6 port – replace. • Noise on transmission wires. • RS232C on hand-held cable.

TABLE 7 – SPECIFIC ERROR TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
—	ER12	N-CTS	No "clear-to-send". • Serial link transmission lines disconnected. • Check JP3 and JP4 on Programmer card. • Noise on 5 volt power supply. • Ground loops or unsuppressed relays. • Programmer card failed, replace.
—	ER13	ORBUF	Receive buffer has overflowed. • Message sent to the drive which exceeds the maximum number of bytes (11 data bytes). • Baud rate set incorrectly. • Byte protocol set incorrectly on external device – refer to Serial Link User's Guide. • Message protocol used incorrectly – refer to Serial Link User's Guide. • Hardware configuration incorrect for serial link operation. • External serial link device failed.
—	ER14	CKSUM	Incorrect checksum received. • The serial link message checksum has been calculated improperly by external Serial Link device – refer to the Serial Link User's Guide. • Baud rates set incorrectly. • Byte protocol set incorrectly on external device - refer to Serial Link User's Guide. • Message protocol used incorrectly – refer to Serial Link User's Guide. • Hardware configuration incorrect for serial link operation. • External serial link device failed.
—	ER15	BDMSG	Something other than a "start of transmission" was received when "S.O.T." was expected. • "Start of transmission" byte of the message sent over the serial link has been determined incorrectly – refer to the Serial Link User's Guide. • Baud rates set incorrectly. • Byte protocol set incorrectly on external device (1 start bit, 8 data bits, and 1 stop bit). • Message protocol used incorrectly – refer to Serial Link User's Guide. • Hardware configuration incorrect for serial link operation. • External serial link device failed.

TABLE 7 – SPECIFIC ERROR TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
—	ER26	N__IFB	Reached 25° retarded firing and no current feedback measured. • DC fuse open. • Generator field loop open. • Check for open SCR or missing gate signals.
—	ER27	N__IFB	Reached 37° retarded firing and no current feedback DC fuse open. • Generator field loop open. • Check for open SCR or missing gate signals.
—	ER28	PRGRM	Parameter values calculated are out of range. • Verify EEROM parameters per Custom Software Description.
—	ER29	TORQP	Torque proving called for by soft jumpers in EEROM. • Verify EEROM parameters per Custom Software Description.

TROUBLESHOOTING "STABILITY" PROBLEMS

There are three categories of drive symptoms broadly classified as "Stability" problems. Technically, two of these are not actually stability problems but, due to their behavior, are included in this troubleshooting procedure. These categories are as follows:

- **"Stability" problems:** "Hunting", oscillations in speed or current, etc. In general all problems in this category are cyclical in nature, with well-defined sinusoidal periods of oscillation. Usually these problems are electrically induced and can be tuned out with proportional and integral gain adjustments.

  There are at least eight regulating loops in most DFM-300 drives, any one of which can be mistuned causing stability problems under the right conditions. They are as follows:

  1. Current regulator
  2. Voltage regulator
  3. Speed regulator – base speed
  4. Speed regulator – weak field speed
  5. CEMF regulator
  6. CEMF limit regulator
  7. Motor field current regulator – base speed
  8. Motor field current regulator – weak field speed

NOTE: *Various forms of position regulators are also available, which, due to their variety are not addressed here.*

- **Mechanical Oscillations** - Problems which are a direct result of mechanical deficiencies but under certain conditions can be aggravated or excited by electrical frequencies. Examples of this are vibration, audible motor noise, gear chatter, etc.
- **"Erratic Operation"** – Intermittent speed or current fluctuations, mechanical "bumping", or other disturbances non-sinusoidal or non-periodic in nature. These may be induced by misfiring SCRs, intermittent grounds, "noise", or mechanical problems.

1. **Before troubleshooting** the drive, first verify the following:

   NOTE: *A programmer is required to make the following parameter checks in this part and adjustments as required in Part II.*

   - All software jumpers, ADR000 through ADR014, ADR251, ADR252, ADR297 and ADR298, are in accordance with drive parameter list.
   - All hardware jumpers are in accordance with the jumper selection chart.
   - All wiring is correct and in agreement with the drive elementaries.
   - All "gain" parameters are in agreement with the parameter list. Specifically, the following parameters should be checked:

NOTE: *The following parameters assume Motor 0 operation. If Motor 1, Motor 2, or Motor 3 stability problems exist, other parameters will affect the speed and current loop.*

Speed Loop:	ADR015-BSLPG
	ADR016-BSLIG
	ADR017-WSLPG
	ADR018-WSLIG
	ADR037-SLEAD
	ADR051-LSPAN
Current Loop:	ADR299-MIAPG
	ADR300-MIAIG
	ADR304-MIAAG (Lead)
Voltage Loop:	ADR301-MSVPG
	ADR302-MSVIG
CEMF Loop:	ADR025-CLPPG
	ADR026-CLPIG
CEMF Limit Loop:	ADR027-CLMPG
	ADR028-CLMIG
Field Current Loop:	ADR031-BFLPG
	ADR032-BFLIG
	ADR033-WFLPG
	ADR034-WFLIG

2. **Next, use Table 8** on the following page to decide which type of problem actually exists, keeping the following pointers in mind.
   - DFM-300 drive current loops are almost always stable.
   - Most stability problems, but not all, are corrected by adjusting speed loop parameters.
   - It is always worthwhile to verify field current stability first.
   - In general, when troubleshooting cascading or series regulating loops, the correct procedure is to stabilize the inner regulator first and then move outward; i.e., current regulator, then speed regulator, then position regulator (if furnished).
   - If at any time the drive becomes stable as a result of a step, record any parameter changes, and do not continue with the remaining procedures.

TABLE 8 – STABILITY TROUBLESHOOTING CHART

SYMPTOM	PROCEDURES
A. Unstable when unloaded but stable when loaded, drive is stable but exhibits gearbox or motor noise or runs erratically.	1. Check for failed SCRs – monitor test point CFB on the Control card with an oscilloscope. 2. "Light-load" instability – increase ADR300-MIAIG. If frequency of instability increases, raise as high as possible without encountering erratic operation. 3. Raise ADR299-MIAPG as high as possible. 4. Verify EE298 Bit 11 is NOT set. 5. If instability still exists, return adjusted parameters to their original setting and skip to Symptom E. 6. Check AN1 (RAM 2152) with Diagram Test 11 to insure current feedback is present.
B. Unstable above and /or below base speed and over a wide speed range (Stabilizing the Field Current Regulator).	1. Confirm Field Current Regulator stability – monitor test point FC on the Control card with an oscilloscope or VOM while in Diagnostic Static (Test 1). If stable, skip to the next appropriate symptom. 2. Field current is unstable – reduce ADR031-BFLPG until FC is stable. Do not reduce below the setting in ADR032-BFLIG. **NOTE:** *ADR031 may already be lower than ADR032 in some applications, maintain this relationship.* Lower ADR032 if necessary. 3. Set ADR033-WFLPG equal to the new value in ADR031-BFLPG. 4. Set ADR034-WFLIG equal to the new value in ADR032-BFLIG. 5. Document any change and proceed to Symptom C or D as necessary.

TABLE 8 – STABILITY TROUBLESHOOTING CHART (continued)

SYMPTOM	PROCEDURES
C. Speed instability below base speed.	1. Confirm that the values in ADR015-BSLPG, ADR016-BSLIG and ADR037-SLEAD agree with the parameter list and satisfy the following equation. BSLIG/BSLPG = .127 (SLEAD) **NOTE:** *If ADR037-SLEAD = 0, skip to Step 4.* 2. If ADR037-SLEAD contains a value (typically "10" or less) readjust BSLPG to satisfy the above equation. DO NOT change SLEAD unless different from the parameter list. 3. Raise or lower ADR016-BSLIG and ADR015-BSLPG maintaining the above ratio. Raising the parameters makes the drive more responsive, lowering makes the drive less responsive. 4. If SLEAD is 0 and the drive is slow and unresponsive – increase ADR015-BSLPG and adjust ADR016-BSLIG upward or downward as necessary. 5. If no improvement, return the adjusted parameters to their original settings. 6. If the drive is a CEMF regulator – lower ADR039-IRCMP to zero. 7. If there is some improvement but instability still exists – document changes and skip to Symptom E.
D. Speed stability problems above base speed.	1. Confirm that the values in ADR017-WSLPG, ADR018-WSLIG and ADR037-SLEAD agree with the parameter list and satisfy the following equation. WSLIG/WSLPG = .127 (SLEAD) **NOTE:** *If ADR037-SLEAD = 0, skip to Step 4.* 2. If ADR037-SLEAD contains a value (typically "10" or less) readjust WSLPG to satisfy the above equation. DO NOT change SLEAD unless different from the parameter list. 3. Raise or lower ADR018-WSLIG and ADR017-WSLPG maintaining the above ratio. Raising the parameters makes the drive more responsive, lowering makes the drive less responsive. 4. SLEAD is 0 and the drive is slow and unresponsive – increase ADR017-WSLPG and adjust ADR018-WSLIG upward or downward as necessary. 5. If no improvement, return the adjusted parameters to their original settings. 6. Drive in CEMF Limit – to confirm, lower ADR079-CROSS (should be approximately 150 to 200 counts lower than ADR080-CMFLM). 7. If there is some improvement but instability still exists – document changes and skip to Symptom E.

TABLE 8 – STABILITY TROUBLESHOOTING CHART (continued)

SYMPTOM	PROCEDURES
E. Drive is unstable: – over a small part of the speed range, **or** – when loaded but stable when unloaded, **or** – such that the oscillating frequency is proportional to speed.	1. Confirm that the problem is not tach related. Correct tach feedback problem (misaligned or loose coupling, belt slippage, wiring or excessive ripple). 2. Verify that the power supplies are clean with an oscilloscope (refer to "Checking Power Supplies" in the Troubleshooting section). 3. Run drive to top speed and push the "RESET" button on the Control card. The drive will **coast** stop. Mechanically induced problems will still be present as the motor decelerates. If problem disappears, continue to next step. 4. Lower settings in ADR015, ADR016, ADR017 and ADR018 as low as possible while maintaining the following ratios: a. $\frac{\text{FINAL ADR016}}{\text{FINAL ADR015}} = \frac{\text{ORIGINAL ADR016}}{\text{ORIGINAL ADR015}}$ b. $\frac{\text{FINAL ADR018}}{\text{FINAL ADR017}} = \frac{\text{ORIGINAL ADR018}}{\text{ORIGINAL ADR017}}$ 5. Reduce ADR037-SLEAD and double ADR015 and ADR016. 6. Determine resonant frequency "Fr" in Hz with an oscilloscope, if possible. Convert to radians per second as follows: "Wr" = 6.28 X "Fr". 7. Adjust ADR037-SLEAD and ADR051-LSPAN a. Set ADR037-SLEAD = .615 X "Wr" b. Set ADR051-LSPAN = 5 and readjust as necessary 8. If no improvement, return the parameters to their original settings and skip to Symptom G.

TABLE 8 – STABILITY TROUBLESHOOTING CHART (continued)

SYMPTOM	PROCEDURES
F. Undirectional Stability Problems	1. Try running the motor in the opposite direction to confirm that stubbing brushes or brush position is not the problem. 2. Inspect the load to make sure that nothing is wrong mechanically. 3. Make certain that all SCRs are firing in both directions by monitoring test point CFB on the Control card with an oscilloscope.
G. Additional Procedures	If the above procedures do not result in stable operation, additional steps may be required including: 1. Checkout of the current loop. 2. Running the Square-Wave Reference Test (Test 25). 3. Running the field regulator open loop to verify stability. If such procedures are required, contact GE DRIVES, Product Service Engineering, for instructions. **(814-875-3851)**

9 – TYPICAL WAVEFORMS

The following are examples of typical waveforms that can be seen on an oscilloscope. Each figure is accompanied by the name of the signal, the test point from which it is derived (in parenthesis), and the proper time setting for the waveform.

(For information on using an oscilloscope, consult the owner's manual.)

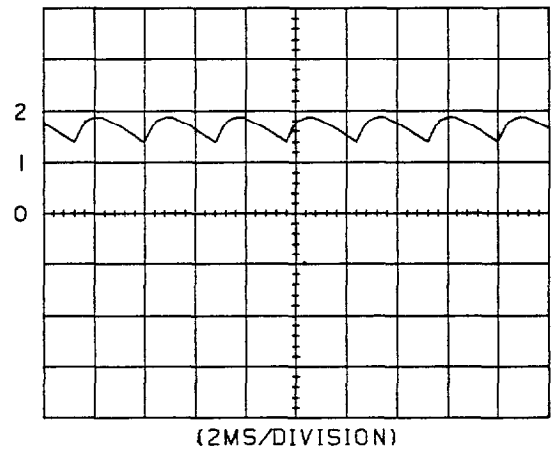


FIGURE 9-1
GENERATOR FIELD CURRENT FEEDBACK (CFB)

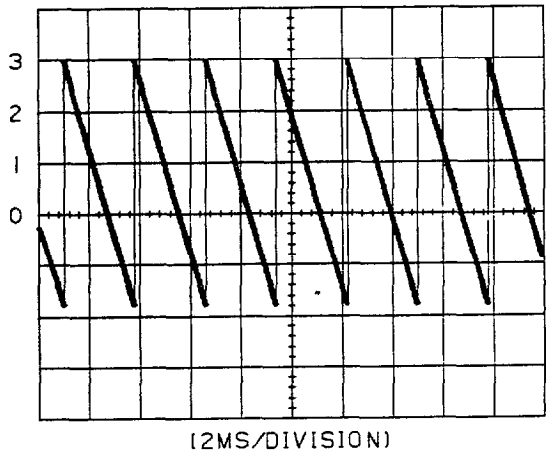


FIGURE 9-2
GENERATOR FIELD VOLTAGE FEEDBACK (VFB)

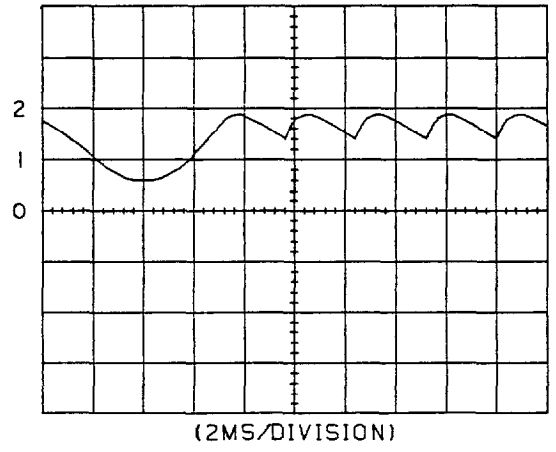


FIGURE 9-3
GENERATOR FIELD CURRENT FEEDBACK (CFB)
WITH ONE SCR OPEN OR DISCONNECTED

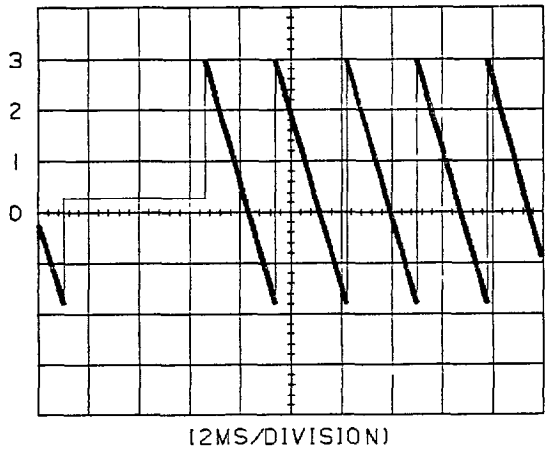


FIGURE 9-4
GENERATOR FIELD VOLTAGE FEEDBACK (VFB)
WITH ONE SCR OPEN OR DISCONNECTED

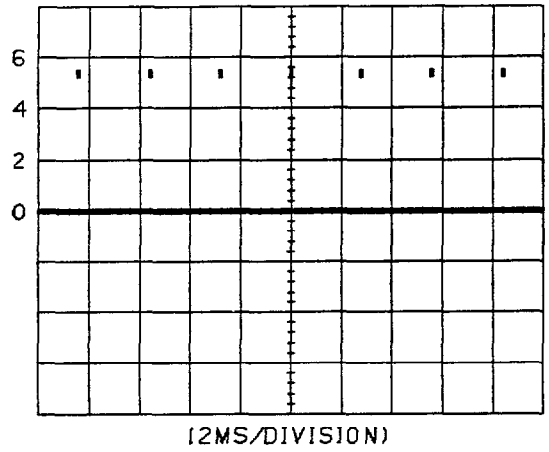


FIGURE 9-5
INITIAL PULSE OF EACH GENERATOR FIELD SCR FIRING PULSE BURST

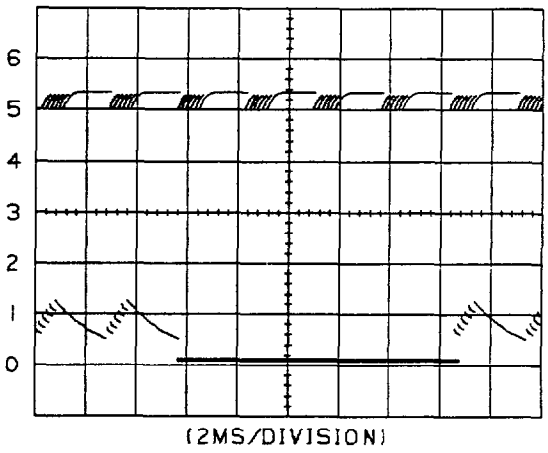


FIGURE 9-6
TOP—DELAYED FIRING POWER (DFP)
BOTTOM—GATE PULSES FOR A1 SCR

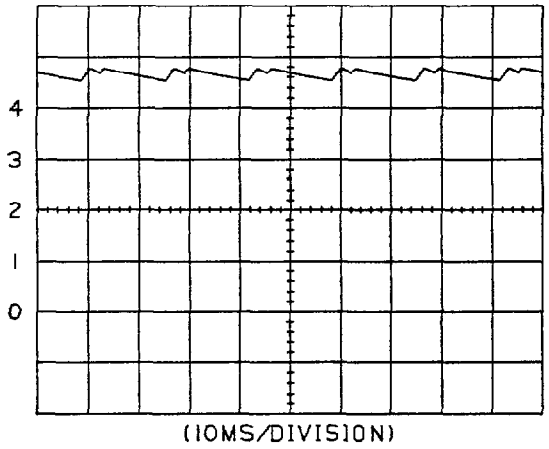


FIGURE 9-7
MOTOR FIELD CURRENT (IFC) MAGNITUDE VARIES WITH FIELD WEAKENING

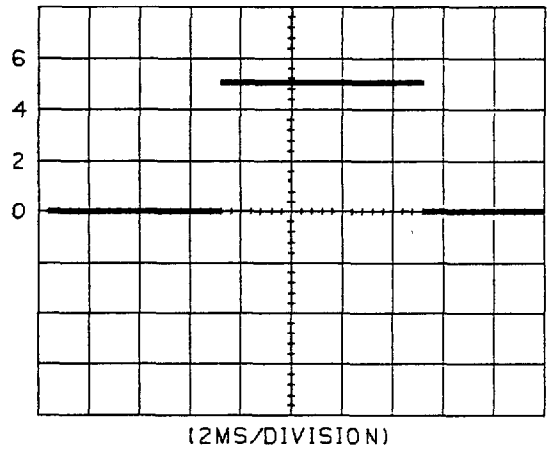


FIGURE 9-8
SYNC SIGNAL (SY) SQUARE WAVE SYNCHRONIZED TO AC LINE FREQUENCY

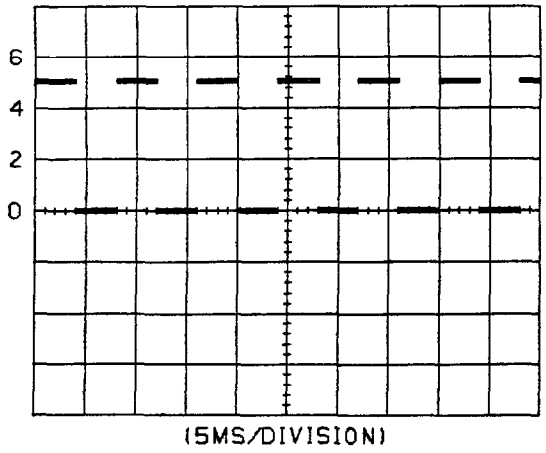


FIGURE 9-9
AC TACH SIGNAL (TPN) AT 450 RPM

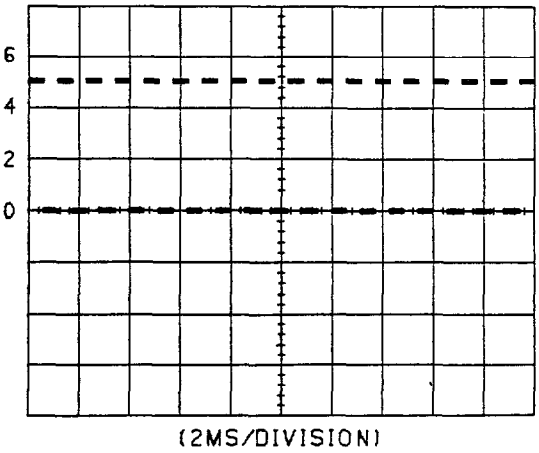


FIGURE 9-10
AC TACH SIGNAL (TPN) AT 3000 RPM

10 – USE OF THE OPTIONAL PROGRAMMER MODULE

The optional Programmer assembly is available in two configurations, as pictured in Figures 4-25 and 4-26. The Local configuration is mounted on the Control card and plugged into 18PL. The Hand-held configuration is contained in a box and plugged directly into 18PL on the Control card.

CAUTION

The programmer cable must be oriented properly or the control and programmer cards will be damaged.

The Hand-held Programmer uses a different cable than the Local Programmer and care must be exercised when the Hand-held Programmer is plugged in.

To properly orient the cable, hold the unit in front of the drive and extend the ribbon cable back to 18PL, so that the cable is not twisted, and plug into 18PL. The red stripe in the ribbon cable should be on the left when looking at the front of the Hand-held Programmer and should be on the left when plugged into 18PL.

The local Programmer cable is also oriented with the red wire on the left side of the cable when the operator is facing the drive.

The Programmer provides an alphanumeric readout of fault codes (instead of the diagnostic LEDs) and operation/adjustment of the drive through three modes – OPERATE, PARAMETER, and DIAGNOSTIC modes.

THREE MODES OF OPERATION

1. **OPERATE** Mode – The "normal" mode, allows the red and black keypad functions to be active (the **RED** keys are dominant). This is the mode the drive will enter when power is applied.
2. **PARAMETER** Mode – Allows the user to examine and change as necessary any of 254 EEROM (E^2) Parameter addresses to "customize" the drive for a given application. In this mode all **BLACK** keys are active plus [INC], [DEC] and [STOP].
3. **DIAGNOSTIC** Mode – Allows the user to utilize any of the internal digital displays, plus run a series of diagnostic checks and self-tuning procedures. In this mode, both the **BLUE** and **BLACK** key functions are active (with **BLUE** keys dominant), plus [INC], [DEC] and [STOP].

CHANGING BETWEEN MODES

NOTE: *Throughout this section, a number or function encased in brackets [ ] indicates a programmer key pressed.*

The drive will normally enter the OPERATE mode as soon as power is applied.

1. To enter the PARAMETER mode, first confirm that the display reads "OPERATE", then press [SET], [DRV#], [7], [7], and [ENTER]. The display should now show "PARAMETER".
2. To enter the DIAGNOSTIC mode, the drive must first be in the PARAMETER mode. If the display shows "PARAMETER", simply press [CLEAR/MODE] and the display will immediately show "DIAGNOSTIC". If a particular parameter is being displayed, press [CLEAR/MODE] causing "PARAMETER" to be displayed then press [CLEAR/MODE] again; "DIAGNOSTIC" will appear.
3. To return to the OPERATE mode, simply press [CLEAR/MODE] once in the DIAGNOSTIC mode, or if the drive is in the PARAMETER mode, press [CLEAR/MODE] either two or three times, as required until the display returns to OPERATE.

NOTE: *The Programmer may appear to "hang up" due to programming errors. In such cases it may be necessary to initiate a "hardware reset" using the RESET switch (SW2) on the Control card.*

WARNING

Potential drive activation and inadvertent equipment movement when testing: Pertaining to the remainder of this section, do not use diagnostic run, square-wave reference test or any programmer "Run" functions on any crane or elevator drive.

OPERATE MODE - (RED Keys dominant)

In the OPERATE mode, any of the **RED** key functions, if activated (if key is not activated, the Programmer will display "FL41 KEYOF" momentarily), can be used as follows:

[RUN] – Run at speed set by RUNRF.

[SLOW] – Run at speed set by SLWRF.

[JOG] – Jog at speed set by JOGRF.

[STOP] – Controlled stop.

[R1] – Run at speed set by R1RF.

[R2] – Run at speed set by R2RF.

[REV] – Reverse RUN or JOG as required by first pressing [REV], then either [RUN], [SLOW], [JOG], [R1], [R2], or [REFR].

[A/M] – Changes drive from AUTO to MANUAL, or Vice-versa, enabling use of either ASR or MSR external speed reference inputs.

[RATIO] – Displays the drive run ratio, or the percentage of speed reference at which the drive will run. Ratio is used to set a draw between multiple drives monitoring the same external speed reference. Normally set at 10,000 indicating ratio of 1:1.

[PRGSV] – For future use.

[REFR] – Run at speed reference at which drive was last run, (includes preprogrammed keypad references or changes caused by using the [INC] and [DEC] keys, MSR or ASR).

[DRV#] – Used to examine the drive number (in most cases "00"). Also part of access code to PARAMETER mode.

[SET] – Used to adjust the settings of various keypad functions; activated by pressing [SET], then [RUN], [SLOW], [JOG], [R1], [R2], or [RATIO], then the desired setting in decimal numbers, then [ENTER]. Also, used for entering Parameter mode. The EEROM must be write "ENABLED" to make these adjustments.

[INC] & [DEC] – Increases or decreases the speed of any of the keypad Run or Jog functions, or the Ratio function; simply enter the desired function and then press [INC] or [DEC] as desired.

[RESET] – Resets the drive after fault or error messages appear. The reset key is referred to as a "Software Reset".

The normal display in the OPERATE Mode will be "M XX% YY%" where "M" indicates Manual mode using the Manual System Reference, MSR (alternatively an "A" will indicate Auto mode using the Auto System Reference, ASR on the optional Microapplication card), XX% indicates percent of rated speed, and YY% indicates percent of rated load.

PARAMETER MODE
(Black keys, plus INC, DEC & STOP active)

1. To enter the PARAMETER mode, press [SET], [DRV#], [7], [7], and [ENTER].
2. To **examine** the present value of any parameter simply key in any three-digit parameter address (do not press [ENTER]). The name for that address will immediately be displayed along with the present contents of that address. Press [CLEAR/MODE] to clear present parameter (displays "PARAMETER") then the address number for the next parameter desired.

   Occasionally, it may become necessary to check a series of parameters. Simply call up the lowest parameter address and press [NEXT] as often as required until the highest parameter number is reached.
3. To change parameters, two conditions must be present:

   - The EEROM (E^2) enable device must be in the "ENABLE" position.

     JP13: 1-2; position; E^2 write "PROTECT"
     2-3; E^2 write "ENABLE"

CAUTION

Do not apply or remove power from the drive or perform a "Hardware Reset" with the E^2 enable device in the "ENABLE" position.

   - Parameter address #3 (designated as ADR003-DGNJP) must contain an **odd** number (1, 3, 5, etc.). If an even number exists, add "1" to the present value and enter the new number.

     Example: Examine and change the drive acceleration and deceleration rates. **Assume** the drive acceleration rate is set at 2 seconds and the deceleration rate is set at 10 seconds. Change the acceleration rate (ADR057-ACELT) from 2 seconds to 7.5 seconds. Reduce the drive deceleration rate (ADR058-DCELT) from 10 to 5.2 seconds.

   a. Confirm that the E^2 enable device is in the "ENABLE" position and that ADR003-DGNJP contains an odd number.

   b. After examining ADR003, press [CLEAR/MODE] the [0], [5], [7]. The display shows "ACELT. 20", corresponding to a 2-second acceleration rate.

   c. Either press [INC] until the display value increases to "75" or press [7], [5]. In either case the display should now show "ACELT. 75".

   d. Press [ENTER], placing the new value into EEROM memory.

   e. Press [NEXT], commanding the drive to go to the next parameter. ADR058-DCELT; the display shows "DCELT. 100", corresponding to a 10-second deceleration rate.

   f. Press [DEC], until the display shows "DCELT. 52" or press [5], [2].

   g. Press [ENTER] placing the new value into memory.

   h. Press [CLEAR/MODE]; the display will now show "PARAMETER".

   i. To examine other addresses simply key in the appropriate three-digit parameter address number.

NOTE: *This next step is **extremely important** to prevent accidental reprogramming of the drive.*

   j. To **seal** the new parameter values, **remove** the "1" from ADR003-DGNJP returning it to the "as shipped" **even** number, and return E^2 enable device to the "PROTECT" position.

4. As in the OPERATE mode, the display may return to "'M.XX%YY%" after a period of inactivity, but the drive is still in the PARAMETER mode (indicated by the apostrophe before the M) awaiting further keypad entries. Simply enter the next parameter address to be examined.
5. To **exit** the PARAMETER mode, press [CLEAR/MODE] either once or twice to go to the DIAGNOSTIC mode. To re-enter the OPERATE mode, press [CLEAR/MODE] again.

DIAGNOSTIC MODE

The Diagnostic Tests listed below will aid in the start-up, tune-up and troubleshooting of the DFM-300 drive. To use these diagnostic features either a Local or Hand-held Programmer is required. Availability of certain tests may be dependent on hardware or software configurations provided.

WARNING

Potential drive activation and inadvertent equipment movement when testing: Do not use diagnostic run, or any programmer run functions on any crane or elevator applications.

STANDARD DIAGNOSTIC TESTS:

Test 1	[DGS] – Diagnostic Static.
Test 2	[DGR] – Not used on DFM-300 controller.
Test 3	[DVM] – Displays Digital Voltmeter.
Test 4	[ARM V] – Displays armature voltage.
Test 5	[ARM I] – Displays armature current in amps.
Test 6	[FIELD I] – Displays motor field current in amps.
Test 7	[RPM] – Displays motor speed.
Test 8	[TORQ] – Displays motor torque in 1 lb-ft.
Test 9	[HP] – Displays motor horsepower.
Test 10	Displays software revision of Control card microprocessors.
Test 11	Displays diagnostic variables (selectable).
Test 51	Revision date of Programmer software.
Test 52	Revision level of Programmer software.
RAM TEST	Displays single diagnostic variable (selectable).
DAC0, DAC8 & DAC12	Digital to Analog converter outputs.
BITS 1, 2 & 3	Status bit outputs.

ADVANCED DIAGNOSTIC TESTS:

Test 20	Electronic Wire-Check.
Test 21	Diagnostic Variable Pairs.
Test 22	Diagnostic Variables through DAC Outputs.
Test 25	Square-Wave Reference.

STANDARD DIAGNOSTIC TESTS

NOTE: *For Tests 1 through 9 simply enter the DIAGNOSTIC mode and press the appropriate key. To enter all other diagnostic tests, press [TEST], [X], where "X" is the number of the particular test to be performed, and then press [ENTER].*

Test 1 – Diagnostic Static (DGS)

Diagnostic Static (DGS) allows observation and adjustment of the reference without actually running the drive. Reference is selected by ADR003-DGNJP and may set either MSR, the Programmer INCREASE/DECREASE keys or the internal programmed references, ADR092 and ADR093 (TSTR1 and TSTR2) as the reference. Refer to the Custom Software Description to determine which reference is used.

Diagnostic Static is initiated from a stopped condition. When initiated, the drive goes from field economy set by ADR050-FLDEC to full motor field current set by ADR076-FLDMX. Diagnostic Static also fires the gate pulse transformers to the power conversion bridge without energizing the MA contactor and running the motor. This allows adjustment of the motor field settings and motor field regulator gain as well as checking the pulse transformer outputs on the Power Connection card.

Pressing [STOP] discontinues Diagnostic Static.

Test 3 – DVM Mode (Requires Optional Process Interface Card) (Also see Diagnostic Mode for Drive Setup)

Test 3 provides the capability of utilizing the Local or Hand-held Programmer and Process Interface card as a digital voltmeter (DVM) for checking various analog test points (TP) on the drive cards. DC voltage in the +/-40 volt range can be measured. This requires a test probe wire (supplied in the miscellaneous parts packet shipped with the drive) from test point DVM on the Process Interface card to the particular test point to be monitored. The DVM may be calibrated by monitoring the +5 volt test point and adjusting P3-DVM TRIM on the Process Interface card.

Pressing [TEST] discontinues DVM mode and displays "CLR TEST".

Tests 4 thru 9 Various Displays (ARM V, ARM I, FIELD I, SPEED, TORQ, HP)

Tests 4 thru 9 are accomplished by setting the drive in DIAGNOSTIC mode and then entering the desired test number. To change to a different test simply push another number from 4 thru 9. To cancel the last test push [TEST] once, the Programmer will display "CLR TEST".

Test 10 – Software Revision

Test 10 will display the revision level of the microprocessors on the Control card. For example, "R20F20T20" says the regulator microprocessor (RUP) is version 2.0 software, the firing microprocessor (FUP) is version 2.0 and the tachometer microprocessor (TUP) is version 2.0. This can be verified by comparing the information with white labels on the individual microprocessor chips on the Control card.

Test 11 – Diagnostic Variable Display (also see Tests 21 and 22)

Test 11 can be used to display, in hexadecimal numbers, certain drive variables for the purpose of comparison. For example, speed reference and speed feedback may be compared. To accomplish this, several things must be done. First, place the drive in PARAMETER mode as described previously. Next determine, from the list which follows, the RAM address of the regulator variables to be monitored. Then put the first variable RAM address number into parameter ADR122-DIAG1 and the second variable RAM address into parameter ADR123-DIAG2. The last step is to put the drive in DIAGNOSTIC mode and select Test 11 as described above. The programmer will then display "DGXXXX. YYYY", where "XXXX" is the hexadecimal value of the RAM address entered in ADR122, and "YYYY" is the hexadecimal value of the RAM address entered in ADR123.

NOTE: *The value displayed is in 2's complement. Values from 0000 to 7FFF are positive magnitude (0 to + 32767). Values from FFFF to 8000 are negative magnitude (FFFF = -1, 8000 = -32768).*

RAM ADDRESSES FOR "ANALOG" SIGNALS

RAM ADR	DESCRIPTION
053	Speed Loop Output
055	Generator Field Current Feedback
061	Current Loop Output
065	Timed Speed Reference
069	Jog Reference
071	Speed Feedback
103	Generator Field Voltage Feedback
105	CEMF
463	Motor Field Current
494	ACOM
501	TOC Timer
2164	Motor Terminal Voltage
2259	Generator Terminal Voltage
2265	Armature Current Feedback

NOTE: *These addresses will contain a variable number, representing an "analog" signal. Refer to "RAM Variable Scaling" Table 4 in the Adjustment Section for the relative hexadecimal values of the displayed variables.*

RAM ADDRESSES FOR LOGIC SIGNALS

RAM ADR	DESCRIPTION
137	Fault
140	Current Limit Flag
141	CEMF Limit Flag
148	Reference Enable
152	Precondition
156	Zero Current Flag
2279	MD Permissive

NOTE: *This list contains addresses of signals which will be either "high" (display "8080") or "low" (display "0000"). If the signal is high, then the statement is true. For example, if "137" is in ADR122, Test 11 is initiated, and a fault condition exists, then the display will show "DG8080.YYYY", (YYYY value will depend on the number in parameter ADR123). If the drive is not faulted then the display will be "DG0000.YYYY".*

Test 51 –
Programmer Card Software Revision Date

Test 51 allows a check of the software **date** of the Programmer microprocessor (PUP).

Test 52 – Programmer Card Software Revision

Test 52 provides a check on the software **level** of the "PUP".

RAM Test – Displays RAM Location Value

RAM test allows a "one-shot" look at a RAM address (refer to Test 11 for variables). The value displayed is in decimal numbers. To display a RAM address, press [RAM], the Programmer will display "***RAM***". Then press the keys corresponding to the desired RAM variable. The drive will display the following: "XXX YYYY", where "XXX" is the RAM address and "YYYY" is the RAM value. Press [CLEAR/MODE] to clear the display.

DAC0, DAC8 and DAC12
(DAC8 and DAC12 Require an Optional Application card)

WARNING

Potential drive activation and inadvertent equipment movement when testing: These outputs may already be utilized by the drive. Refer to the drive elementaries and custom software description.

Three selectable analog outputs, DAC0, DAC8 and DAC12, are available for test purposes and are programmed similar to Test 11. To program DAC0, enter the Parameter mode and call up ADR090-DAC0. Enter the signal to be monitored from Table 4, "RAM Variable Scaling". Putting that RAM Address in ADR090 will cause that signal to appear at the DAC0 test point. DAC8 and DAC12 perform in a similar manner, but the signal RAM address is loaded into ADR096-DAC8 or ADR097-DAC12 and the output appears at the DAC8 or DAC12 terminal on 4TB.

Status Bits 1, 2 and 3
(Requires Optional Application card)

WARNING

Potential drive activation and inadvertent equipment movement when testing: These outputs may already be utilized by the drive. Refer to the drive elementaries and custom software description.

ADR098-BIT#1, ADR099-BIT#2 and ADR100-BIT#3 are programmed in the same manner as DAC8 and DAC12. They are different in that they are logic outputs rather than analog outputs.

Signals listed in Test 11, Logic Signals, can be used to select addresses to enter in ADR098, ADR099 and ADR100. Variables listed under Test 11, "Analog Signals", may also be used in the bit outputs to indicate the polarity of the signal (positive or negative). These outputs also appear on 4TB as "B01", "B02", and "B03" respectively.

ADVANCED DIAGNOSTIC TESTS

Test 20 – Electronic Wire-Check

WARNING

Potential drive activation and inadvertent equipment movement when testing: These outputs may already be utilized by the drive. Refer to the drive elementaries and custom software description.

WARNING

Potential drive activation and inadvertent equipment movement when testing: Signal level detectors and bit outputs may control external devices. Consult the system elementaries before using this feature.

Test 20 allows five different types of wire-checking and output forcing test as follows:

1. Signal Level Detector (SLD) forcing.
2. Status Bit monitoring.
3. Fault Code testing.
4. Logic input monitoring (Microapplication card).
5. Logic input monitoring (Control card).
6. Microapplication card input monitoring (future).

To perform all tests, it is first necessary to be in Diagnostic mode with the EEROM write enabled (write protect the EEROM after this test is completed). Then key in [TEST], [2], [0], and [ENTER]. This causes Test 20 to be active and the display to show "'1.'.00111111". The left-most digit indicates the test level in this case, the '1.' indicates level 1 or Signal Level Detector (SLD) forcing.

To change the type of test, press the [INC] or [DEC] key until the correct test number is displayed in the left-most digit. For example, if Test 20 is called for and the initial display is shown above, press [INC] once to run the next level (20-2), Status Bit monitoring. The programmer display will be similar to "'2.'.11111100". Each test will be described below.

NOTE: *Test 20-1 temporarily changes the value of Parameter ADR009-SLDJP. If the AC control power is removed while keys 0 thru 5 are depressed, ADR009-SLDJP may be changed and will not contain the correct value when AC power is reapplied. The value of SLDJP should be noted prior to initiating this test.*

Test 20-1 – Signal Level Detector (SLD) Forcing (Requires Microapplication card)

This display will show "'1.'.00111111". SLD0 is energized if the right-most digit is 1. and dropped out if it is a zero. Scanning from right to left, the status of all six SLDs (SLD0 through SLD5) can be seen. That is, the second digit from the right is SLD1, the third digit from the right is SLD2, etc. To change the state of SLD0, depress [0] key.

NOTE: *SLDs cannot be toggled while the drive is running or the drive will fault. While the [0] key is depressed, SLD0 will be maintained in the opposite state it was originally. When the [0] key is released, SLD0 returns to its original state. Keys 1-5 operate in the same manner for SLD1 thru SLD5.*

Test 20-2 – Status Bit Outputs (Microapplication card)

Status bits B01, B02 and B03 (ADR098, ADR099 and ADR100 respectively) are programmed as previously described under "STANDARD DIAGNOSTIC TESTS". If test 20-2 is called for, B01 will appear as the right-most character in the display, B02 will be one to the left of B01, and B03 is one to the left of B02. If Bit#1 is "high" the right-most character will be "1"; if it is "low", it will be displayed as [0].

Test 20-3 – Fault Code Outputs (Microapplication card)

To test remote fault code lines and their effects on external equipment, place the drive in Test 20-3. Next enter the desired fault code by keying in [0]. The programmer then displays "FAULT". Next key in number of fault code desired and press [ENTER].

NOTE: *Fault 63 will cause FC5 to be high.*

This test does **not** cause the drive to fault, unless a number greater than 127 is entered. This test is a way of forcing the fault code output lines on the Microapplication card to check interfaces with external equipment using the fault output lines. The Diagnostic LEDs will also display the appropriate fault code.

Test 20-4 – Logic Input Monitoring (Microapplication card)

If Test 20-4 is selected the display will show "'4.'.XXXXXX" where Xs will be either "1" or "0". A "1" will be displayed if a particular input is tied to REF24 and the drive is programmed for **negative** logic (JP19 and JP20 on the Control card are in 1-2 and 3-4 position respectively; and JP5 and JP6 on the Microapplication card are in the 1-2 and 3-4 position respectively). From right to left the digits are MODE 0 thru MODE 7.

Test 20-5 – Logic Input Monitoring (Control card)

Test 20-5 is similar to Test 20-4 except that the left four digits do not apply. Starting at the first digit from the right, the status of four logic inputs will be displayed as:

"5.XXXXDCBA".

A = Status of RUN command inverted ("0" indicates RUN is initiated).

B = Status of JOG command inverted ("0" indicates JOG is initiated).

C = Status of POL command inverted ("0" indicates POL is initiated).

D = Status of XSTOP command ("1" indicates XSTOP initiated).

Test 20-6 –
Microapplication Card Input Monitoring (Microapplication card required)

Test 20-6 is reserved for future use.

Test 21 – Diagnostic Variable Pairs

NOTE: *Test 21 will cause ADR122-DIAG1 and ADR123-DIAG2 to change.*

Test 21 performs in a similar manner to Test 11 discussed under "STANDARD DIAGNOSTIC TESTS", except that several pairs of RAM addresses are already pre-programmed in memory. Test 21 is entered from the Diagnostic mode by keying in [TEST], [2], [1], and [ENTER]. The programmer will then display "TST21". Next key in [0], [1], [2], [3], or [1], [0] (corresponding to Tests 21-0, 21-1, 21-2, 21-3, or 21-10 respectively), and [ENTER], to display the pairs of RAM addresses shown in the table below.

As in Test 11, the display will show "DGXXXX.XXXX". The left and right four characters are as selected from the listing. Test 21-3 will display the variables programmed into ADR096-DAC8 and ADR097-DAC12.

If [1], [0] is entered, calling for Test 21-10, then the display will prompt with "ADDR1". At this time, key in the first RAM address you wish to monitor and press [ENTER] (typical RAM addresses are listed under Test 11). The display will then prompt with "ADDR2". Key in the second RAM address and press [ENTER]. The display will now be DGXXXX.XXXX", where the left four digits are the value of the signal address stored in ADDR2.

Test 22 –
Diagnostic Variables Through DAC Outputs

Test 22 operates similarly to Test 21, with the addition that the various signal pairs are automatically sent to the DAC8 and DAC12 terminals on 4TB. Calling for Test 22-3 causes +5V to appear at the DAC8 and DAC12 terminals in order to quickly check the integrity of DAC8 and DAC12.

NOTE: *Test 22 temporarily changes the value of ADR096-DAC8, ADR097-DAC12, ADR122-DIAG1 and ADR123-DIAG2. If AC control power is removed while conducting Test 22 then DAC8, DAC12, DIAG1 and DIAG2 will not contain the original value when power is reapplied. The value of DAC8, DAC12, DIAG1 and DIAG2 should be noted prior to conducting this test.*

RAM VARIABLE PAIRS FOR TESTS 21 AND 22

TEST NUMBER	LEFT 4 DIGITS (RAM-Addr)		RIGHT 4 DIGITS (RAM-Addr)	
0	Speed loop output	(053)	Current loop output	(061)
1	Timed Speed Ref.	(065)	Speed feedback	(071)
2	Position Error	(471)	In position flag	(154)
3	DAC8	(ADR096)	DAC12	(ADR097)
10	Diag. Var. #1	(ADR122)	Diag. Var. #2	(ADR123)

Test 25 - Square Wave Test Reference

WARNING

Potential drive activation and inadvertent equipment movement when testing: Do not use the square wave test reference on any crane or elevator applications.

Test 25 causes the drive to run and follow an internal drive square wave reference set up by the values in ADR092-TSTR1, ADR093-TSTR2, and ADR094-TSTT1. TSTR1 and TSTR2 are run reference levels and TSTT1 is the time duration of each half cycle. Run references are 0 to 127 with 100 equal to 100% speed with positive reference. Negative reference is accomplished with values from 128 to 255 with the magnitude equal to 256 minus the setting (for example, 156 corresponds to - 100%). The time period is set with TSTT1 and will have a 50% duty cycle for powers of 2 (1, 2, 4, 8, 16, etc.). The cycle time will be 262* TSTT1 milliseconds. That is, with TSTT1 set to 8, the test will cycle between TSTR1 and TSTR2 with approximately 1 sec for each reference. Pressing [STOP] will end the test.

NOTE: *Test 25 temporarily changes the value of ADR003-DGNJP. If AC control power is removed while conducting this test then DGNJP will not contain the correct value when AC power is reapplied. The value of DGNJP should be noted prior to conducting this test.*

Example:

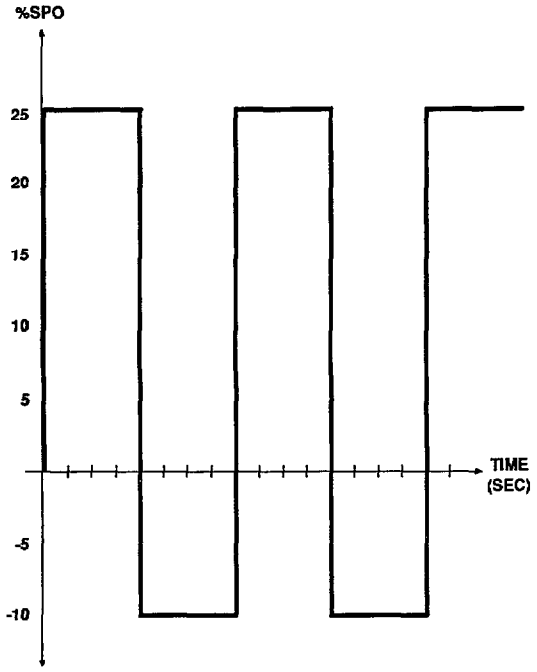
Program the square wave reference to run 25% of top speed forward for approximately 4 seconds and 10% reverse for approximately 4 seconds.

ADR092-TSTR1 = 25 (25% speed)

ADR093-TSTR2 = 246 (-10% speed)

ADR094-TSTT1 = 4000 ms/262 ≈ 16

The reference will look like the sketch below:



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11 – PARTS REPLACEMENT

REPLACEMENT OF SCR MODULES
(Figures 11-1 and 11-2)

1. Lower the card carriers to expose the rear of the unit.
2. Disconnect the wires from the damaged module and remove the bus bar and mounting screws.
3. Clean the mounting surface of old lubricant and dirt.
4. Apply a thin film of General Electric GL322 Versilube® Plus (Part No. 104X214AB059) to both contacting surfaces, install the module and tighten screws.
5. Reconnect wires and bus as before.

CARD REPLACEMENT PROCEDURE
(Figures 11-3 and 11-4)

CAUTION

Treat all cards with static sensitive handling techniques. Use "Grounding" strap when changing cards and store cards in the anti-static bags they are shipped in.

1. Carefully disconnect all cables.
2. Release all hold-down tabs and remove board.
3. Set all jumpers on the replacement board exactly as the jumpers on the failed board.
4. Install the new board verifying that all tabs snap tightly into position.
5. Reconnect all cables verifying that they are properly seated at both ends.
6. Set all pots in the same physical position as the pots on the failed card and fine tune as required per instructions in Table 1 of the Adjustments section.
7. If the Control card is changed, remove the EEROM from the failed card and install in the new card. If the EEROM from the failed card is suspected to be damaged, it will be necessary to reprogram a new EEROM to the custom parameter list. See Figure 11-4 for typical EEROM removal tool and grounding techniques.

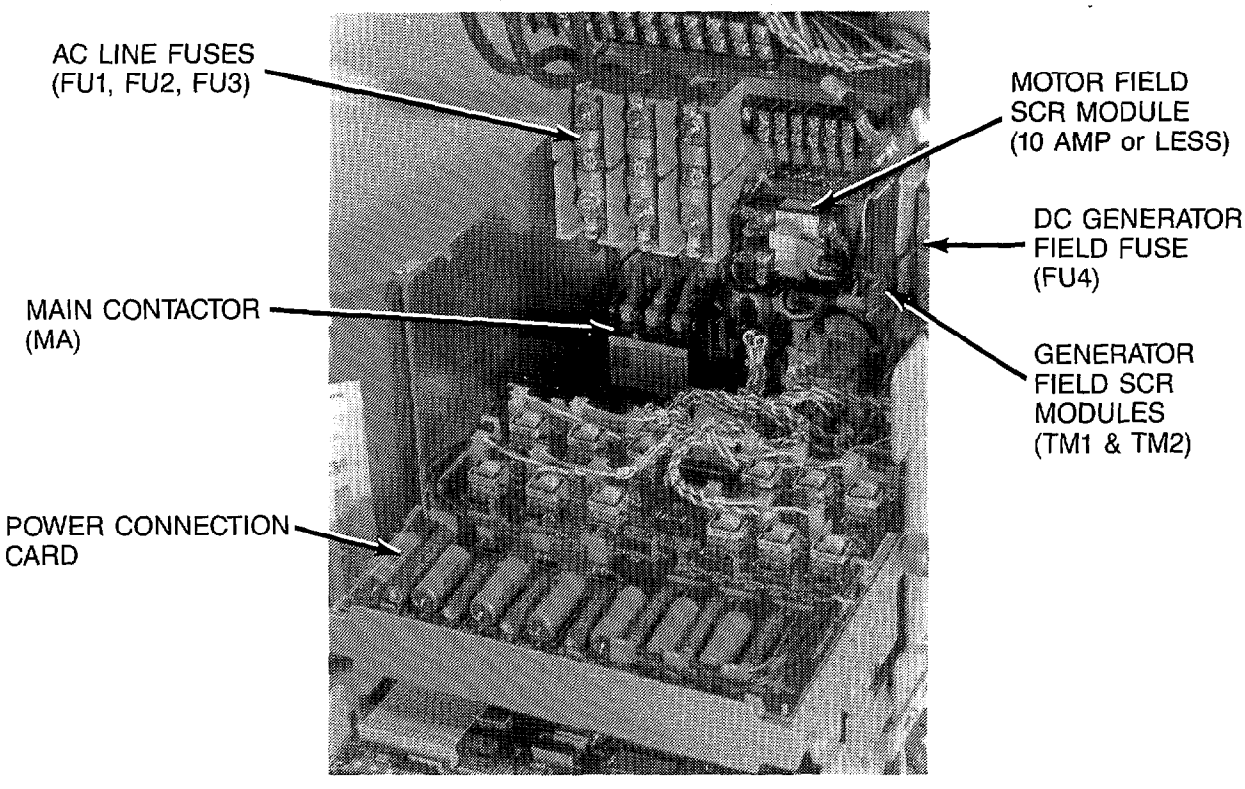


FIGURE 11-1
SCR MODULE LOCATION

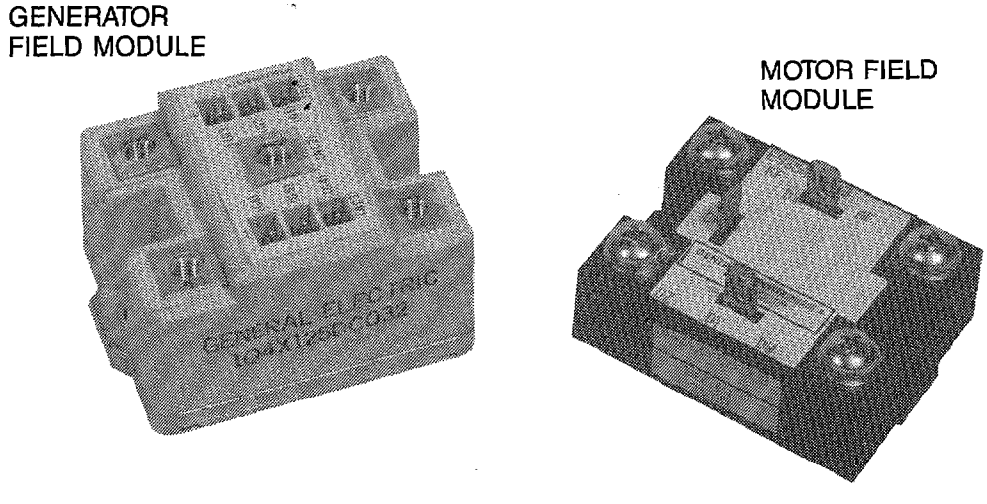


FIGURE 11-2
TYPICAL SCR MODULES

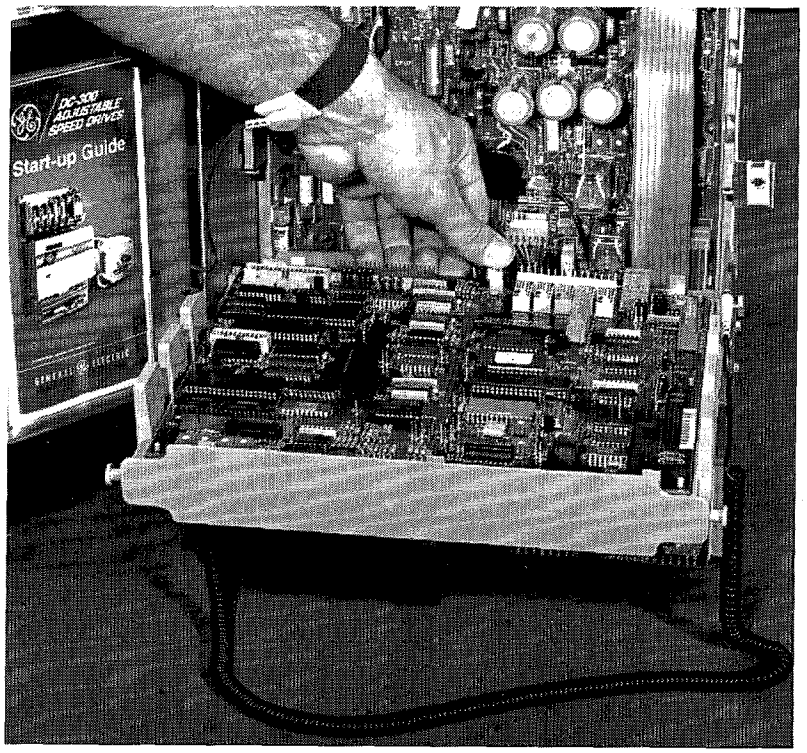


FIGURE 11-3
SIX CARD RELEASE TABS
PUSH TABS AWAY FROM CARD TO REMOVE

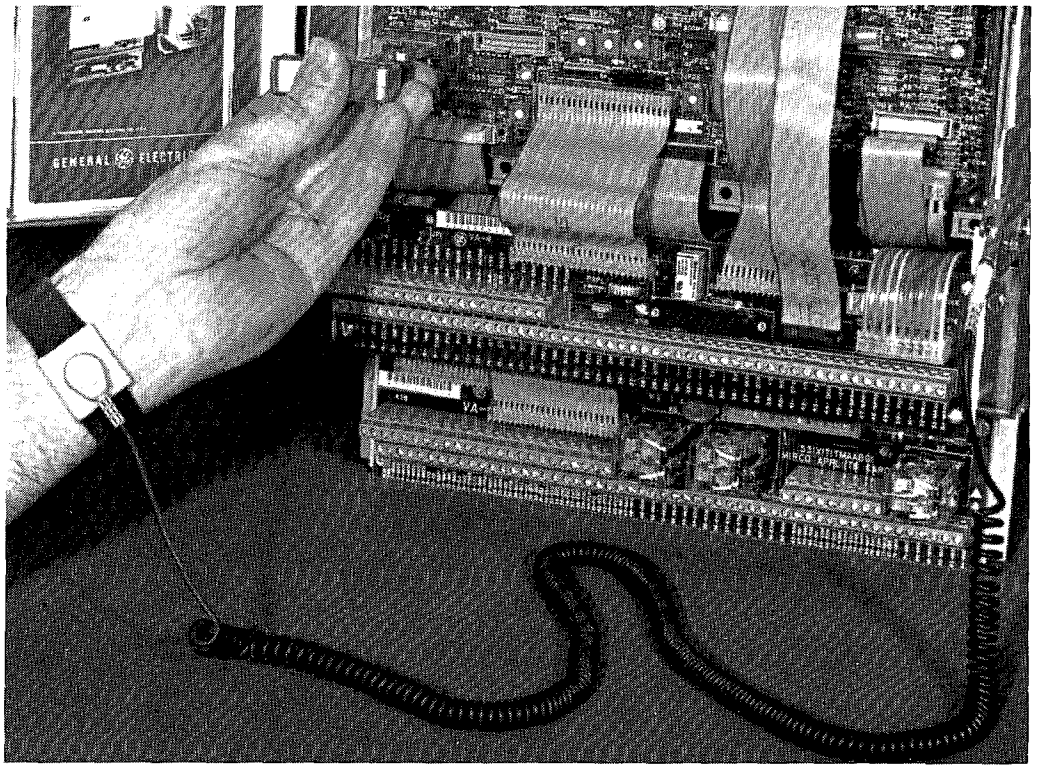


FIGURE 11-4
TYPICAL EEROM REMOVAL AND/OR REPLACEMENT EQUIPMENT
EXERCISE CARE WHEN REMOVING OR INSERTING EEROM
TO PREVENT BENDING PINS

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12 – SPARE AND RENEWAL PARTS

Renewal parts information for this equipment is included in the **CUSTOM INSTRUCTION BOOK** provided for the drive. The renewal parts list is generated specifically for the equipment furnished on each order and provides information on parts relative to the particular rating and options incorporated in the drive at the time of shipment. If the renewal parts information is missing, a replacement list may be obtained by contacting your local **GE Sales Office** or **GE DRIVES Authorized Distributor** with the drive **MODEL and REQUISITION numbers.** Parts lists may also be obtained direct from **GE DRIVES Technical Services, (814) 875-2075.**

It is recommended that the parts list be reviewed and a set of spare parts be maintained to minimize down time should repairs be required. Renewal parts may be ordered by contacting your local **GE Sales Office** or an **Authorized GE Parts Distributor in your area.**

PART NUMBER IDENTIFICATION

The parts listed are divided into four categories, each with its own part number structure:

Order-Specific Assemblies
Common Assemblies
Component Parts
Circuit Cards

Order-Specific Assemblies are generally major assemblies consisting of a group of common assemblies that make up the specific controller provided. Other items which are obtained specifically for the order may also be provided with a similar part number structure.

Example:

7VXYZ999CD01

7V	– assembly provided by GE DRIVES DEPARTMENT
XYZ999	– identification number (original order)
CD	– assembly type (CD = Core Drive)
01	– unit identification number

Common Assemblies are subassemblies, consisting of a group of component parts, which are in common use in GE DRIVES products and are not specific to a particular order. The part number consists of the number "36" followed by an alpha character and may be up to 14 characters in total length.

Example:

36C77345AAG01 is the part number for an external Motor Field Control Module for a DFM-300 controller.

Component Parts are the basic components used in the assemblies and generally represent the lowest discrete level. Component part numbers consist of a combination of alphanumeric characters, defining the class of component and the specific item.

Example:

104X109AD038

104X	– component part
109	– component type (fuse)
AD038	– specific item (500 V, 3.2A FNQ)

Circuit Cards are identified by a catalog number giving the product class, card function and revision levels. It should be noted that some cards contain software which is designated by the suffix "Mn" where "n" is the software number. These cards are also available less software with suffix "Gn" designation. The factory may substitute later versions of cards based on availability and design enhancements.

Example:

531X300CCHADM3

531X	– card for GE DRIVES PRODUCTS OPERATION microprocessor based controller
300CCH	– Control card for DFM-300 controller
ADM3	– revision level, including software

As an aid in expediting card identification, a partial listing of the cards, functions and associated root part numbers follows:

SPARE AND RENEWAL PARTS LIST

PRINTED CIRCUIT BOARDS	CATALOG NUMBER
Control card (with microprocessors) (1)	531X300CCHA__M3
Microapplication card (1)	531X139APMA__M4__M7
Power Connection card (4 Quadrant)	531X121PCRA__G3
Power Supply card	531X111PSHA__G1/G2 (with external field control)
Digital Field Auxiliary card	531X196DFAA__G1
	531X196DFAB__G1
PRINTED CIRCUIT CARD – OPTIONAL	
Programmer card	531X135PRGA__M1
Process Interface card	531X133PRUA__G1
Terminal Board card – 3TB	531X170TBSA__G1
Terminal Board Relay card – 4TB (1)	531X137TRGAAG1
Microapplication TB card – 4TB (1)	531X171TMAA__G1
Programmer card Assembly	36B605548A__G01
Hand-held Programmer Assembly	6VHHP10A1
SUB ASSEMBLIES AND COMPONENTS	
Motor Field Module	104X125DC012
Motor Field Control Module (24 Amp)	36C774345AAG01
K6, K7, K8 Relays (4TB) (2)	104X166AA077
Thermoswitch (optional)	104X152AC067
Control Power Transformer (230 VAC)	36B605573AEG01
Control Power Transformer (460 VAC)	36B605573AEG02
Suppressor	36A358207AAG02
Fan (small – 4-1/2 inch)	14X215CA003
Current Transformer Assembly	36B605537ACG01
Fuses (Control Power) 3 amps (CFU6)	104X109AD004
1.5 amps (CFU7)	104X109AD088
0.5 amps (CFU8)	104X109AD009
3.2 amps (CFU4 & 5)	104X109AD038
15 amps (CFU1, 2 & 3)	104X109AD076
AC Power Fuse	104X109AG006
DC Power Fuse	104X109AG007
Generator Field SCR Modules	104X125DC032
MA Contactor	104X105CB003
MA Aux Interlock	104X105CB007

NOTE:

1. *Check old part to verify catalog number before ordering renewal part.*
2. *Once relay contacts are used on high current or voltage signals, they cannot be used for low level signals.*

INTERCONNECTION CABLES

CONNECTOR	CATALOG NUMBER
1PL	36A358208AAG02
2PL	104X154CB111
3PL	36A358208AAG04
4PL-3	36A358208AAG07
5PL	36A358208AAG01
6PL	36A358208AAG08
7PL	36A358208AAG05
8PL-10	36A358208ABG01
9PL	36A358208CXG01
10PL-3	36A358208AAG09
11PL	36A358208AAG11
12PL	36A358208AAG10
13PL	36A358208AAG12
14PL	36A358208EYG01
16PL (6 foot)	36A358208AAG18
16PL (8 foot)	36A358208AAG19
16PL (10 foot)	36A358208ARG01
16PL (15 foot)	36A358208ARG02
17PL	36A358208AAG14
18PL	36A358208ADG01

NOTE: *Connector used may vary with options provided.*

NOTE: *16 PL connects the Microapplication card to a discrete external connection device to provide a 16 bit parallel input from outside sources. Inputs are compatible with ±24 volt sources, with a threshold of approximately 2.5 volts.*

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13 – WARRANTY PARTS AND SERVICE

1. Consult the **GE Drive Products Operation Terms and Conditions** for details of the product warranty, including warranty period and parts or service coverage.

2. Warranty replacement parts or service assistance may be obtained direct from the factory during regular working hours **(8 a.m. - 5 p.m.; M-F)** by calling:

   GE Drive Products Operation
   Product Service Engineering
   Erie, PA
   (814) 875-3851

3. An **EMERGENCY HOT-LINE** has been established by **GE Drives Products Operation** to provide a 24-hour telephone answering service to our customers.

   GE Drive Products Operation
   EMERGENCY HOT-LINE
   (814) 875-3219

   For a **Prompt Response The Next Working Day** the answering service operator will record the required information and your **Name, Company and Phone Number.** In cases of emergency, where assistance must be obtained immediately, the operator may be requested to contact a Customer Service Representative (for parts) or Product Service Engineer (for technical assistance) at home.

4. So that we may serve you more expediently, please have the following information ready whenever contacting the factory:

   Drive Serial Number & Model Number
   Part Number & Description

5. **GE Drive Product Operation** may request that failed parts, replaced under warranty, be returned to the factory. A **Return Authorization** tag or number is to be obtained and included with any parts being returned. When returning parts, pack them carefully to prevent damage in transit and ship to:

   General Electric Company
   Drive Products Operation
   1100 Lawrence Parkway
   Erie, PA 16531
   Attention: Parts Repair and Exchange

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14 – GLOSSARY OF TERMS

Base Speed	—	Maximum speed of a motor at rated armature volts and field amps.
DFP	—	Delayed Firing Power to armature SCRs.
DFA	—	Digital Field Auxiliary.
EEROM	—	Electrically Erasable Read-Only Memory, also called EE or E^2 (E squared). Contains the drive Parameters.
EE Enable Device	—	Device used to "Enable" or "Protect" EEROM programming. A hardware jumper (JP13) located on the Control card.
Field Weakening	—	Lowering of field current at rated armature volts to allow higher speeds. Actual armature voltage where field weakening begins is called the crossover point (adjusted by Parameter ADR079-CROSS).
FUP	—	Firing Microprocessor, U5 on the Control card.
"Hard" Reset	—	Hardware Reset, resets the drive cards. Used to clear some faults and allow certain parameter changes. Activated by depressing SW2 on the Control card or connecting 3TB69 to 3TB66.
K1	—	"RUN" Relay located on Power Supply card.
K2	—	MA Auxiliary Relay, pilot relay for MA contactor. K2 is located on the Power Supply card.
K3	—	Fault Relay located on the Power Supply card.
LED	—	Light Emitting Diode.
MA	—	Main AC Contactor (see General Description).
MD	—	Armature circuit DC contact (see General Description)
MFC	—	Motor Field Control (see General Description).
MOV	—	Metal Oxide Varistor.
MUP	—	Microapplication Microprocessor, U33 on Microapplication card.
Parameters	—	Adjustable software settings used to program and tune the drive regulators.
PI	—	Process Interface (card).
PL	—	Plug (may be plug or receptacle).
PSC	—	Power Supply card.
PUP	—	Programmer Microprocessor, U1 (and U3 PROM) on Programmer card.
RAM	—	Random Access Memory.
RS232C	—	Serial Link Communication Standard.
RS422	—	Serial Link Communication Standard.
RUP	—	Regulator Microprocessor, U4 (and U1 PROM) on Control card.
SCR	—	Silicon Controlled Rectifier (Thyristor).
SLD	—	Signal Level Detector, SLD0 thru SLD5, (RX1 or K1 thru RX5 or K5) on optional Microapplication card and SLD5 on 3TB.
"Soft" Reset	—	Software Reset initiated by pressing the RESET key on the optional Programmer.
TUP	—	Tachometer Microprocessor, U6 on the Control card.



GE DRIVE PRODUCTS OPERATION

GENERAL ELECTRIC COMPANY
DRIVE PRODUCTS OPERATION
1100 LAWRENCE PARKWAY
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